



EXPANDED PROGRAM ON IMMUNIZATION (EPI)



Save the Children

Learning objectives

At the end of this lesson the students will be able to:

- Define immunization, immunity and vaccine
- Discuss about classification of immunity.
- Identify immunization schedule.
- Recognize C/I, S/E, and their measure to be taken



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Introduction

- **Immunity:-** is the resistance of the body against a disease producing agent.
- **Immunization:-** The production of immunity by artificial means or the process of becoming immune.
- **Vaccines :-** is an antigen design to induce immunity against the particular pathogen / production of Abs/.
- **Toxioid :-** is a modified bacterial toxin that has been rendered nontoxic, and retains the ability to stimulate the formation of antibodies to the toxin



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Why immunization?

- The most successful and cost effective health intervention
- 2 to 3 million deaths averted annually by vaccination



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The immune system

- The main function of the immune system is to prevent or limit infections by microorganism such as bacteria, virus, fungi and parasites
- The foetal immune system develops in a sterile and protected environment, and therefore **lacks antigenic experience**
- Soon after birth, the newborn is exposed to the "**hostile world**" of bacteria, viruses, fungi, and parasites, and must immediately defend itself



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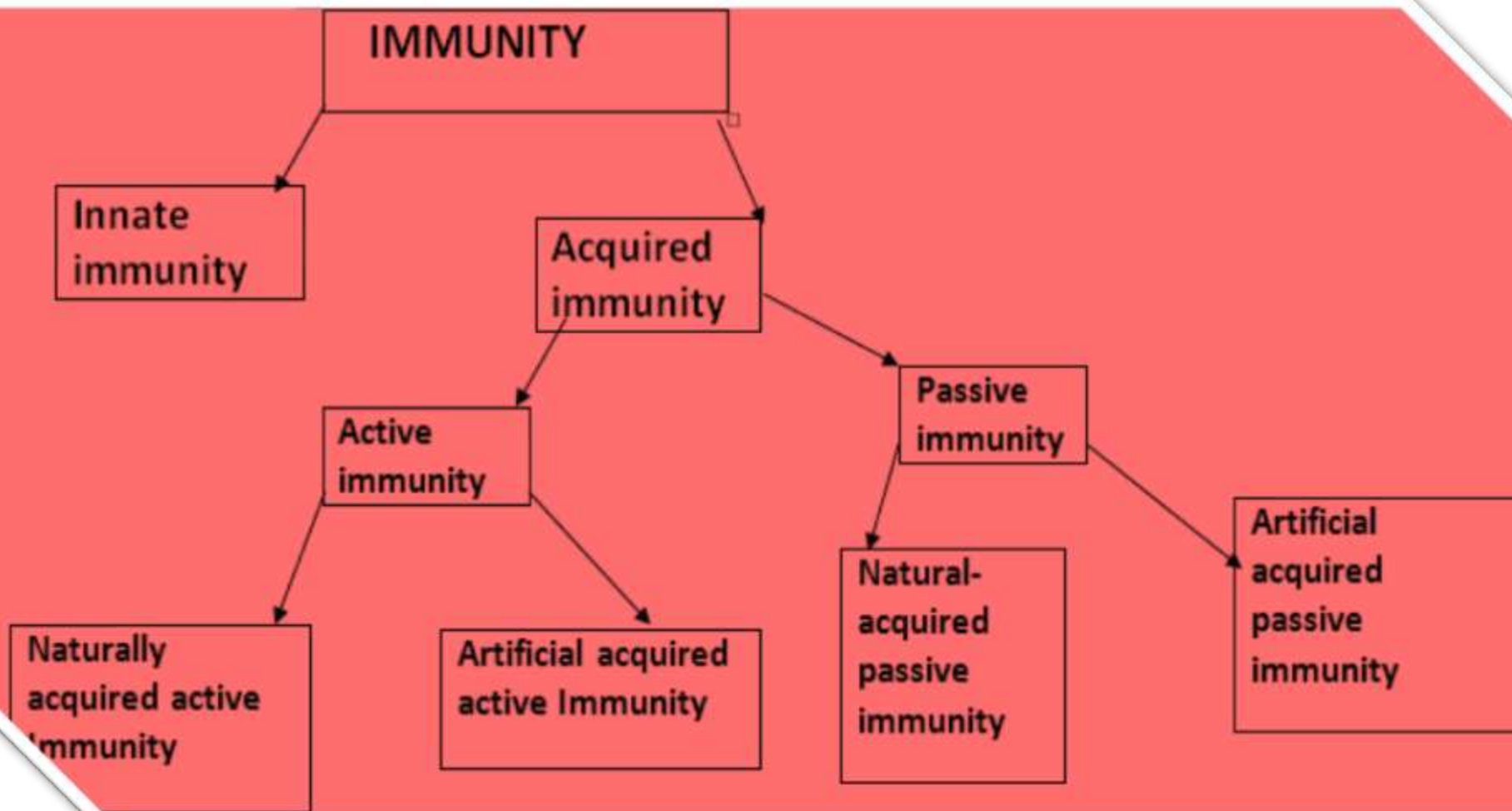
The immune system

- The immunologic competence of the neonate progresses rapidly in the first three months of life, as the cells involved in acquired immunity mature and gain antigenic experience
- During this period, the neonate mainly depends upon components of the innate or antigen-independent immune system'
- The overall performance of the immune system in the neonatal period is diminished in several important respects
- As a result, very young infants are more susceptible than older infants to serious bacterial infection, as well as some viral and fungal infections



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Classification of immunity



Immunity divided in to two

I. Congenital or innate

- ✓ Non-specific immunity.
- ✓ It is the natural resistance of body
- ✓ The natural immune process have no memory ex. Skin, mms, WBC etc.

II. Acquired

- ✓ specific immunity
- ✓ Active acquired immunity
 - The child body makes its own antibodies.
 - Long lasting immunity.
 - Characterized by long term memory
- ✓ Passive acquired immunity
 - The child gets ready-made antibodies (Ab). Short live immunity.



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Active acquired immunity divided in to two.

A. Naturally acquired active Immunity.

The child makes immunity after Exposure to a disease.

B. Artificial acquired active Immunity.

The child makes anti bodies after administration of antigen vaccine .



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Passive acquired immunity

A. Natural-acquired passive immunity

- Trans-placentally transferred Maternal antibodies in the body of the child. E.g TT vaccine for the mother.

B. Artificial acquired passive immunity.

- Formed antibodies are administered to the child.

e.g T.A.T



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Active immunity

- Is acquired when a person's own immune system generates the immune response
- lasts for many years or for a lifetime
- This long-term protection is the result of immunologic memory
- This immunity can occur after exposure to natural pathogens or after exposure to vaccines
- Vaccines mimic the characteristics of the natural antigen
- The immune system mounts a response and establishes an immunologic memory as it would for an infection



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Passive immunity

- Is produced when the immunoglobulins of one person are transferred to another.
- This immunity lasts only weeks or months
- Passive immunity can be obtained by injection of exogenous immunoglobulin
- It can also be transferred from mothers to infants via colostrum or the placenta



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Introduction to EPI Program



Innovation of vaccination/immunization

- On May 14, 1796, Edward Jenner, performed an experiment that would revolutionize public health.
 - He made two small cuts on the arm of an eight-year-old boy and inserted material taken from a sore on a woman infected with cowpox, a mild disease common to dairy workers.
 - Six weeks later, Jenner injected the boy with fluid from a smallpox lesion, and James did not contract smallpox.
- Inoculation of a person with relatively harmless disease material could protect the person from a more dangerous disease.



He called this process “vaccination”, derived from the Latin name for cowpox, vaccinia.



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Milestones in Vaccine Development

- **1885:** First use of live attenuated viral vaccine (rabies) in humans
- **1909:** First live attenuated bacterial vaccine (Bacillus Calmette Guerin, or BCG) created for use against tuberculosis.
- **1921:** Diphtheria toxoid developed
- **1924:** Tetanus toxoid produced
- **1930:** Pertussis vaccine developed
- **1932:** Yellow fever vaccine developed



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Cont...

- **1940s:** Diphtheria-tetanus-pertussis (DTP) **combination** introduced
- **1955:** Inactivated polio vaccine introduced
- **1963:** Live attenuated oral polio vaccine introduced
- **1963:** Measles vaccine introduced
- **1986:** First **recombinant** vaccine (hepatitis B) introduced
- **1990:** First polysaccharide **conjugate** vaccine (Haemophilus influenzae type b) introduced PCV (2011) & Rota 2012

EPI in Ethiopia

- EPI was launched in 1974 by the WHO to control vaccine preventable diseases
- The objective of the National Immunization Policy was to reduce mortality and morbidity in children from the EPI target diseases through the immunization of all children under the age of one.



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EPI in Ethiopia ...

- The Ethiopian immunization policy was updated in 2007.
 - Children of under-one year of age and women of reproductive age group (15-49 years age) are the targets for the EPI vaccines.
- There are no booster doses recommended in routine EPI for childhood immunization
- However, there are periodical **supplemental** doses for measles and polio.
- The policy recommends the use of **static** sites, **outreach** sites and **mobile** teams as appropriate strategies for delivering immunization services.



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EPI in Ethiopia ...

- Health Facilities conducting Routine Immunization
 - Health Posts
 - Health Centers
 - Hospitals
 - (NGO clinics – if appropriate)
 - (Private Clinics – if appropriate)
- EPI Staff
 - All HEWs
 - All HWs who have pre-service training on conducting EPI are responsible to conduct/lead EPI program in the assigned Health Facility.



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EPI Delivery Strategies

- **Static:** immunization performed as part of routine activity of the Health units.
- **Outreach:** an immunization approach in which the staffs of health unit go out and administer vaccine to mothers and children in their catchments areas.
- **Mobile:** an immunization approach only for **single dose vaccination** (measles, BCG) in nomadic, settlement areas and mostly used for controlling epidemics of measles.
- **Campaign:** an immunization approach conducted by mobilizing the community, example polio and measles vaccination.



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The Vaccines



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Introduction

Common observed Problems:

- A. Inappropriate dose and route of administration.**
- B. Misunderstanding of schedules for routine EPI.**
- C. False contraindications to vaccination**



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Nature and Type of vaccines

Types of Vaccine

- **Live attenuated**

- Virus, e.g., oral polio vaccine (OPV), measles, yellow fever
- Bacteria, e.g., BCG

- **Inactivated**

Whole

- Virus, e.g., inactivated polio vaccine (IPV)
- Bacteria, e.g., whole-cell pertussis

Fractional

- Protein-based
 - Subunit, e.g., acellular pertussis
 - Toxoid, e.g., diphtheria and tetanus
- Polysaccharide-based
 - Pure, e.g. meningococcal
 - Conjugate, e.g., *Haemophilus influenzae* type b (Hib)

- **Recombinant**, e.g., hepatitis B



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DTP-HepB-Hib (Pentavalent)



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DTP-HepB-Hib combination vaccine

- ▶ **Type of vaccine** – Fully liquid vaccine
- ▶ **Number of doses/child** – Three (*DTP-HepB-Hib*)
- ▶ **Schedule** - 6, 10, 14 weeks of age
- ▶ **Dosage** - 0.5ml
- ▶ **Injection site** - Outer mid-thigh
- ▶ **Injection type** – Intramuscular
- ▶ **Storage** - Store between 2° C to 8° C. Never freeze



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SAFETY AND SIDE EFFECT- Pentavalent vaccine (DPT-HepB-HIB)

- 95% effective for Hep B, Hib, Diphtheria, 80% for Pertussis
- Mild reactions following immunization are common
 - Pain and swelling, redness at the site of injection,
 - Fever, irritability, malaise
 - Self- limiting, hardly requiring even symptomatic treatment
- Reassure parents about such events so that they know about it
- Side effects are not contraindications for vaccination
- Contraindications – There are no contraindications to DTP except that
- It should not be given to children who have suffered a severe reaction to a previous dose.



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Measles Vaccine



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Measles vaccine

- ▶ **Type of vaccine** - Live attenuated
- ▶ **Number of doses** - two dose,
 - ▶ Second opportunity should be given in campaigns/Routine
- ▶ **Schedule** - At 9 –11 months of age
- ▶ **Dosage** - 0.5ml
- ▶ **Injection site** - Outer upper arm
- ▶ **Injection type** - Subcutaneous
- ▶ **Storage** - Store between 2° C–8° C (vaccine will not be damaged by freezing but the diluent should never be frozen).
- ▶ Soreness and pain at injection site- 24 hours
- ▶ 5% - fever 5-12 days after injection
- ▶ 5% mild rash 5-12 days after injection –last 2 days



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Measles vaccine ...

- ▶ **85% Efficacy**
- ▶ **Contraindications**
 - ▶ Severe reaction to previous dose
- ▶ **Adverse reactions**
 - ▶ Malaise, fever, rash 5–12 days later; idiopathic thrombocytopenic purpura; rarely, encephalitis, anaphylaxis
- ▶ **Severe reactions are rare:**
 - ▶ **Anaphylaxis 1/1,000,000 doses**
 - ▶ **Encephalitis :1/1,000,000 doses**



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Oral Polio Vaccine (OPV)



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Oral polio vaccine (OPV)

- ▶ **Type of vaccine** - Live attenuated
- ▶ **Number of doses** - Four in endemic countries (including birth dose)
- ▶ **Schedule** - At birth, 6, 10, 14 weeks
 - ▶ Additional dose: given in campaigns
- ▶ **Adverse reactions**
 - ▶ Headache , diarrhea muscle pain- <1%
 - ▶ VAPP very rare (approximately 2 to 4 cases per million children vaccinated)
- ▶ **Dosage** - 2 drops into the mouth
- ▶ **Storage** - Store between 2° C–8° C (maybe frozen for long-term storage).



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Oral Polio Vaccine (OPV)

- **Contraindications – There are no contraindications to polio vaccination**
- ✓ However, if a child has diarrhea when OPV is given, an extra dose should be administered
- ❖ If the OPV0 dose is not given within 14 days of birth, it should not be given at all, and the primary series should begin with OPV1 at six weeks of age.



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Inactivated polio vaccine

- On 16 December 2015 Ethiopia introduce Inactivated Polio Virus (IPV) vaccine against poliomyelitis.
- The introduction of IPV in Ethiopia is a step forward towards global polio eradication efforts
- Combining IPV and OPV provides stronger protection to the current oral polio vaccine (OPV) against polio.



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Inactivated polio vaccine

- The IPV will help provide additional protection to children against polio disease – and will give the child the benefits of both vaccines.
- IPV strengthens immunity in the blood while OPV strengthens immunity in the gut.
- Administered by IM, at 14 weeks, 0.5 ml



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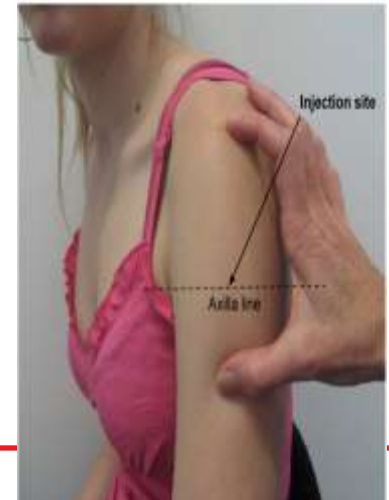
Tetanus Toxoid Vaccine (TT)



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TT vaccine

- ▶ **Type of vaccine** - Toxoid
- ▶ **Contraindications** - Anaphylactic reaction to previous dose
- ▶ **Adverse reactions** –
 - ▶ Soreness, swelling at the injection site 1-3 days
 - ▶ Mild fever -10%
- ▶ **Dosage** - 0.5ml
- ▶ **Injection site** - Outer upper arm(left arm)
- ▶ **Injection type** – Intramuscular
- ▶ **Storage** - Store between 2° C–8° C.



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Tetanus Toxoid vaccination schedule and period of protection

Dose	When to give	Level of protection	Period of protection
TT1	First contact	Nil	None
TT2	At least 4 weeks after TT1	80%	1-3 years
TT3	At least 6 months after TT2	95%	At least 5 years
TT4	At least 1 year after TT3	99%	At least 10 years
TT5	At least 1 year after TT4	99%	For all child bearing yrs and possibly longer



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TT vaccine

- Women who have received three doses of DTP during childhood or additional doses of tetanus toxoid containing vaccine during their school years don't need all five doses of TT in adulthood for protection
- They must retain their vaccination card for health workers to determine additional doses needed



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BCG Vaccine



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BCG vaccine

- ▶ **Type of vaccine** - Live attenuated bacterial
- ▶ **Number of doses** - One
- ▶ **Schedule** - At or as soon as possible after birth
- ▶ **Contraindications** –
 - ▶ Symptomatic HIV infection
- ▶ **Adverse reactions** –
 - ▶ Local abscess, regional lymphadenitis;
 - ▶ Rarely, distant spread to osteomyelitis, disseminated disease



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BCG vaccine

- ▶ **Special precautions –**
 - ▶ Correct intradermal administration is essential.
 - ▶ A special syringe and needle is used for the administration of BCG vaccine
- ▶ **Dosage - 0.05ml**
- ▶ **Injection site - Outer upper right arm**
- ▶ **Injection type – Intra-dermal**
- ▶ **Storage - Store between 2° C–8° C. (vaccine may be frozen for long-term storage but not the diluents)**
- ▶ **Protection: variable, 90% only for disseminated disease**



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Safety and side effects- BCG

- Most children will have :
 - Small raised lump at site of injection- disappear within 30 min
 - Red sore appear for two weeks.
 - Leaving a scar
- Generalized infection: 5/1,000,000 in immuno-compromised



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Pneumococcal Conjugate Vaccine (PCV)



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Pneumococcal Conjugate Vaccines (PCV)

- Streptococcus pneumoniae (pneumococcus) is the most common cause of pneumonia, sepsis, meningitis, and otitis media in young children
- The two available pneumococcal vaccines are
 - Pneumococcal conjugate vaccine (PCV) and
 - Pneumococcal polysaccharide vaccine (PPSV).
- PPSV contains 23 strains of S. pneumoniae.
- It does not provoke an immune response in children younger than 2 years of age
- PPSV is given to children older than 2 years of age who are at high risk for pneumococcal sepsis



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Pneumococcal Conjugate Vaccines (PCV)

- ▶ Conjugated to protein carrier to increase immunogenicity in infants
- ▶ PCV presentations:
 - ▶ 7-valent: 4, 6B, 9V, 14, 18C, 19F, 23F
 - ▶ **10-valent: 7-valent + 1, 5, 7F**
 - ▶ **Fully liquid in 1-dose and 2-dose vial without preservative**
 - ▶ 13-valent: 10-valent + 3, 6A, 19A
 - ▶ Fully liquid in 1-dose vial
 - ▶ 13 strains of *S. pneumoniae*



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PCV 10 - Synflorix™

- 2 dose vial
- No preservative
- VVM on cap; VVM 30
- Storage temp: + 2° to + 8° C (Do not freeze!)
- Administration at 6, 10 and 14 weeks
- Intramuscular injection in right upper outer thigh
- Pentavalent injection first in left thigh followed by PCV injection right upper thigh



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Pneumococcal vaccine reactions:

▶ Vaccine Reactions

▶ Mild reactions

▶ Local reactions

- Swelling and tenderness at injection site are common (50%)

▶ Systemic reactions

- Irritability and crying are very common; and transient fever $>39^{\circ}\text{C}$ (5%).

▶ Severe reactions:

- ▶ Rare allergic reactions (dermatitis) which occur in 0.1%

▶ Contraindications:

- ▶ PCV should not be administered to children with a known severe hypersensitivity reaction to a previous dose of the vaccine

- ▶ Infants with moderate or severe illness(temperature above 39°C) should not be vaccinated until their condition improves



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Rotavirus Vaccine (Rotarix)

VIAL = ONE DOSE



Cardboard boxes of x50; Standard 1.5ml tube

Rotarix...

Formulation	Rotarix vaccine
Administration	Oral Squeeze tube
Dose size	1.5 mls
Dose per tube	1
Storage Temperature	+2 C ⁰ to +8C ⁰
Shelf life	36 months from manufacturing date stored at appropriate temperature
Vaccine Vial Monitor	VVM 14
Packed volume per dose	17.1cm ³ per dose



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Efficacy of Rotarix Vaccination

- 50% efficacy in preventing significant deaths from **diarrhea**,
- 85% protection against **severe rotavirus gastroenteritis** and
- 100% protection against most **severe dehydrating rotaviral gastroenteritis episodes**



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Absolute contraindications to Rotarix

- Hypersensitivity after previous administration of rotavirus vaccines
- Severe immune deficiency
- Administration of Rotarix™ should be postponed in subjects suffering from diarrhea or vomiting and in need of rehydration therapy
- Mild illness such as an upper respiratory tract infection or mild diarrhea is **not** a contraindication



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Conditions which are **not** contraindications to immunization

- ▶ **Allergy or asthma** (except if there is a known allergy to a specific component of the vaccine mentioned above);
- ▶ **Any minor illness**, such as respiratory tract infections or diarrhoea with temperature below 38.5° C;
- ▶ **Family history of adverse events following immunization;**
- ▶ **Family history of convulsions, seizures, or fits;**
- ▶ **Treatment with antibiotics;**
- ▶ **Suspected HIV infection** with no signs and symptoms of AIDS;



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Cont...

- ▶ **Child being breastfed;**
- ▶ **Chronic illnesses** such as chronic diseases of the heart, lung, kidney, or liver
- ▶ **Stable neurological conditions**, such as cerebral palsy or Down's Syndrome;
- ▶ **Premature or low-birth weight** (vaccination should not be postponed);
- ▶ **Recent or imminent surgery;**
- ▶ **Malnutrition;** and
- ▶ History of **jaundice** at birth.



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Absolute Contraindications to Immunization

There are **only three contraindications** to vaccination:

- Do not give BCG or yellow fever vaccine to a child with **symptomatic HIV** infection or AIDS, but give the other vaccines
 - Do not give DPT-2 or -3 to a child who has had convulsions or shock within 3 days of the most recent dose
 - Do not give DPT to a child with recurrent convulsions or an active disease of the central nervous system
 - A child with diarrhoea who receive oral polio vaccine should be given a dose, but this dose should not be counted in the schedule
- Make a note on the child's immunization record that it coincided with diarrhoea, so that the health worker will give the child an extra dose.



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Immunization schedule for children under one of age, Ethiopia

Vaccine	Age				
	Birth	6 weeks	10 weeks	14 weeks	9 months
BCG	X				
Oral polio	X	X	X	X	
DPT-HepB-Hib		X	X	X	
PCV		X	X	X	
Rota		X	X		
Measles					X

The EPI schedule focuses on provision of immunization services against the EPI target diseases.



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Strategies to Boost immunization coverage

- Vaccinate children who come for other illness.
- Educate mother very well to reduce the no of defaulters.

Mass Immunizations

- Used in regions or countries where the immunization rate is low.
- Control an unexpected out-breaks.



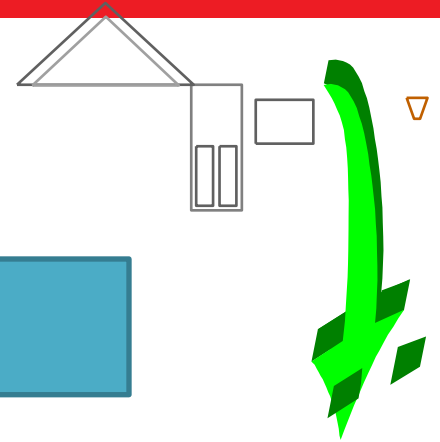
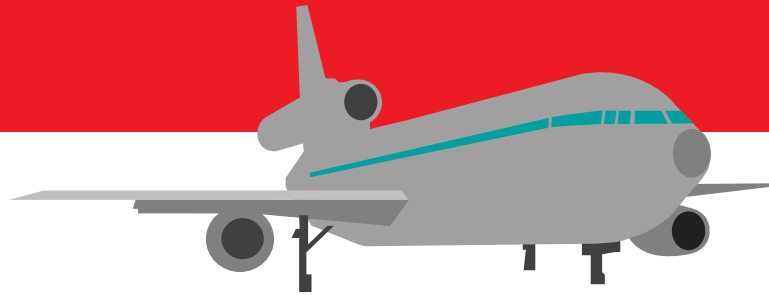
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Summary of common minor vaccine reactions

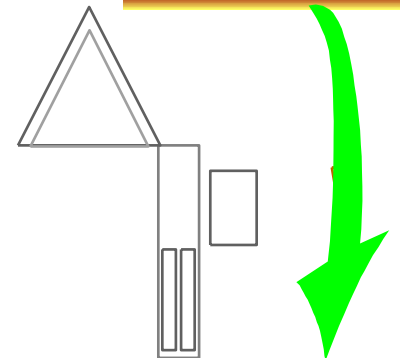
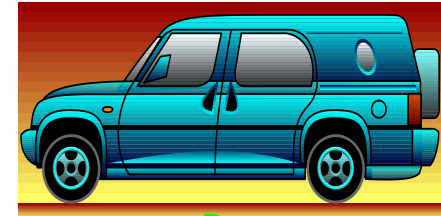
Vaccine	Local reaction (pain, swelling, redness)	Fever	Irritability, malaise and non-specific symptoms
BCG	common	-	-
Hib	5–15%	2–10%	-
Hepatitis B	adults up to 30% children up to 5%	1–6%	-
Measles/MMR	up to 10%	up to 5%	up to 5%
Oral polio (OPV)	none	less than 1%	less than 1% ^{a)}
Tetanus/DT	up to 10% ^{b)}	up to 10%	up to 25%
DTP ^{c)}	up to 50%	up to 50%	up to 60%



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The Cold Chain



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The Cold Chain

Cold chain is The system used for keeping & distributing vaccines in good condition

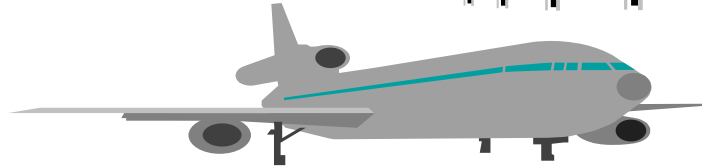
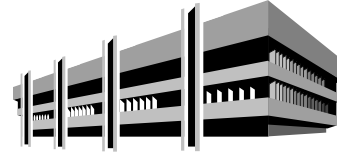
- Vaccines are sensitive to:
 - ✓ Heat
 - ✓ Freezing
 - ✓ Light
- So vaccines must be kept at a correct temperature.
- The system used for keeping (storing) and transporting (distributing) them in good condition (*correct temp*) is called *cold chain*.



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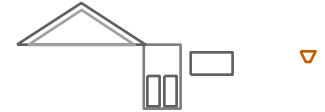
Cold Chain

Manufacturer



Cold chain = system to ensure the potency, the safety of vaccines during distribution to the point of use.

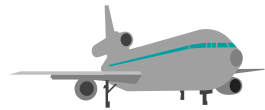
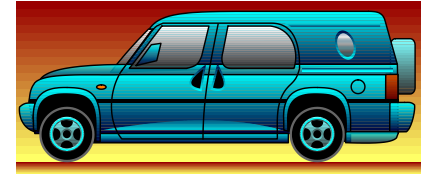
Central Cold room



Mobile strategy



**Regions /
Zone/District**



Out reach strategy

Health facility



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Major Cold Chain Equipments

1. Refrigerator
2. Cold boxes
3. Vaccine Carrier
4. Ice-Packs (0.6 lit & 0.4 lit)
5. Foam Pads



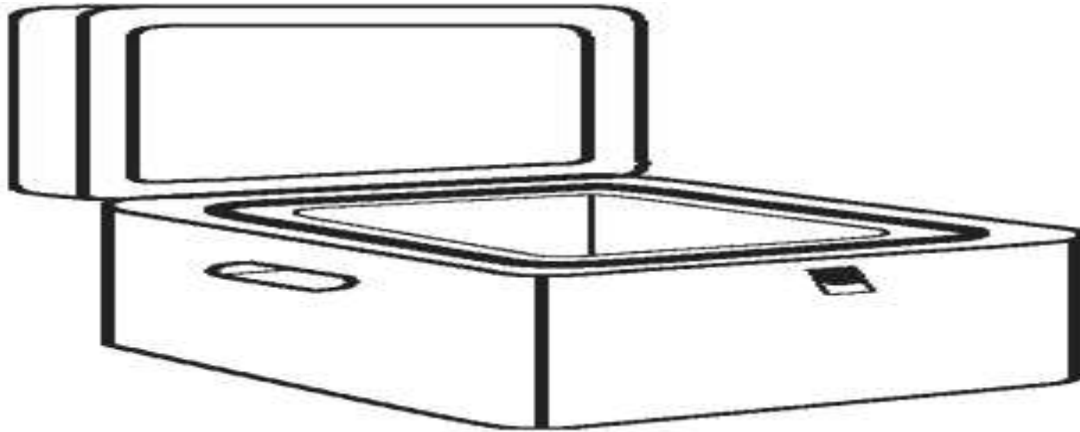
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Refrigerator in a HF should be able to hold

- ✓ A one-month supply of vaccines and diluents;
- ✓ Frozen ice-packs in the freezer compartment; bottles of water or unfrozen ice packs in the refrigerator compartment(bottom)
- ✓ Half the total space in the refrigerator should be left empty to allow air to circulate.

Cold boxes

- An insulated container
- Used to collect vaccines during transportation, refrigerator break
- Its cold life is from two to seven days



Vaccine carriers

- Are insulated containers
- Its cold life is maximum for 48 hours with the lid closed
- Used to transport vaccines and diluents to outreach sites



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Foam pads

- Is a piece of soft foam that fits on top of the ice-packs in a vaccine carrier
- Serves as a temporary lid to keep unopened vaccines
- To hold, protect and keep cool opened vaccine vials



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Foam pad



During an immunization session, vials are protected from heat for a longer period of time if they are inserted in a foam pad.



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Ice-packs

- Are flat, square plastic bottles that are filled with water and frozen
- Used to keep vaccines cool inside the vaccine carrier or cold box
- Every health facility should have sets of ice-packs for each of their cold boxes and vaccine carriers:



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Cold Chain Monitoring Equipment

- Vaccine vial monitors
- Vaccine cold chain monitor card
- Thermometers
- Freeze indicators (Freeze Watch, Freeze-tag)
- Fridge Tag



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Temperature Monitoring Device

The Vaccine Vial Monitor says...

if the expiry date is not passed,

 **USE** the vaccine

 **USE** the vaccine
FIRST

 **DO NOT USE**
the vaccine

 **DO NOT USE**
the vaccine

VVM

Thermometers



Freeze-tag



Fridge-tag



Fridge-tag®

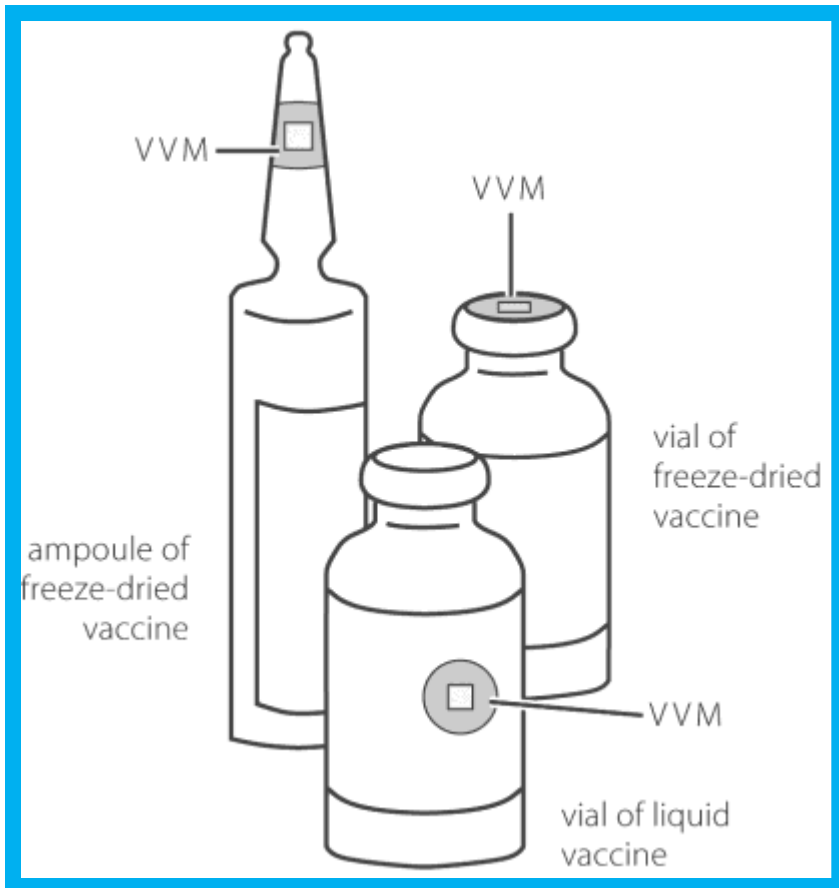


Fridge-tag®2



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Vaccine Vial Monitors (VVMs)



How it works?

- VVMs show the heat exposure only for the vial to which it is attached
- The VVM on liquid vaccine is on the label (but PCV), while the VVM on reconstituted vaccine is on the caps (measles, BCG, YF, some formulations of Hib)



How To Read A Vaccine Vial Monitor (VVM)

The Vaccine Vial Monitor says...

if the expiry date is not passed,



USE the vaccine



USE FIRST the vaccine



DO NOT USE the vaccine



DO NOT USE the vaccine



Inner square lighter than outer circle.
If the expiry date has not been passed, USE the vaccine.



At a later time, inner square still lighter than outer circle. *If the expiry date has not been passed, USE the vaccine.*



Discard point:
Inner square matches colour of outer circle.
DO NOT use the vaccine. Inform your supervisor.



Beyond the discard point:
Inner square darker than outer circle.
DO NOT use the vaccine. Inform your supervisor.



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Category of VVM

Category: (Vaccines)	No. days to end point at +37°C	No. days to end point at +25°C	Time to end point at +5°C
VVM30 HIGH STABILITY	30	193	> 4 years
VVM14 MEDIUM STABILITY	14	90	> 3 years
VVM7 MODERATE STAB.	7	45	> 2 years
VVM2 LEAST STABILITY	2	NA*	225 days
*VVM reaction rates determined at two temperature points			



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Arrangement of vaccines



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Vaccines in Inclined refrigerator



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Correct conditions for storing EPI vaccines



Recall- What Are Vaccines Sensitive To ?



Heat



Cold



Light

Irreversible Damage- Physical Appearance may remain unchanged but potency might be lost.



DILUENTS CAN PLAY ROLE IN POTENCY LOSS OF VACCINE

5



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Sensitivity to Heat

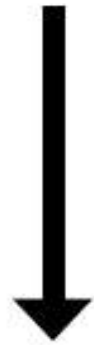
- All vaccines are sensitive to heat to some extent, but some are more sensitive than others.
- All freeze-dried vaccines becomes much more heat sensitive after they are reconstituted.



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Heat sensitivity

Most sensitive



Least sensitive

- Live oral polio vaccine (OPV)
- Measles (Lyophilized) *
- Pertussis and Mumps (Lyophilized)
- Hepatitis B
- Adsorbed Diphtheria-Pertussis-Tetanus vaccine (DPT)
- Adsorbed Diphtheria-Tetanus vaccine (DT, Td)
- BCG (Lyophilized) *
- Tetanus Toxoid (TT)

* Note: These vaccines become much more heat sensitive after they have been reconstituted with diluent.



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Sensitivity to cold (Freezing)

Some vaccines are sensitive to cold and loss potency at temperature below 0°C.

Range	Vaccine
most sensitive	DTP-HepB-Hib
	PCV
	TT
least sensitive	Rotarix



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Sensitivity to light

- ▶ Some vaccines are also very sensitive to strong light (Ultraviolet light) loss of potency.
- ▶ **BCG, Measles, (supplied in dark brown glass)**



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Actions to take when temperature is incorrect

Temperature ranges			
2 – 8 °C	below 0 °C	8 – 10 °C	Above 10 °C
Situation normal No action necessary	Vaccine at risk Take immediate action to correct the situation Carry out a shake test Make a report	If there has been a temporary power failure, no further action is necessary Check the refrigerator is working Monitor the situation closely	Vaccine at risk Take action to implement the agreed contingency plan Make a report



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Shake test and contingency plan

- ✓ Whenever it is suspected that vaccine has been frozen, at least one member of the duty staff should know how to perform and interpret a 'shake test'
- ✓ Most importantly, all responsible staff should know when and how to respond in the event of equipment failure (Contingency plan)



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Shake Test

Frozen test vial



Frozen control vial



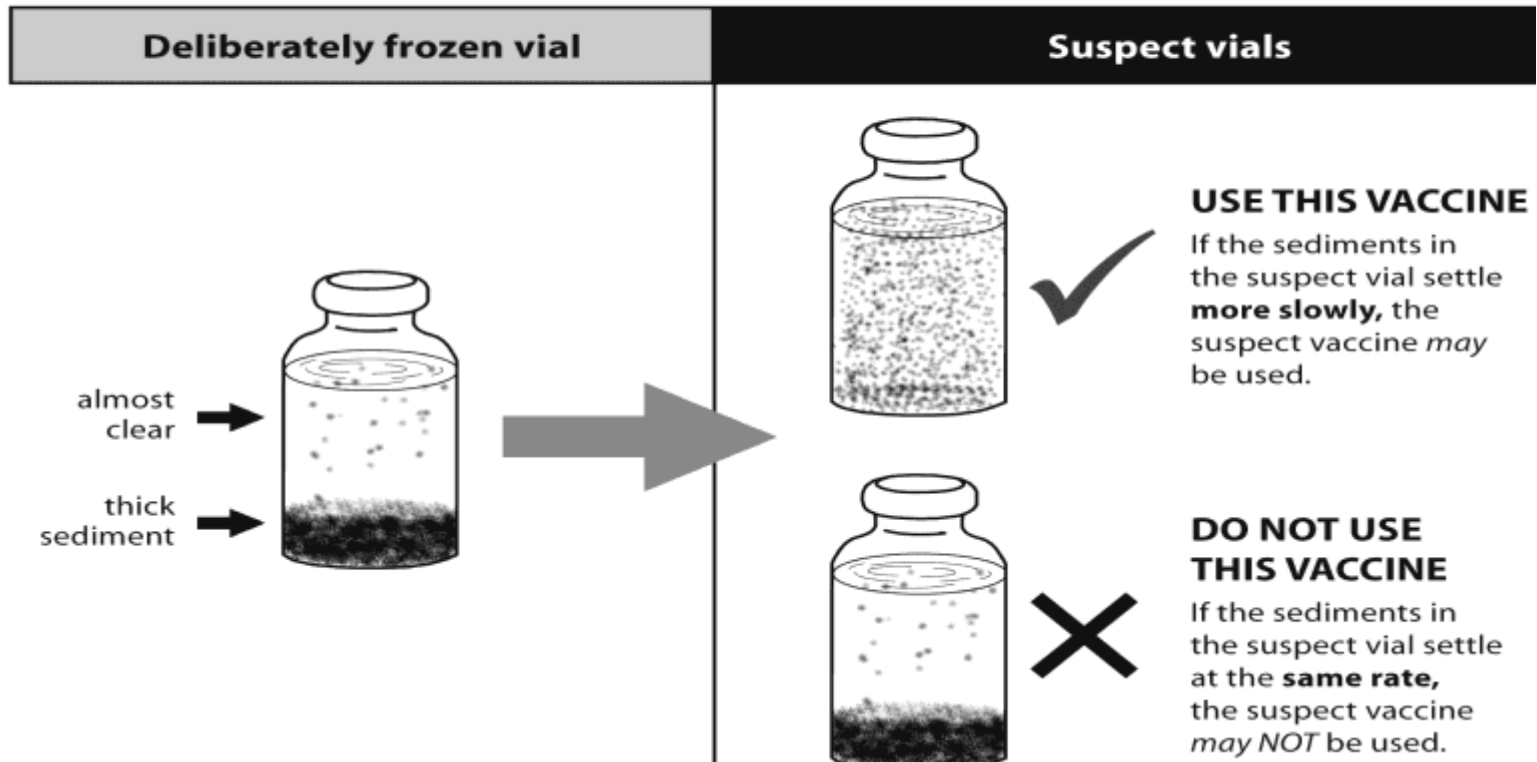
Non-frozen test vial



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The shake test

Compare the deliberately frozen vial next to the suspect vial



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Transportation freeze sensitive vaccines; Prevention of Exposure to Freezing

- ✓ Chill the water packs in the bottom of the refrigerator at list for 24 hours and use the chilled water pack.
- ✓ If you only have frozen ice packs – condition it
- ✓ Don't use ice for the storage or transportation of freeze sensitive vaccines
- ✓ Remember the vaccines are both heat and freeze sensitive.



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Cont...

Vaccine Transportation

- Freeze sensitive vaccines be transported in separate cold boxes with conditioned ice packs/chilled water from center to regions and zones
- Freeze sensitive vaccines be transported with conditioned ice packs/chilled water from zonal to Wereda, health facility and outreach level in 2-8 ° C.



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After vaccines reach the health facility you must:

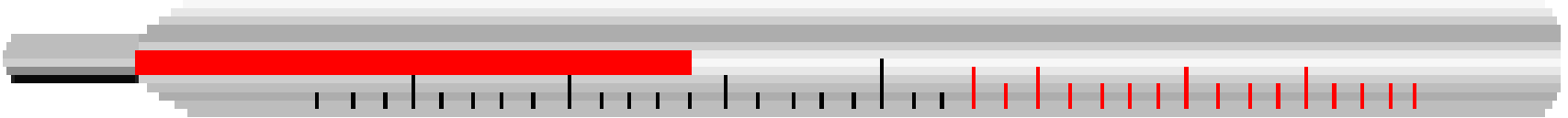
- Keep them between 2°C and 8°C in your health facility refrigerator.
- Carry them to the immunization session in a vaccine carrier with chilled water packs or conditioned ice packs
- Keep the vaccines cool using a foam pad in the vaccine carrier while you immunize the children.



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Vaccine Cooling

For storage the vaccines, the temperature should be stable within a range between $+2^{\circ}\text{C}$ and $+8^{\circ}\text{C}$



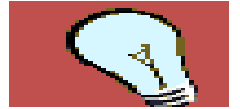
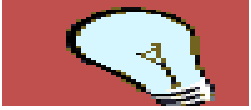
Do not store any vaccines, until the appliance has performing the required stable operation temperature.

Perform a daily check and record the temperature when the unit is in operation.



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VACCINES STORAGE



IMPORTANT

- Store your vaccines such that air can circulate between the individual packages.
- Keep the appliance tidy and dry inside



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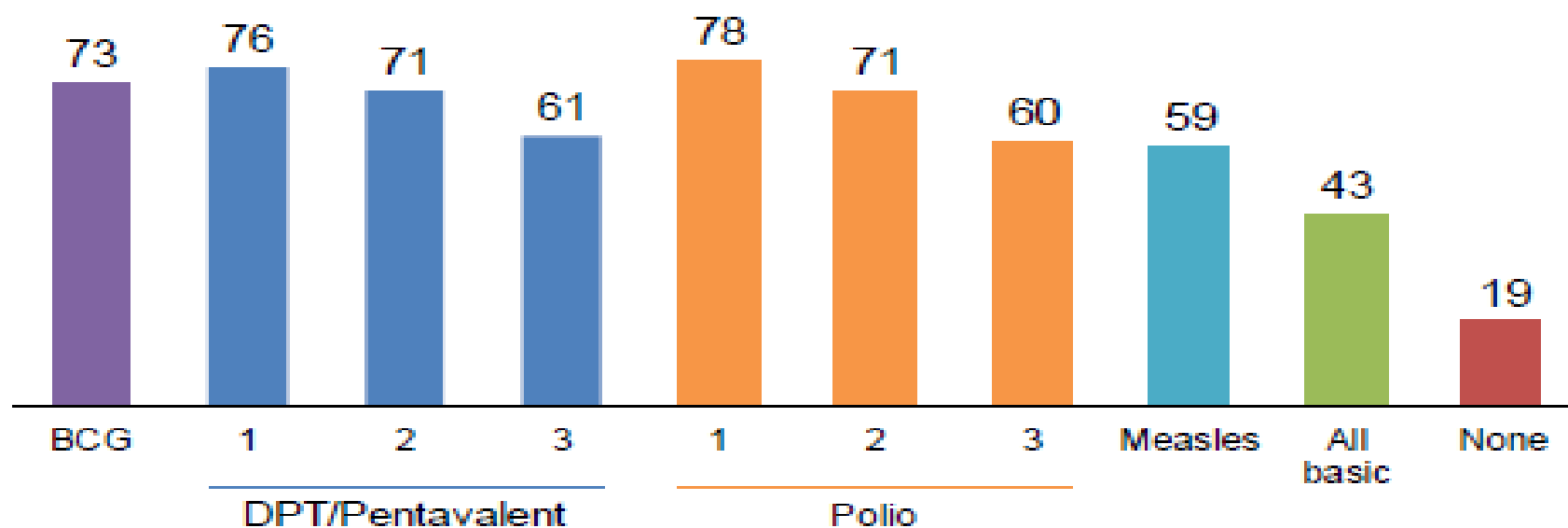
IMMUNIZATION COVERAGE

- Immunization coverage is the proportion of the target population that has been vaccinated
- The difficulty of calculating and interpreting immunization coverage rates often stems from problems related to estimating the size of the target population
- Thus, in turn, are the result of inaccurate or outdated census counts, population migrants, and unforeseen changes in birth rates or infant mortality
- At the end of the month, health staff add the daily tallies of vaccinations and converts them into a percentage called coverage.



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Percentage of children age 12-23 months vaccinated at any time before the survey



SOURCE: EMDHS 2019



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Calculating annual coverage for DPT1, DPT2, DPT3 and measles vaccines 2010

❖ **Penta1: coverage:**

Number of immunized by 12 months with **Penta1** in 2011 * 100

Number of surviving infants < 12 mos. Of age in 2011

❖ **Penta 3 coverage:**

Number of immunized by 12 months with **Penta 3** in 2011 * 100

Number of surviving infants < 12 mos. Of age in 2011

❖ **Measles coverage :**

No of immunized by 12 months with measles in 2011 * 100

Number of surviving infants < 12 mos. Of age in 2011

Estimating the number of surviving infants

- Total population : 5,500, 000
- Crude birth rate (CBR): 30/1000
- Infant mortality rate (IMR): 80/1000
- Number of surviving infants= Total population * CBR * (1- IMR)
- $= 5, 500,000 * 30/1000 * (1- 0.080)$
- $= 5,500, 000 * 0.03 * 0.920$
 $= 151, 800$



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- If coverage rates indicates that people are not using immunization services, **health workers and their supervisors need to find the causes and take appropriate action**
- Comparing coverage rates for different vaccines sometimes points toward the likely **problem as well as possible solutions**
- Eg, the data coverage shows BCG 80%, penta1 60%, penta3 55% measles 55%
- What is the issue here? Why the decrease between BCG and penta1 vaccines?
- What is the possible reasons?



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Vaccine Wastage

- **Some causes of Vaccine wastage**
 - Poor reconstitution practices
 - Cold chain failures
 - Small session size
 - Lack of knowledge how to read and interpret VVMs
 - Contamination



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Calculating Vaccine usage and wastage Rates

Vaccine Usage Rate/VUR/

To Calculate usage rate /UR/ we should have the following

1. Children Immunized= CI
2. Beginning Balance= BB
3. Received during the month= RDM
4. Ending Balance =EB

$$\text{❖ VUR} = \frac{\text{CI}}{(\text{BB} + \text{RDM}) - \text{EB}} * 100$$

$$\text{❖ Vaccine Wastage Rate/VWR/}$$
$$\text{VWR} = 100 - \text{VUR}$$



Calculating Vaccine usage and wastage.....

Examples

Vaccine Usage Rate/VUR/ for PCV

CI = 200 BB= 100

RDM = 200 EB= 90

$$\text{VUR} = \frac{200}{(100+200)-90} * 100$$

$$\text{VUR} = \frac{200}{210} * 100 = 95.2 \%$$

Vaccine Wastage Rate

VUR= 95.2% (PCV)

Calculation

$$\text{VWR} = 100 - 95.2 = 4.8\%$$



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Calculating Vaccine Wastage Factor

Indicate how much additional vaccine should be ordered in order to allow for the given wastage rate

Example:

Wastage rate: 30%

To Calculate Wastage Factor/WF/ we should have wastage rate

$$\text{WF} = \frac{100}{(100 - \text{Wastage rate})}$$

“wasted rate” – is the number of doses (in %) wasted

$$\text{WF} = \frac{100}{(100 - 30)}$$

$$\text{WF} = \frac{100}{(100 - \text{Wastage rate})}$$

$$\text{WF} = 1.43$$

1.43 times more vaccine should be ordered so as to cover the estimated 30% wastage vaccine



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Calculate coverage, Drop out rate and unimmunized children;

e.g. Children receiving Penta1, Penta2, Penta3 and Measles are 288, 250, 192 and 160 respectively and 800 doses of penta vaccine was used. If SI of the kebele is 320;

1. What is Penta1, Penta3 and Measles coverage?
2. What is Penta1 –penta3 and Penta1 – Measles drop out rate?
3. What is unimmunized children with Penta3 and Measles?



Save the Children

Calculate vaccination coverage, Drop out rate and unimmunized children;

A. Penta1, Penta3 and Measles coverage (%)

$$\text{\%Penta1} = 288/320 * 100 = 90\%$$

$$\text{\%Penta3} = 192/320 * 100 = 60\%$$

$$\text{\%Measles} = 160/320 * 100 = 50\%$$



Save the Children

Calculate vaccination coverage, Drop out rate and unimmunized children;

B. Dropout Rate

Penta1-Penta3 dropout rate -

$$\frac{(\text{Penta 1}-\text{Penta 3}) \times 100}{\text{Penta 1}}$$

Penta 1

$$(288-192)/288*100 = 33\%$$

Penta1-Measles dropout rate -

$$\frac{(\text{Penta 1}-\text{Measles}) \times 100}{\text{Penta 1}}$$

Penta1

$$(288-160)/288*100 = 44\%$$



Save the Children

Calculate vaccination coverage, Drop out rate and unimmunized children;

C. Un-immunized children

- **No. of unimmunized children with DPT- HepB-Hib3** = Surviving infants - No. of children vaccinated with DPT-HepB-Hib3
 - $320 - 192 = 128$
- **No. of unimmunized children with measles** = Surviving infants - No. of children vaccinated with measles
 - $320 - 160 = 160$



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EPI TARGETED DISEASES (Vaccine Preventable Disease/VPD)



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TARGETED DISEASES FOR EPI

LEARNING OBJECTIVES

At the end of this lesson the students will be able to:

- Describe vaccine preventable diseases
- Prevention methods of these diseases
- Goals for eradication or elimination of some EPI target diseases covered
- Discuss the definition ,c/m and the management of EPI target disease.
- Recognize the Case definitions



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Introduction

- Globally, each year 130 million children are born, 91 million of which are in the developing countries
- However, around 10 million children under the age of five years die every year
- Over 27 million infants in the world do not get full routine immunization
- About 41% of these were in Sub-Saharan Africa and 34% in South Asia.



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Introduction

- Immunization being one of the most cost effective public health interventions which is directly or indirectly responsible to prevent the bulk of mortalities in under-fives
- WHO estimates 29% of deaths among children birth to 59 months of age are **vaccine-preventable**



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Diseases for which vaccination is routinely recommended

1. Tetanus
2. Diphtheria
3. Pertusis
4. Polio
5. Measles
6. Pneumococcal diseases
7. TB
8. Hepatitis B
9. Haemophilus influenzae type b (Hib)
10. Rota
11. Meningococcal disease
12. Mumps
13. Rubella
14. Varicella (chickenpox)
15. Hepatitis A
16. Herpes zoster (shingles)
17. Human papillomavirus (HPV)
18. Influenza



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1. Tetanus

- It is an acute, spastic paralytic neurotoxin illness.
- Greek words -“**tetanos and teinein**”, meaning rigid and stretched, which describe the condition of the muscles affected by the toxin, **tetano spasmin**,
- Tetanus was first described in Egypt over 3000 years ago (Edwin Smith Papyrus)
- **Causative agent:** *clostridium tetani*
- It produces a powerful toxin that affects the nervous system
- **Incubation period** : typically is 2–14 days, but it may be as long as months after the injury.



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Epidemiology

- Potent neurotoxin produced during the growth of bacteria in dead tissue such as the umbilical cord.
- The disease is common and serious in newborns
- Most babies who get the disease die
- Neonatal tetanus is common in rural areas where the deliveries are at home



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Epidemiology

- Tetanus spores are found throughout the environment, usually in soil, dust, and animal waste.
- Tetanus is acquired through contact with the environment



The usual locations for the bacteria to enter the body

- Contaminated wounds: major or minor
- Deep puncture wounds, crush wounds(such as those caused by rusty nails, splinters, or insect bites.
- May follow elective surgery, burns,, otitis media , dental infection, animal bites, abortion,IV drug access sites are also potential entryways for the bacteria.



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Mode of spread

- **Not** transmitted from person to person
- A person becomes infected when dirt enters a wound or cut
- Grows in deep cut wounds
- Women have additional risk due to child birth/abortion
- A new baby may get infected when the umbilical cord is contaminated with dirty hands, knives, razor, application of substances on stump (e.g. cow dung, soil), etc.



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Clinical manifestations

- Muscular stiffness of the jaw(‘trismus’) is the first symptom
- Stiffness of the neck, abdominal muscles, sweating, headache, fever, etc.
- Seizure
- Sardonian smile/risus sardonius
- Opisthotonos
- Airway obstruction and asphyxiation - laryngeal and respiratory muscle spasm



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Clinical manifestations

Trismus

- In generalized tetanus, the presenting symptom in about half of cases is trismus (masseter muscle spasm, or lockjaw)



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Clinical manifestations

- Risus sardonicus
- **Sardonic smile** of tetanus (risus sardonicus) results from intractable spasms of facial and buccal muscles



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Clinical manifestations

Opisthotonos

- **Opisthotonos** is an equilibrium position that results from unrelenting total contraction of opposing muscles, all of which display the typical board like rigidity of tetanus.
- The smallest disturbance by sight, sound, or touch may trigger a tetanic spasm



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Neonatal tetanus



- Neonatal tetanus (**tetanus neonatorum**), When a newborn is infected by tetanus during the first 28 days of life.
- When a woman delivers her baby in unsanitary conditions, a newborn baby may become infected.
- Newborn babies are normal at birth but stop sucking between 3 – 28 days



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Neonatal tetanus

- Stop sucking and bodies become stiff and death often occurs
- Typically, an apparently healthy baby will stop sucking after a couple of days due to lockjaw, developing stiffness, arching of the body and convulsions.



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Case definition

- **Suspected case:** Any neonatal death between three and 28 days of age in which the cause of death is unknown; any neonate reported as having suffered from neonatal tetanus between three and 28 days of age and not immunized
- **Confirmed case:** Any neonate with normal ability to suck and cry during the first 2 days of life **and**-cannot suck normally between 3 and 28 days of age, **and** becomes stiff and/ or has spasms (i.e. jerking of the muscles)



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Maternal Tetanus

- Maternal tetanus is When tetanus strikes a woman during pregnancy or within 42 days following the delivery
- Commonly occurs when a woman delivers her baby in unsanitary conditions, for example, on soil or a dirt floor or during unsafe abortion practices.
- It is most prevalent where poverty, lack of education and poor living conditions exist and where babies are delivered by untrained attendants.



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Diagnosis

- Diagnosis is based on clinical findings and the immunization history
 - Nonimmunized or partially immunized patient.
 - History of skin wound.
 - Spasms of jaw muscles (trismus).
 - Stiffness of neck, back, and abdominal muscles, with hyperirritability and hyperreflexia.
 - Episodic, generalized muscle contractions.



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Management

- **General measures:** IV fluid, O2, quite dark room,
- **Antibiotics:** to eradicate the source of tetanus toxin
- **Antitoxin (TIG or TAT):** to neutralizes circulating toxin
- **Control of muscle spasms:** Diazepam
- **Respiratory care:** gentle suctioning, tracheostomy



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Control of muscle spasms

- The patient should be admitted to a quiet, darkened room where all possible auditory, visual, tactile, or other stimuli are minimized.
- Diazepam controls spasms better and safer than other options (The initial dose of 0.1–0.2mg/kg every 3–6 hr given intravenously is subsequently titrated to control the tetanic spasms);



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Control of muscle spasms

- Other drugs which can be used in combination with diazepam include:
 - Chlorpromazine (1-5mg/kg/dose q8hr)
 - Phenobarbitone (loading dose 20mg/kg, then 2.5mg/kg/dose q12hr, increased to max 5mg/kg/dose q12hr)



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Antitoxin therapy

- Tetanus immunoglobulin (TIG)/human tetanus immunoglobulin should be given intramuscularly in a single dose (3,000 to 6,000 IU)
- If human serum immunoglobulin is unavailable, **tetanus antitoxin (TAT)** should be given, assuming sensitivity reactions to horse serum are negative
- The antitoxin is given intravenously and intramuscularly (half of the dose via each route)-
NEONATE



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Antimicrobial therapy

- Metronidazole (30 mg/kg/day, given at six hour intervals; maximum 4 g/day) or
 - Parenteral penicillin G (100,000 U/kg/day) is an alternative.
 - Treatment for 10 to 14 days is recommended
- ## Wound treatment
- After the patient has been sedated and received antitoxin, the - Wound should be thoroughly cleansed and debrided



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Supportive treatment

- Oxygen should be available.
- During early stages, oral feeding should be avoided because of the danger of aspiration
- A continuous intra- venous infusion can provide fluid



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Respiratory care

- Meticulous nursing care is imperative
- Gentle suctioning of oropharyngeal secretions is done although tracheotomy need not be considered a routine procedure, it should be done prior to the development of severe asphyxia



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Prognosis

- Mortality is about 90%
- Bad prognostic signs include
 - I. Onset in the first week of life
 - II. Interval between lock jaw (trismus) and onset of muscle spasms less than 48 hrs
 - III. High fever
 - IV. Tachycardia
- ❖ An attack of tetanus does not confer immunity so active immunization following recovery is a must



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Prevention

- Universal immunizations of pregnant mothers between 16 and 36 weeks of pregnancy with 2 doses of tetanus toxoid can prevent neonatal tetanus
- The second dose should be given at least 2 weeks before the expected date of delivery
- Immunization of infants with three doses of pentavalent
- Immunizing women of child bearing age women
- Those who recover from infection don't develop **natural immunity** and therefore, should be immunized
- Wound Management
- Tetanus toxoid (TAT) 3,000-5,000 U may be given intramuscularly after testing for hypersensitivity.



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Prevention

- Additionally, sterile delivery practices need to be Emphasized such as
 - Conducting delivery in a clean room
 - Sterile gloves to be worn while conducting delivery
 - Cutting the umbilical cord with sterile equipment
 - Sterile cord ties are to be used to the cord
 - Not to apply anything over the umbilical stump
 - ❖ These are also known as the following 5 cleans: **clean surface, clean hands, clean blade, clean cord and clean cord tie**

2. DIPHTHERIA

- It is an acute bacterial infection mainly disease of the tonsils, pharynx, larynx & nose.
- Diphtheria is Greek for leather
- **Causative Agent:** Corynebacterium species
- ✓ Typically Corynebacterium diphtheriae
- ✓ Rarely toxigenic strains of Corynebacterium ulcerans.
- ❖ Both toxigenic and nontoxigenic C. diphtheriae cause skin and mucosal infection, and rarely can cause focal infection after bacteremia.



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Case definition

WHO-Recommended Case Definition: Diphtheria

- **Clinical description:** An illness characterized by laryngitis or pharyngitis or tonsillitis and the presence of an **adherent membrane of the tonsils, pharynx, and/or nose**
- **Laboratory criteria for diagnosis:** Isolation of *Corynebacterium diphtheriae* from a clinical specimen

Source: WHO, 1998



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Mode of spread

- Person to person through close physical and respiratory contact
- Droplets and secretions from the nose, throat, and eyes of an infected person
- Skin diphtheria is spread through direct contact with diphtheria ulcers or through clothing and other articles that have been contaminated with fluid from the ulcers.
- **Diphtheria can also be transmitted through unpasteurized milk products.**



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Clinical manifestations

- The manifestations of *C. diphtheriae* infection influenced by :
 - Anatomic site of infection
 - Immune status of the host, and
 - Production and systemic distribution of toxin
- **Incubation period** - Average 2–4 days



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Respiratory tract diphtheria

TONSILLAR AND PHARYNGEAL DIPHTHERIA

- The primary focus of infection is the **tonsils or pharynx** (94%)
- Sore throat is the universal early symptom:
- **Within the 1st few days of respiratory tract infection (usually in the pharynx)** a gray-brown, leather-like adherent pseudomembrane
- Removal is difficult and reveals a bleeding edematous submucosa
- **Malaise & mild fever**
- Paralysis of the palate and hypopharynx
- A bull-neck appearance



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Diphtheria: This is a picture of the throat of a child who has diphtheria. Notice the thick gray coating over the back of the throat. If not treated, this child could die from suffocation.



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Diphtheria: This child has bullneck diphtheria.



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Nose and larynx

- **Next 2 most common sites**
- Common among infants
- Local signs and symptoms of inflammation develop
- Infection of the anterior nares, causes purulent, erosive rhinitis with membrane formation.
- Shallow ulceration of the external nares and upper lip



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Diagnosis :- C/m

Throat swab for culture & smear

DIPHTHERIA



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Management

- **Antitoxin** : Diphtheria antitoxin
- **Antibiotics**: Erythromycin is marginally superior to penicillin for eradication of nasopharyngeal carriage.
- 40–50 mg/kg/day QID PO or IV
- **Fluid diet**
- **Bed rest** usually for ≥ 2 wk
- **I.V fluid**
- Cutaneous wounds are cleaned thoroughly
- Cases are isolated



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COMPLICATIONS.

- Respiratory tract obstruction
- Cardiomyopathy, and/or demyelination of nerves.(Most common)
- Kidney tubule necrosis
- Thrombocytopenia

Prevention

- Vaccination
- Droplet precautions and contact precautions.
- Mgt of carrier : contact with diphtheria patients should also be given antibiotics
- Diphtheria toxoid can also be used to immunize adults during out break



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3. Pertussis (whooping cough)

- Pertussis or Whooping cough is a disease of the respiratory tract caused by bacteria
- Many children that contract pertussis have coughing spells that last for six weeks
- The disease is most dangerous in infants
- Is an acute contagious respiratory tract infection involving the mucous membrane lining the air passage .

ETIOLOGY

- *Bordetella pertussis* is the sole cause of epidemic pertussis



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Case definition

- **WHO-Recommended Case Definition:**

Pertussis

- **Clinical description:** A person with a cough lasting at least two weeks with at least one of the following: Fits of coughing ,Intake of breath accompanied by a whooping sound,Vomiting immediately after coughing and without any other apparent cause
- **Laboratory criteria for diagnosis:** Isolation of Bordetella pertussis or detection of genomic sequences by polymerase chain reaction (PCR).

Source: WHO, 1998



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Mode of spread

- Spreads very easily from child to child by droplets produced by coughing or sneezing, close contact
- In many countries the disease occurs in regular epidemic cycles of three to five years
- The disease spreads easily among susceptible people who live in crowded conditions.



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Clinical manifestation

- Incubation period is 5 -10 days
- Classically, pertussis is a prolonged disease, divided into
 - Catarrhal : 1-2 weeks
 - Paroxysmal (**Cough stage**): 2-6 weeks
 - Convalescent stages: weeks to months



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I. The catarrhal stage (1–2 wk)

- ✓ Begins insidiously after an incubation period ranging from 3–12 days .
- ✓ Characterized by the insidious onset of coryza (runny nose), sneezing, low-grade fever, and a mild, occasional cough, similar to the common cold
- ✓ The cough gradually becomes more severe, and after 1–2 weeks



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II. Paroxysmal stage (2–6 wk)

- ❖ Coughing marks the onset of the stage
- ✓ Cough begins as a dry, intermittent, irritative hack and evolves into the inexorable paroxysms that are the hallmark of pertussis.
- **Beginning a machine-gun burst of** uninterrupted coughs, chin and chest held forward
- Tongue protruding, eyes bulging and watering, face purple, until coughing ceases and a loud whoop follows as inspired air traverses the still partially closed airway.



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III. Convalescent stage

- Exhaustion is universal.
- At the peak of the paroxysmal stage, patients may have more than 1 episode hourly
- As the paroxysmal stage fades into the convalescent stage, the number, severity, and duration of episodes diminish.



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Pertussis: This child has pertussis (whooping cough). He has severe coughing spasms, which are often followed by a “whooping” sound. It is difficult for him to stop coughing and catch his breath.



Pertussis: This child has broken blood vessels in his eyes and bruising on his face because of coughing from pertussis.



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Complication

- The most common complication, and the cause of most pertussis-related deaths, is secondary **bacterial pneumonia**
- Young infants are at highest risk for acquiring pertussis



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Management

- Goals of therapy are to limit the number of paroxysms
- Antibiotics. Macrolides are preferred agents
- An antimicrobial agent is always given when pertussis is suspected or confirmed primarily to limit the spread of infection and secondarily for possible clinical benefit.
- Provide mental & physical rest
- Provide warm humidity.
- Sedative
- Good nutrition



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PREVENTION.

- Universal immunization of children with pertussis vaccine, beginning in infancy with periodic reinforcing doses, is central to the control of pertussis



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4. Poliomyelitis

- **Poliomyelitis**, literally meaning “gray spinal cord inflammation
- **Causative agent:** poliovirus (Genus Enterovirus)
- Type-1, 2, & 3 ;- all types can cause paralysis
- Type -1 most frequently for epidemics

Transmission

- Feco-oral
- It is contagious: usually spread from person to person.
- **Incubation period:** ranges from 6 to 20 days



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Polio: This child has a severely deformed leg caused by polio.



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Case definition

- **WHO-Recommended Case Definition:**
- **Clinical case definition:** Any child under fifteen years of age with acute flaccid paralysis (AFP)/**floppiness of one or more of the extremities** or any person with paralytic illness at any age when polio is suspected.

Source: WHO, 1998



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Clinical manifestation

- **Acute stage: Generally lasts 7 to 10 days.**
- Many include fever, pharyngitis, headache, anorexia, nausea, and vomiting.
- Illness may progress to aseptic meningitis and meningoencephalitis in 1% to 4% of patients.
- These patients develop a higher fever & severe headache with stiffness of the neck and back.



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Clinical manifestation

- ❖ Paralytic disease occurs 0.1% to 1% of those who become infected with the polio virus.
- **Flaccid paralysis of muscles**
- **Paralytic is asymmetrical leg, arms, trunk in that order.**
- ❖ Paralysis occurs twice as often in the lower extremity as in upper extremity
- ❖ Weakness is clinically detectable only when more than 60% of the nerve cells supplying the muscle have been destroyed.
- ❖ Paralysis of the respiratory muscles or from cardiac arrest if the neurons in the medulla oblongata are destroyed.



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DX

- Isolation of the virus from stool or throat swab



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Management

- **No specific Rx**
- Close monitoring of respiratory and cardiovascular functioning is essential during the acute stage of poliomyelitis along with fever control and pain relievers for muscle spasms.
- Mechanical ventilation, respiratory therapy may be needed depending of the severity of patients
- **Bed rest**
- **Nutrition**
- **Change position frequently**
- **Physiotherapy**
- **Avoid urine retention**



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Prevention

- A. Immunization against polio: The two vaccines**
- B. Isolation of cases**
- C. proper disposal of excreta**
- D. Active surveillance**



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5. MEASLES

- Measles is one of the communicable diseases still causing preventable mortality and morbidity in the country
- Is a highly infectious disease caused by a virus (paramyxovirus).
- Kills more children than any other vaccine preventable diseases
- Tends to occur in epidemic form causing more complications among malnourished children
- **ETIOLOGY: Measles virus**



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- Incubation period - 7–18 days

RISK FACTORS

- Severe malnutrition
- Low serum retinol levels
- Immuno compromised persons are associated with increased morbidity and mortality



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Mode of spread

- Measles is extremely infectious
- **Airborne droplets**
- **Contact** with nose and throat secretions
- Cases can infect others for several days before and after they develop symptoms
- Spreads easily in over crowded areas (schools, military barracks, health facilities, etc)



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Clinical manifestations

1. Prodromal phase
2. Exanthemata phase
3. Complete recovery



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Clinical manifestations

1. Prodromal phase

- Follows an incubation period of 10-12 days
- Lasts 3-5 days
- Characterized by:
 - Fever, cough, coryza, conjunctivitis
 - Malaise, myalgia, photo-phobia and peri-orbital edema
 - Koplik spot appear as 1 to 2 mm grayish white spots (discrete), is the pathognomonic sign of measles and appears 1 to 4 days prior to the onset of the rash



Save the Children

Clinical manifestations

- Koplik spots appear & disappear rapidly, usually within 12-18 hours.
- Koplik spots have been reported in 50–70% of measles cases but probably occur in the great majority

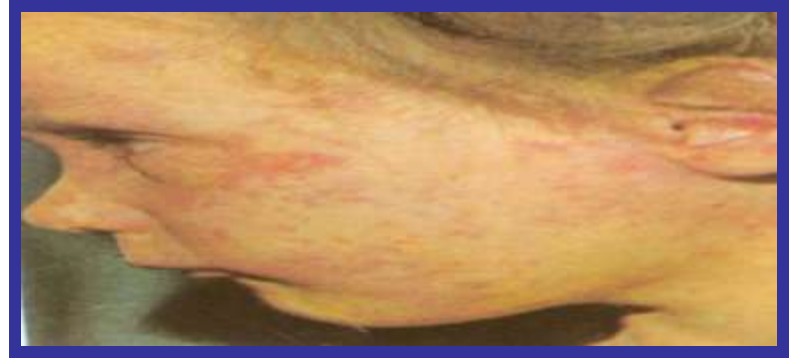


Save the Children

Clinical manifestations

2. Exanthemataus phase

- Characterized by:
 - High grade fever
 - Rash – begins at the hair line posteriorly and spreads caudally over the next 3 days with resolution of prodromal symptoms
 - Rash lasts 6 days and fades from head downwards (desquamations may be present)



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Clinical manifestations

3. Complete recovery – usually within 7-10 days from the onset of rash



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Case definition

➤ **Clinical case definition:** any person in whom a clinician suspects measles infection or any person with = fever + maculopapular rash + one of the following:

- Cough
 - Coryza (runny nose)
 - Conjunctivitis
- } The 3 c's

➤ At least a fourfold increase in antibody titer, or isolation of measles virus, or the presence of measles – specific IgM antibodies



Save the Children

Complication

- **Otitis media -the most common complication**
- Hemorrhagic measles-into GIT, mms,and CNS
- Thrombocytopenia
- Myocarditis
- Croup ,tracheitis, and bronchiolitis are common complications in infants and toddlers
- Reactivation of underlying mycobacterium tuberculosis
- Malnutrition
- Pneumonia
- Diarrhea
- Corneal scaring causing blindness



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Management

- Hydration
- Antipyretics
- Antibiotics (for secondary bacterial infection)
- Vitamin A (therapeutic dose): 1st, 2nd, and 14 days
- Admission for those with severe complications



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Management

■ Vitamin A in Measles Case Management

- Measles seriously depletes vitamin A in children, making them more susceptible to complications.
- For example, until recently, half of the blindness in children in one East African country resulted from corneal scarring following measles infection.
- Vitamin A supplementation for these children would have prevented this outcome.



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Strategies For Reduce Measles Morbidity And Mortality Reduction

- Routine immunization of $> 90\%$ of children aged 9 to 11 months
 - Provide a second opportunity for measles vaccination
 - Case-based measles surveillance
 - Improved case management including provision of vitamin A.
- ❖ Since 2002, Ethiopia adopted these regional goals and strategies and has been taking important steps to control and ultimately to eliminate measles by 2020.



Save the Children

Prevention

- Isolation – for infectious period
- Routine immunization
- Immunization campaigns



Save the Children

6. Pneumococcal Diseases

- A group of diseases caused by Pneumococcus
- Main pneumococcal diseases
 - Severe forms: Pneumonia, Meningitis & Bacteremia
 - Less severe forms: Otitis media, Sinusitis, Bronchitis,
 - Less common: arthritis, osteomyelitis, peritonitis, sinusitis



Save the Children

Streptococcus Pneumoniae (Sp) (Pneumococcus)

- SP is an exclusively human pathogen that colonizes the nasopharynx of normal hosts.
- The noses and throats of up to 70% of healthy people contain pneumococcus at any given time.

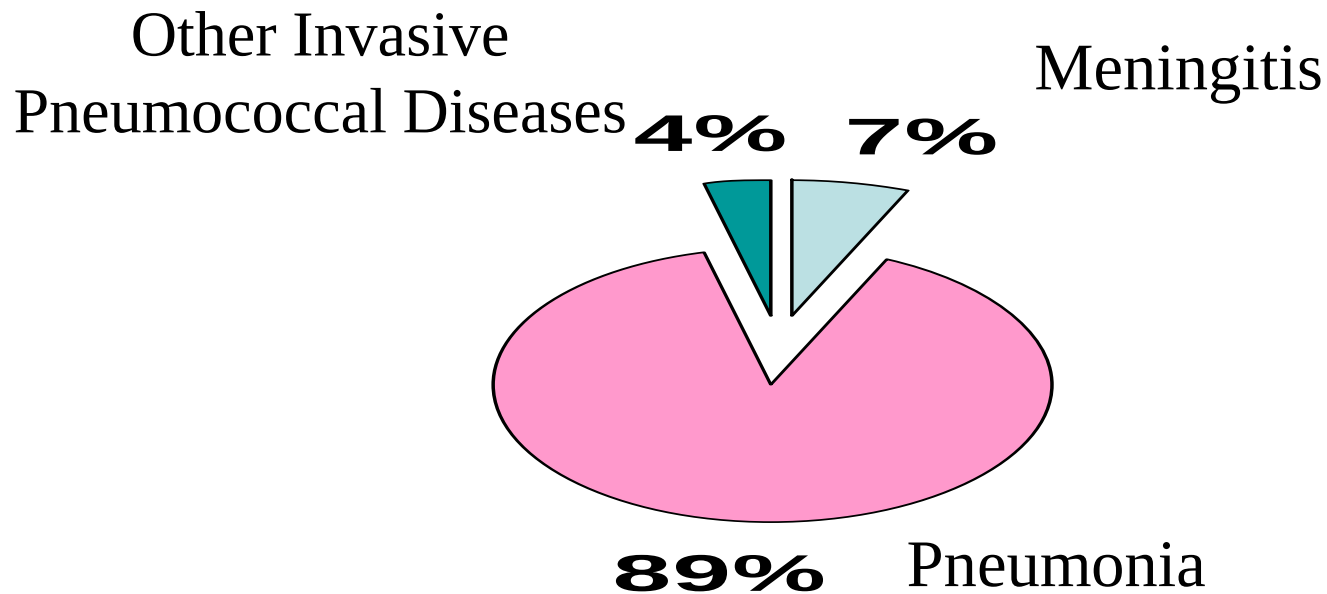


Spreads mainly by coughing or sneezing with air droplets



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Distribution of Pneumococcal Deaths by Syndrome



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Other causes of Pneumonia and Meningitis

- Pneumococcal infection is the leading cause of pneumonia death
- Pneumococcus **is not the only cause** of pneumonia and meningitis among children
- ✓ Hib (**Haemophilus influenzae type b**) is another important cause of meningitis and pneumonia
- ✓ Other causes of pneumonia among children may be: viruses (Respiratory Syncytial Virus, influenza virus) or bacteria (S. aureus, or Klebsiella pneumoniae)



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Pneumonia: Leading Child Killer

Pneumococcal disease

- Pneumococcus is the leading cause of child pneumonia deaths (~40%)
- About 1 in 10 child deaths is due to pneumococcal disease
- Ethiopia ranks 3rd among the top 10 countries with the largest number of pneumococcal deaths



Save the Children

Strategies For Preventing And Treating Pneumonia

- Protection
 - Breastfeeding promotion
 - Hand washing promotion
 - Zinc supplementation
 - Adequate nutrition
 - Reduce indoor air pollution
- Prevention
 - Vaccination
 - New: Pneumococcal, Hib
 - Routine: Measles, pertussis
 - HIV prevention
- Treatment
 - Improve health care seeking behaviour
 - Community case management (ICCM)
 - Health facility case management (IMNCI)
 - Antibiotics



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Key Points To Remember

- **One bacteria, many diseases- Pneumococcus can cause:**
 - Pneumonia
 - Meningitis
 - Bacteraemia
 - Other: middle ear infection (otitis media), sinusitis & bronchitis.
- **One disease, many causes - Pneumonia is caused by:**
 - Bacteria e.g. Streptococcus, Staphylococcus, Haemophilus
 - Viruses e.g. RSV, influenza
- **One vaccine, many diseases – Pneumococcal vaccines will prevent**
 - Pneumonia
 - Meningitis
 - Bacteraemia
 - Other: middle ear infection (otitis media), sinusitis & bronchitis



Save the Children

7. Hepatitis B

- Hepatitis B is a viral infection of the liver.
- Acute infection either resolves or progresses to chronic infection, which may lead to cirrhosis or liver cancer several decades later
- When it resolves, patients develop lifelong immunity.
- Caused by a virus that affects the liver
- Adults who get hepatitis B usually recover
- Infected infants and adolescents become chronic carrier



Save the Children

Mode of transmission

- Unsafe injection or needle stick
- Mother to child during birth
- To children during contacts through cuts, scratches, bites etc.
- Sexual intercourse

Sign and symptoms

- IP: 6 weeks mostly (but can be up to 6 months)
- Jaundice, bleeding symptoms with Hepatic failure & death
- Complications
 - Chronic hepatitis, cirrhosis, liver cancer

Prevention

Vaccine: Hep B, Pentavalent Vaccine



Save the Children

Hepatitis B Vaccine

Management of an infant born to Hep B positive mother:

- Infants born to HBsAg-positive mothers should be immunized within 12 hours after birth with hepatitis B vaccine and hepatitis B immune globulin (HBIG) at separate sites, regardless of gestational age or birth weight;
- The birth dose in such infants should not be counted in the series of three hepatitis B vaccines, and the full three-dose series should be administered starting at age of 6 wks



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Summary : VPD

Disease	Agent	reservoir	Spread	Duration of Immunity by infection	Risk factors for infection in Developing countries
Diphtheria	Toxin producing bacterium corynebacterium diphtheria	Humans	Close respiratory contact or contact with infectious material	Usually life long	Crowding
Meningitis and pneumonia caused by Hib	Haemophilus influenza type b, bacterium	Humans	Close respiratory contact	Usually life long	Not being breastfeeding Crowding

Summary : VPD

Measles	Virus	Humans	Close respiratory contact and aerosolized droplets	Lifelong	Crowding
Pertussis	Bordetella pertussis	humans	Close reparatory contact	Usually lifelong	Crowding
TB	M. Bacterium Tb	Humans	Airborn droplets	Not known; reactivation of old infection is common	crowding, immune deficiency , malnutrition



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Summary : VPD

Hepatitis B	Virus	Humans	Mother to Newborn, child to child, blood, sexual Transmission at birth or early childhood is dominant	If infection resolves, life long immunity; otherwise life long carrier status	Infected mother Unsafe injections Unsafe blood transfusions Multiple sexual partners
Polio	Poliomyelitis virus – serotypes 1, 2, 3)	humans	Fecal- oral Close respiratory contact	Life long type specific immunity	Poor environmental hygiene
Tetanus	Toxin producing bacterium (/c. Tetani)	Soil, animal, intestine	Spores enter body Umbilical cord of newborns	None	Home delivery Unsafe birth Exposure to animal feces, untreated wounds

Integrated Management of newborn & Childhood Illnesses (IMNCI)

Muhammed D.

(BSc, in Paediatrics and Child Health Nursing)

Email: ouseid867@gmail.com



Save the Children

Objectives

At the end of this session, students will be able to:

- Describe the advantage of IMNCI
- Assess, classify and treatment of common childhood illness
- Recognize general danger signs



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Introduction to IMNCI

- Children who come to health institutions are often suffering from more than one condition
- Making it impossible to give a single diagnosis.
- IMNCI is an integrated strategy and is based upon the combined treatment of the major childhood illnesses
- The IMNCI guidelines are designed for the management of sick children from birth up to five years old



Save the Children

IMNCI

- **What is it?**



A strategy to provide comprehensive & continuous care to a sick child

- **Purpose**



- To improve quality of care
- To reduce mortality among under five children

- **Approach?**



- Targets major killer diseases
- Use simple and effective tools

- **Prerequisites?**



- Commitment
- Skilled and motivated personnel
- Priority diseases selected



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IMNCI ...

- More than 70% of these child deaths are due to five diseases namely
 - Pneumonia
 - Diarrhoea
 - Malaria
 - Measles
 - Malnutrition



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IMNCI...

- These diseases are also the reasons for seeking care for at least **three out of four** children who come to health facilities.
- As children usually present with more than one of these conditions, **it was recognised that there was a need for an integrated approach in order to manage the child in a holistic manner**
- This lead to the development of the IMNCI strategy



Save the Children

IMNCI...

- IMNCI is an integrated approach to child health that focuses on the wellbeing of the whole child
- IMNCI aims to reduce death, illness and disability, and to promote improved growth and development among children under five years of age
- IMNCI includes both preventive and curative elements that are implemented by families and communities as well as by health facilities

IMNCI....

- IMNCI strategy includes three main components:
 1. Improving case management skills of healthcare staff
 2. Improving the health systems
 3. Improving family and community health practices



Save the Children

IMNCI...

- In health facilities, the IMNCI strategy:
 - Promotes the accurate identification of childhood illnesses in out-patient settings
 - Ensures appropriate combined treatment of all major illnesses
 - Strengthens the counselling of caregivers
 - Speeds up the referral of severely ill children.



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IMNCI...

- In the home setting, IMNCI:
 - Promotes appropriate care-seeking behaviours
 - Helps to improve nutrition and preventative care
 - Supports the correct implementation of prescribed care



Save the Children

IMNCI...

- The IMNCI case management process is presented on two different sets of charts:
 - Managing sick young infants aged from birth up to two months and
 - Managing sick children aged from two months up to five years



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IMNCI...

- First decide which chart to use depending on the age of the child
- Up to five years means the child has not yet had his or her fifth birthday
- If the child is not yet two months of age, the child is considered a young infant
- A child who is two months old would be in the group two months up to five years, not in the group birth up to two months



Save the Children

The IMNCI case management process

- The charts describe the following steps:
 - Assess the young infant or child
 - Classify the illness
 - Identify treatment
 - Treat the young infant or the child
 - Counsel the mother
 - Give follow-up care
 - Assess---Classify---Treatment----Counsel + follow up



Save the Children

Assessment

- Assess a child by checking first for general danger signs (or possible bacterial infection in a young infant)
- Asking questions about common conditions



Save the Children

Assessment (2-59 months)

- Look for general danger signs
- Check for cough or difficult breathing
- Check for diarrhoea
- Check for fever
- Check for ear problem
- Check for acute malnutrition
- Check for Anemia
- Check for HIV infection



Save the Children

Assessment (birth – 2 months)

- Check for very severe diseases and local bacterial infection
- Check for jaundice
- Check for diarrhoea
- Check for HIV infection
- Check for feeding problem or low weight for age



Save the Children

Classification

- Classify a child's illnesses using a colour-coded classification system
- Because many children have more than one condition, each illness is classified according to whether it requires:
 - Urgent pre-referral treatment and referral (**pink**), or
 - Specific medical treatment and advice (**yellow**), or
 - Simple advice on home management (**green**).



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Classification

- **Pink row** needs urgent attention and referral or admission for inpatient care.
- This is a severe classification
- A classification in a **yellow row** means that the young infant or the child needs an appropriate antibiotic or other treatment.
- The treatment includes teaching the mother how to give the oral drugs or to treat local infections at home.



Save the Children

Classification

- The HCPs advises her about caring for the young infant or child at home and when she should return
- A classification in a **green row** means the young infant or child does not need specific medical treatment such as antibiotics.



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Identify treatment

- After classifying all conditions, identify specific treatments for the child
- If a child requires urgent referral, give essential treatment before the patient is transferred
- If a child needs treatment at home, develop an integrated treatment plan for the child and give the first dose of drugs in the clinic
- If a child should be immunized, give immunizations



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Follow-up care

- When a child is brought back to the health institution as requested, give follow-up care and
- If necessary, reassess the child for new problems
- If this is the child's first visit for this episode of an illness or problem, then this is an **initial visit**
- If the child was seen a few days before for the same illness, this is a **follow-up visit**
- A follow-up visit has a different purpose from an initial visit



Save the Children

General danger signs (GDS)

- Since IMNCI takes a holistic approach to assessing, classifying and treating childhood illnesses;
- It is important to look for general danger signs as well as symptoms and signs of specific childhood illnesses
- The general danger signs are signs of serious illness that are seen in children aged two months up to five years and
- ✓ Will need immediate action/urgent attention to save the life of the child



Save the Children

General danger signs (GDS)

- There are five general danger signs
 1. Unable to drink or breastfeed
 2. Vomit every thing
 3. Had convulsing/ history of convulsions during the current illness
 4. Lethargic or unconsciousness
 5. Convulsing now



Save the Children

General danger signs (GDS)

Box 1.2 Checking for general danger signs

For **ALL** sick children ask the mother about the child's problem, then
CHECK FOR GENERAL DANGER SIGNS



CHECK FOR GENERAL DANGER SIGNS

ASK:

- Is the child able to drink or breastfeed?
- Does the child vomit everything?
- Has the child had convulsions?

LOOK:

- See if the child is lethargic or unconscious.
- See if the child is convulsing now.

A child with any general danger sign needs **URGENT** attention; complete the assessment and any pre-referral treatment immediately so referral is not delayed



Save the Children

General danger signs (GDS)

- A child with a general danger sign has a serious problem
- Most children with a general danger sign need urgent referral to hospital
- They may need life saving treatment with injectable antibiotics, oxygen or other treatments that may not be available in the health post



Save the Children

ASK: Is The Child Able To Drink Or Breastfeed?

- ✓ A child has the sign 'not able to drink or breastfeed' if the child is not able to suck or swallow when offered a drink or breast milk
- ✓ When you ask the mother if the child is able to drink, make sure that she understands the question
- ✓ If the mother replies that the child is not able to drink or breastfeed, ask her to describe what happens when she offers the child something to drink.
- ✓ For example, is the child able to take fluid into his mouth and swallow it?



Save the Children

Cont...

- ✓ If you are not sure about the mother's answer, ask her to offer the child breast milk or a drink of clean water
- ✓ Look to see if the child is swallowing the breast milk or water.
- ✓ A child who is breastfed may have difficulty sucking when his nose is blocked
- ✓ If the child's nose is blocked, clear it
- ✓ If the child can breastfeed after the nose is cleared, the child does not have the danger sign, 'not able to drink or breastfeed'.



Save the Children

ASK: Does The Child Vomit Every Thing?

- ✓ A child who is not able to hold anything down at all has the sign '**vomits everything**'.
- ✓ A child who vomits everything will not be able to hold down food, fluids or oral drugs
- ✓ A child who vomits several times but can hold down some fluids does **not have this general danger sign**
- ✓ When you ask the question, use words the mother understands
- ✓ Give her time to answer



Save the Children

Cont...

- ✓ If the mother is not sure if the child is vomiting everything, help her to make her answer clear.
- ✓ For example, ask the mother how often the child vomits
- ✓ Also ask if each time the child swallows food or fluids, does the child vomit? If you are not sure of the mother's answers, ask her to offer the child a drink. [See if the child vomits](#)



Save the Children

Ask: Has The Child Had Convulsions?

- During a convulsion, the child's arms and legs stiffen because the muscles are contracting or if the child has repeated abnormal movements
- The child may lose consciousness or not be able to respond to spoken directions
- Ask the mother if the child has had convulsions during this current illness.
- Use words the mother understands.
- For example, the mother may know convulsions as 'fits' or 'spasms'. **See also if the child is convulsing now.**



Save the Children

Look If The Child Is Lethargic Or Unconscious

- A lethargic child is not awake and alert
- The child is drowsy and does not show interest in what is happening around him
- Often the lethargic child does not look at his mother or watch your face when you talk
- The child may stare blankly and appear not to notice what is going on around him



Save the Children

Cont...

- An unconscious child cannot be wakened
- He does not respond when he is touched, shaken or spoken
- Ask the mother if the child seems unusually sleepy or if she cannot wake the child
- Look to see if the child awakens when the mother talks or shakes the child or when you clap your hands



Save the Children

Cont...

- However, if the child is sleeping and has cough or signs of difficult breathing, you must count the number of breaths first before you try to wake the child because it is easier to count the exact breathing rate when the child is calm
- When you have completed the above steps, you should record what you have found on the sick child case recording form
- You must circle any general danger signs that are found, process for general danger signs in practice.
Yes___ No___



Save the Children

Cont...

MANAGEMENT OF THE SICK CHILD AGE 2 MONTHS UP TO 5 YEARS

Child's Name: Fatuma Age: 18 months Weight: 11.5 kg Temperature: 37.5°C

ASK: What are the child's problems? cough, trouble breathing Initial Visit? ☒ Follow-up Visit? ☐

ASSESS (Circle all signs present)

CLASSIFY

CHECK FOR GENERAL DANGER SIGNS

NOT ABLE TO DRINK OR BREASTFEED

LETHARGIC OR UNCONSCIOUS

VOMITS EVERYTHING

CONVULSIONS

CONVULSING NOW

General danger sign present?

Yes ☒ No ☐

Remember to use danger
sign when selecting
classifications



Save the Children

SICK CHILD AGE 2 MONTHS UP TO 5 YEARS



Save the Children

Does The Child Have Cough Or Difficult Breathing?

ASK THE MOTHER WHAT THE CHILD'S PROBLEMS ARE

- Determine if this is an initial or follow-up visit for this problem.

- If follow-up visit, use the instructions on 'GIVE FOLLOW UP CARE' chart.

- If initial visit, assess the child as follows:

CHECK FOR GENERAL DANGER SIGN

ASK:

- ▶ Is the child able to drink or breastfeed?
- ▶ Does the child vomit everything?
- ▶ Has the child had convulsions?

LOOK:

- ▶ See if the child is lethargic or unconscious.
- ▶ Is the child convulsing now?

URGENT attention

A child with any general danger sign needs urgent attention, complete the assessment and any pre-referral treatment immediately so referral is not delayed

THEN ASK ABOUT MAIN SYMPTOMS:

Does the child have cough or difficult breathing?

IF YES, ASK:

LOOK, LISTEN, FEEL:

- Count the breaths in one minute
- Look for chest indrawing
- Look and listen for stridor
- Look and listen for wheezing

CHILD MUST BE CALM

Classify COUGH or DIFFICULT BREATHING

- For how long?
- Contact with TB patient

If wheezing with either fast breathing or chest indrawing: Give a trial of rapid acting inhaled bronchodilator for up to three times 15-20 minutes apart. Count the breaths and look for chest indrawing again, and then classify.

If the child is:

Fast breathing is:

2 months up to 12 months ≥50 breaths per minute more

12 months up to 5 years ≥40 breaths per minute more

**If pulse oximeter is available, determine oxygen saturation and refer if < 90%.*

*** If referral is not possible, manage the child as described in the pneumonia section of the national referral guidelines or as in Ethiopian pocket book for hospital care for children.*

**** If inhaled bronchodilator is not available, oral salbutamol may be tried but not recommended for treatment of severe acute wheeze.*

***** Assess for TB infection (see page 32)*

▶ Refer Page 60 for Wheezing

SIGNS	CLASSIFY AS	TREATMENT (Urgent pre-referral treatments are in bold print)
• Any general danger sign	VERY SEVERE DISEASE	▶ Give diazepam if convulsing now ▶ Quickly complete the assessment ▶ Give appropriate pre-referral treatment immediately ▶ Treat to prevent low blood sugar ▶ Keep the child warm ▶ Refer URGENTLY.
• Any general danger sign OR • Stridor in calm child.	SEVERE PNEUMONIA OR VERY SEVERE DISEASE	▶ Give first dose of IV/IM Ampicillin and gentamycin* ▶ Refer URGENTLY to hospital**
• Fast breathing OR • Chest indrawing	PNEUMONIA	▶ Give oral Amoxicillin for 5 days ▶ If wheezing (or disappeared after rapidly acting bronchodilator) give an inhaled bronchodilator for 5 days*** ▶ Soothe the throat and relieve the cough with a safe remedy ▶ If Chest indrawing in HIV exposed child, give first dose of amoxicillin and refer ▶ If coughing for ≥ 14 days or there is contact with TB patient do assessment for TB**** ▶ Advise mother when to return immediately ▶ Follow-up in 2 days
No signs of: • Very severe disease OR • Pneumonia	COUGH OR COLD	▶ If wheezing (or disappeared after rapidly acting bronchodilator) give an inhaled bronchodilator for 5 days*** ▶ Soothe the throat and relieve the cough with a safe remedy ▶ If coughing for > 14 days or there is contact with TB patient assess for TB**** ▶ Advise mother when to return immediately ▶ Follow-up in 5 days if not improving



Save the Children

Does The Child Have Diarrhea

Does the child have Diarrhoea?

IF YES, ASK

- For how long?
- Is there blood in the stool?

LOOK AND FEEL:

- Look at the child's general condition is the child:
 - Lethargic or unconscious?
 - Restless and irritable?
- Look for sunken eyes.
- Offer the child fluid. Is the child:
 - Not able to drink or drinking poorly?
 - Drinking eagerly, thirsty?
- Pinch the skin of the abdomen. Does it go back:
 - Very slowly (> 2 seconds)?
 - Slowly?

For
Dehydration

Classify
DIARRHOEA

and if
diarrhoea
14 days or
more

and if
blood
in stool

SIGNS	CLASSIFY AS	TREATMENT (Urgent pre-referral treatments are in bold print)
Two of the following signs: • Lethargic or unconscious • Sunken eyes • Not able to drink or drinking poorly • Skin pinch goes back very slowly	SEVERE DEHYDRATION	If child has no other severe classification: - Give fluid for severe dehydration (Plan C). OR If child also has another severe classification: • Refer URGENTLY to hospital with mother giving frequent sips of ORS on the way. • Advise the mother to continue breastfeeding. ▶ If child is 2 years or older, and there is cholera in your area, give antibiotic for cholera.
Two of the following signs: • Restless, irritable • Sunken eyes • Drinks eagerly, thirsty • Skin pinch goes back slowly	SOME DEHYDRATION	▶ Give fluid, Zinc supplements and food for some dehydration (Plan B) If child also has a severe classification: - Refer URGENTLY to hospital with mother giving frequent sips of ORS on the way. Advise the mother to continue breastfeeding. ▶ Advise mother when to return immediately. ▶ Follow-up in 5 days if not improving.
• Not enough signs to classify as some or severe dehydration	NO DEHYDRATION	▶ Give fluid, Zinc supplements and food to treat diarrhoea at home (Plan A) ▶ Advise mother when to return immediately ▶ Follow-up in 5 days if not improving
• Dehydration present	SEVERE PERSISTENT DIARRHOEA	▶ Treat dehydration before referral unless the child has another severe classification ▶ Give Vitamin A ▶ Refer to hospital
• No dehydration	PERSISTENT DIARRHOEA	▶ Advise the mother on feeding recommendation for a child who has PERSISTENT DIARRHOEA ▶ Give Vitamin A, therapeutic dose ▶ Give Zinc for 10 days ▶ Advise mother when to return immediately ▶ Follow-up in 5 days
• Blood in the stool	DYSENTERY	▶ Treat for 3 days with Ciprofloxacin ▶ Advise mother when to return immediately ▶ Follow-up in 2 days



Save the Children

Plan A: Treat Diarrhoea At Home

- **Counsel the mother on the 4 Rules of Home Treatment:**
- **Give Extra Fluids, Give Zinc Supplements, Continue Feeding, When to Return**
- **GIVE EXTRA FLUIDS (as much as the child will take)**
- Tell the mother:
- Breastfeed frequently and for longer at each feed.
- If the child is exclusively breastfed, give ORS in addition to breast milk.
- If the child is not exclusively breastfed, give one or more of the following: ORS
- Solution, food-based fluids (such as soup, rice water and yoghurt drinks), or clean water.



Save the Children

Cont...

- **It is especially important to give ORS at home when:**
- The child has been treated with plan B or plan C during this visit.
- The child cannot return to a clinic if the diarrhoea gets worse.
- Teach the mother how to mix and give ors.
- Give the mother 2 packets of ors to use at home.
- Show the mother how much fluid to give in addition to the usual fluid in-take 10 ml/kg
- Up to 2 years 50 to 100 ml after each loose stool
- 2 years or more 100 to 200 ml after each loose stool



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Cont...

Give zinc supplements :

- **Tell the mother how much zinc to give:**
 - 0-6 months- 1/2 tablet for 10 days
 - 6 months or more - 1 tablet for 10 days
- Show the mother how to give zinc supplements
- Infants- dissolve tablet in a small amount of expressed breast milk, ors or clean water in a cup



Save the Children

Plan B: Treat Some Dehydration With ORS

- **Give in clinic recommended amount of ORS over 4 hour period**
- Use the child's age only when you do not know the weight.
- The approximate amount of ORS required (in ml) can also be calculated by multiplying the child's weight (in kg) times 75
- If the child wants more ORS than shown, give more.
- For infants under 6 months who are not breastfed, also give 100-200 ml use the child's age only when you do not know the weight.



Save the Children

Plan B: Treat Some Dehydration With ORS

- **Show the mother how to give ors solution:**
- Give frequent small sips from a cup.
- If the child vomits, wait 10 minutes.
- Then continue, but more slowly.
- Continue breastfeeding whenever the child wants.



Save the Children

Plan B: Treat Some Dehydration With ORS

After 4 hours:

- Reassess the child and classify the child for dehydration.
- Select the appropriate plan to continue treatment.
- Begin feeding the child in clinic.

If the mother must leave before completing treatment:

- Show her how to prepare ors solution at home.
- Show her how much ors to give to finish 4-hour treatment at home.
- Give her enough ors packets to complete rehydration.
- Also give her 2 packets as recommended in plan a.
- **Explain the 4 rules of home treatment:**



Save the Children

➤ Plan C: Treat Severe Dehydration Quickly

➤ FOLLOW THE ARROWS. IF ANSWER IS "YES", GO ACROSS. IF "NO" GO DOWN.

START HERE

Can you give
Intravenous (IV) Fluid
immediately?

YES

NO

Is IV treatment
available nearby
(within 30 minutes)?

YES

NO

Are you trained to
use a naso-gastric (NG) tube
for rehydration?

NO

YES

Can the child drink?

NO

Refer URGENTLY
to hospital for IV or NG
treatment

- Start IV fluid immediately. If the child can drink, give ORS by mouth while the drip is set up. Give 100ml/kg Ringer's Lactate Solution (or, if not available, normal saline), divided as follows:

AGE	First give 30 ml/kg in:	Then give 70 ml/kg in:
Infants (under 12 months)	1 hour*	5 hours
Children (12 months up to 5 years)	30 minutes*	2 ½ hours

- * Repeat once if radial pulse is still very weak or not detectable
- Reassess the child every 1-2 hours. If hydration status is not improving, give the IV drip more rapidly.
- Also give ORS (about 5 ml/kg/hour) as soon as the child can drink: usually after 3-4 hours (infants) or 1-2 hours (children).
- Reassess an infant after 6 hours and a child after 3 hours. Classify dehydration. Then choose the appropriate plan (A, B, or C) to continue treatment.

- Refer URGENTLY to hospital for IV treatment.
- If the child can drink, provide the mother with ORS solution and show her how to give frequent sips during the trip

- Start rehydration by tube (or mouth) with ORS solution give 20 ml/kg/hour for 6 hours (total of 120 ml/kg)
- Reassess the child every 1-2 hours:
 - If there is repeated vomiting or increasing abdominal distension, give the fluid more slowly.
 - If hydration status is not improving after 3 hours, send the child for IV therapy.
- After 6 hours, reassess the child. Classify dehydration. Then choose the appropriate plan (A, B, or C) to continue treatment.

NOTE:

If the child is not referred to hospital, observe the child at least 6 hours after rehydration to be sure the mother can maintain hydration giving the child ORS solution by mouth.

Does The Child Have Fever?

IF YES:

- Decide Malaria Risk: High/Low or No. If "no" malaria risk, then ask:
- Has the child traveled outside this area during the previous 30 days?
- If yes has he been to a malarious area?

Do blood film or RDT, if malaria risk is High/Low or history of travel to a malarious area, AND there is no Severe Classification***

High/Low
Malaria
Risk

THEN ASK:

- For how long has the child had fever?
- If more than 7 days, has fever been present every day?
- Has the child had measles within the last 3 months?

LOOK AND FEEL:

- Look or feel for stiff neck.
- Look or feel for bulging fontanelts (< 1 year of age)
- Look for any bacterial causes of fever**
- Look for runny nose.
- Look for signs of MEASLES
 - Generalized rash, AND one of these: cough, runny nose or red eyes.
- Look for mouth ulcers:
 - Are they deep or extensive?
 - Are they not deep or extensive?
- Look for pus draining from the eye.
- Look for clouding of the cornea.

If the child has measles now OR within the last 3 months

No
Malaria Risk
and No
travel to
Malarious area

IF MEASLES now or within the last 3 months, Classify

SIGNS	CLASSIFY	TREATMENT (Urgent pre-referral treatments are in bold print)
- Any general danger sign, OR - Stiff neck, OR - Bulging fontanelts (< 1 yr)	VERY SEVERE FEBRILE DISEASE	- Give first dose Artesunate or Quinine for severe malaria - Give first dose of IVIM Ampicillin and Gentamycin - Treat the child to prevent low blood sugar - Give Paracetamol in health facility for high fever ($\geq 38.5^{\circ}\text{C}$) - Refer URGENTLY to hospital
- Positive blood film/RDT, OR - If blood film/RDT not available, any fever (by history, or feels hot, or temp $\geq 37.5^{\circ}\text{C}$)	MALARIA	- Treat with Artemeter-Lumefantrine for P. falcip. or mixed or no confirmatory test done - Treat with Chloroquine for confirmed P. vivax - Give Paracetamol in health facility for high fever (38.5°C or above) - Give an appropriate antibiotic for identified bacterial cause of fever - Advise mother when to return immediately - Follow-up in 2 days if fever persists - If fever is present every day for more than 7 days, refer for assessment
- RDT negative, OR - Blood film negative - Other cause of fever present	FEVER: NO MALARIA	- Give one dose of Paracetamol in health facility for high fever ($\geq 38.5^{\circ}\text{C}$) - Give an appropriate antibiotic for identified bacterial cause of fever - Advise mother when to return immediately - Follow-up in 2 days if fever persists - If fever is present every day for more than 7 days, refer for assessment
- Any general danger sign, OR - Stiff neck, OR - Bulging fontanelts (< 1 year of age)	VERY SEVERE FEBRILE DISEASE	- Give first dose of IVIM Ampicillin and Gentamycin - Treat the child to prevent low blood sugar - Give Paracetamol in health facility for high fever ($\geq 38.5^{\circ}\text{C}$) - Refer URGENTLY to hospital
Any fever	FEVER	- Give one dose of Paracetamol in health facility for high fever ($\geq 38.5^{\circ}\text{C}$) - Give an appropriate antibiotic for identified bacterial cause of fever - Advise mother when to return immediately - Follow-up in 2 days if fever persists - If fever is present every day for more than 7 days refer for assessment
- Any general danger sign, OR - Clouding of cornea, or - Deep or extensive mouth ulcers	SEVERE COMPLICATED MEASLES ****	- Give Vitamin A, first dose - Give first dose of IVIM Ampicillin and Gentamycin - If clouding of the cornea or pus draining from the eye, apply Tetracycline eye ointment - Refer URGENTLY to hospital
- Pus draining from the eye or - Mouth ulcers (not deep or extensive)	MEASLES WITH EYE OR MOUTH COMPLICATIONS ****	- Give Vitamin A, therapeutic dose - If pus draining from the eye, treat eye infection with Tetracycline eye ointment - If mouth ulcers, treat with gentian violet - Advise mother when to return immediately - Follow-up in 2 days
Measles now or within the last 3 months	MEASLES	- Give Vitamin A, therapeutic dose - Advise mother when to return immediately

* These temperatures are based on axillary temperature. Rectal temperature readings are approximately 0.5°C higher.

** Look for local tenderness; oris sores; refusal to use a limb; hot tender swelling; red tender skin or boils; lower abdominal pain or pain on passing urine in older children.

*** If no malaria test available: High/Low malaria risk and no known cause of fever-classify as MALARIA.

****Other important complication of measles-pneumonia, stridor, diarrhoea, ear infection, and acute



Save the Children

Does The Child Have An Ear Problem?

IF YES, ASK:

- Is there ear pain?
- Is there ear discharge?
If yes, for how long?

LOOK, AND FEEL:

- Look for pus draining from the ear
- Feel for tender swelling behind the ear

Classify
EAR
PROBLEM

SIGNS	CLASSIFY AS	TREATMENT (Urgent pre-referral treatments are in bold print)
• Tender swelling behind the ear	MASTOIDITIS	▶ Give first dose of Ampicillin and Chloramphenicol IVIM ▶ Give first dose of Paracetamol for pain ▶ Refer URGENTLY to hospital
• Ear pain, OR • Pus is seen draining from the ear and discharge is reported for less than 14 days	ACUTE EAR INFECTION	▶ Give Amoxicillin for 5 days ▶ Give Paracetamol for pain ▶ Dry the ear by wicking ▶ Follow-up in 5 days
• Pus is seen draining from the ear and discharge is reported for 14 days or more	CHRONIC EAR INFECTION	▶ Dry the ear by wicking ▶ Treat with topical Quinolone eardrops for 2 weeks ▶ Follow-up in 5 days
• No ear pain and • No pus seen draining from the ear	NO EAR INFECTION	▶ No additional treatment



Save the Children

Then Check For Anaemia

LOOK

- Look for palmar pallor, is it;
 - Severe palmar pallor?
 - Some palmar pallor?
 - No palmar pallor?

Classify
ANEMIA

SIGNS	CLASSIFY AS	TREATMENT (Urgent pre-referral treatments are in bold print)
• Severe palmar pallor	SEVERE ANEMIA	▶ Refer URGENTLY to hospital
• Some palmar pallor	ANEMIA	▶ Assess the child's feeding and counsel the mother on feeding according to the FOOD box on the COUNSEL THE MOTHER chart ▶ Give Iron** ▶ Do blood film or RDT for malaria, if malaria risk is high or has travel history to malarious area in last 30 days. ▶ Give Mebendazole or Albendazole, if the child is ≥ 2 years old and has not had a dose in the previous six months ▶ Advise mother when to return immediately ▶ Follow-up in 14 days
• No palmar pallor	NO ANEMIA	▶ No additional treatment ▶ Counsel the mother on feeding recommendation

**If child has severe acute malnutrition and is receiving RUTF, DO NOT give iron because there is already adequate amount of iron in RUTF.



Save the Children

Acute Malnutrition, In Infants < 6 Months

LOOK AND FEEL

1. Check for presence of oedema of both feet (or sacrum) Does the child have oedema?

2. Check the weight and Length

- What is the weight for length Z score?

3. Check for signs of medical complications:

- Any General Danger Sign
- Any severe classification
- Pneumonia
- Dehydration*
- Persistent diarrhoea
- Dysentery
- Fever $\geq 38.5^{\circ}\text{C}$
- Measles (now or with eyelid complications)
- Low body temperature ($<35^{\circ}\text{C}$ axillary)
- Dermatosis+++

SIGNS	CLASSIFY AS	TREATMENT
- WFL $< -3Z$ score, and presence of complications OR - Oedema of both feet	COMPLICATED SEVERE ACUTE MALNUTRITION	- Give first dose of Ampicillin and Gentamycin IM - Treat the child to prevent Low Blood Sugar - Advise mother on the need of referral - Refer Urgently to Hospital
- WFL $< -3Z$ score AND no complications AND - No oedema of both feet	UNCOMPLICATED SEVERE ACUTE MALNUTRITION	- Counsel on breast feeding and care - Undertake appropriate counseling and feeding advise in cases where a child is orphaned with no other option for breastfeeding - Assess for TB infection (Refer table on page 32)*
- WFL $\geq -3Z$ to $< -2Z$ score, AND - No oedema of both feet	MODERATE ACUTE MALNUTRITION	- Assess feeding and advise the mother on feeding - Assess for TB infection (Refer table on page 32)* - Follow up in 5 days if feeding problem - Follow up in 30 days
- WFL $\geq -2Z$ score AND - No oedema of both feet	NO ACUTE MALNUTRITION	- Assess feeding and advise the mother on feeding - Follow up in 5 days if feeding problem - If no feeding problem-praise the mother

• Diagnosis of dehydration in SAM is mainly used on the history rather than on patient's examination alone.



Save the Children

Check For Acute Malnutrition, In Children 6 - 59 Months

LOOK AND FEEL

1. Check for presence of oedema of both feet (orsacrum)

- ◆ Does the child have oedema**? (+, ++, +++)

2. Check the weight and height

3. Check MUAC

4. Check for signs of medical complications:

- Any General Danger Sign
- Any severe classification
- Pneumonia
- Dehydration*
- Persistent diarrhea
- Dysentery
- Fever $\geq 38.5^{\circ}\text{C}$
- Measles [now or with eye/mouth complications]
- Low body temperature ($<35^{\circ}\text{C}$ axillary)
- Dermatitis+++

Do Appetite test (Passed, Failed)

• Appetite test should be done ONLY when there is:

- NO medical complication, and
- NO +++ oedema, and
- NO +++ dermatosis**, and
- NO marasmic kwashiorkor ****

SIGNS	CLASSIFY AS	TREATMENT
• WFL/H $< -3\text{Z}$ score or MUAC < 11 cm or Oedema of both feet (+, ++), and any of the following: • Any one of the medical complications, or • Failed Appetite test • +++ Oedema OR • Marasmic Kwashiorkor (WFL/H $< -3\text{Z}$ with oedema or MUAC < 11 cm with oedema)	COMPLICATED SEVERE ACUTE MALNUTRITION	▶ Give 1st dose of Ampicillin and Gentamycin IM ▶ Treat the child to prevent low blood sugar ▶ Advise the mother to feed and keep the child warm ▶ Advise mother on the need of referral ▶ Refer Urgently to Hospital or admit to inpatient care
• WFL/H $< -3\text{Z}$ score or MUAC < 11 cm or oedema of both feet (+, ++) - AND • No medical complication and • Pass appetite test	UNCOMPLICATED SEVERE ACUTE MALNUTRITION	▶ If Outpatient Treatment Program (OTP) is available, manage as follows: ▶ Give RUTF for 7 days, ▶ Give oral Amoxicillin for 7 days ▶ Give single dose of 5 mg folic acid for those with anemia ▶ Counsel on how to feed RUTF to the child ▶ Advise the mother when to return immediately ▶ Assess for TB infection (Refer table on page 32) ▶ Follow-up in 7 days ▶ If OTP is not available, refer to a facility with OTP service ▶ If there is any social problem at home treat as inpatient
• WFL/H $\geq -3\text{Z}$ to $< -2\text{Z}$ score or MUAC 11 cm to < 12 cm - AND • No oedema of both feet	MODERATE ACUTE MALNUTRITION	▶ Refer to Supplementary Feeding Program if available ▶ Assess for feeding and counsel the mother accordingly ▶ Assess for possible TB infection (Refer table on page 32) ▶ If feeding problem, follow up in 5 days
• WFL/H $\geq -2\text{Z}$ score or MUAC ≥ 12 cm - AND • No oedema of both feet	NO ACUTE MALNUTRITION	▶ Assess feeding and advise the mother on feeding ▶ Follow up in 5 days if feeding problem ▶ If no feeding problem-praise the mother

* Diagnosis of dehydration in SAM is mainly used on the history rather than on patient's examination alone.

** Oedema grading: bilateral oedema below ankles (+); below the knees & the elbows (++); generalized oedema involving the upper arms & face (+++).

*** Dermatoses grading: few discolored or rough patches of skin (+); multiple patches on arms and/or legs (++); flaking skin, raw skin or fissures (openings in the skin) is grade +++ dermatosis.

**** Child with WFL $< -3\text{Z}$ plus oedema, or with MUAC < 11.5 cm plus oedema.



Save the Children

HIV Exposure And Infection, In Children 2 - < 18 Months

ASK:

- What is the HIV status of the mother?
 - Positive
 - Negative
 - Unknown
- What is the HIV antibody test result of the sick child?
 - Positive
 - Negative
 - Unknown
- What is the DNA/PCR test result of the sick child? *
 - Positive
 - Negative
 - Unknown
- Is child on breastfeeding?
 - Yes
 - No
- If no, was child breastfed in the last 6 weeks?
 - Yes
 - No

Classify
for
HIV
Infection

SIGN	CLASSIFY	TREATMENT
Child DNA PCR positive	HIV INFECTED	- Give Cotrimoxazole prophylaxis - Assess feeding and counsel - Assess for TB infection (Refer table on page 32) - Ensure mother is tested & enrolled in HIV care & treatment - Advise on home care - Refer to ART clinic for ART initiation/care & treatment
- Mother positive, and child Antibody or DNA/PCR negative, and breastfeeding - OR - Mother positive, and child antibody & DNA/PCR unknown - OR - Child antibody positive	HIV EXPOSED	- Give Co-trimoxazole prophylaxis - Assess feeding and counsel - If child DNA/PCR is unknown, test as soon as possible - Ensure mother is tested & enrolled in mother-baby cohort follow up at ANC/PMTCT clinic
Mother and child not tested	HIV STATUS UNKNOWN	- Counsel the mother for HIV testing for herself & the child - Advise the mother to give home care - Assess feeding and counsel
- Mother negative, - OR - Mother positive, and child DNA PCR negative, and not breastfeeding, - OR - Mother HIV status unknown, and child antibody negative	HIV INFECTION UNLIKELY	- Advise on home care - Assess feeding and counsel - Advise on HIV prevention - Encourage mother to be tested - If mother HIV status is unknown, advise her on HIV testing

Note:

- If DNA PCR isn't available, AND child antibody is positive, AND two of the following are present (Oral thrush, Severe pneumonia or Very Severe Disease); Consider this child to have "PRESUMPTIVE SEVERE HIV DISEASE". And this child should be referred to ART clinic and treated as "HIV INFECTED" child.



Save the Children

HIV Exposure And Infection, In Children 18 - 59 Months

ASK:

- What is the HIV status of the mother?
 - Positive
 - Negative
 - Unknown
- What is the HIV antibody test result of the sick child?
 - Positive
 - Negative
 - Unknown
- Is child on breastfeeding?
 - Yes
 - No
- If no, was child breastfed in the last 6 weeks?
 - Yes
 - No

Classify
for
HIV
infection

SIGN	CLASSIFY	TREATMENT
• Child antibody positive	HIV INFECTED	▶ Consider Cotrimoxazole prophylaxis ▶ Assess feeding and counsel ▶ Advise on home care ▶ Refer to ART clinic for ARV initiation ▶ Ensure mother is tested & enrolled in HIV care & treatment
• Mother positive, AND • Child antibody negative or unknown, and breastfeeding	HIV EXPOSED	▶ Give Cotrimoxazole prophylaxis ▶ Assess feeding and counsel ▶ If child antibody test is unknown, test as soon as possible ▶ If child antibody test is negative, repeat 6 wks after complete cessation of breastfeeding ▶ Ensure mother is tested & enrolled in mother-baby cohort follow up at ANC/PMTCT clinic
• Mother and child not tested	HIV STATUS UNKNOWN	▶ Counsel the mother for HIV testing for herself and the child ▶ Advise the mother to give home care ▶ Assess feeding and counsel
• Mother negative and child not known	HIV INFECTION UNLIKELY	▶ Advise on home care ▶ Assess feeding and counsel ▶ Advise on HIV prevention ▶ If possible, do HIV antibody test for the child
• Child antibody negative at least 6 weeks after complete cessation of breastfeeding	HIV UNINFECTED	▶ Advise on home care ▶ Assess feeding and counsel ▶ Advise on HIV prevention



Save the Children

Assess And Classify The Child For Tuberculosis

THEN ASK:

- Cough of ≥ 14 days
- Fever and night sweats*
- Contact history with TB patient **

LOOK AND FEEL:

- Swelling or discharging wound***
- Signs of acute malnutrition ****

Classify

DO THE FOLLOWING IF AVAILABLE:

- AFB or Gene Xpert if there is sputum production
- Chest X-ray*****

* Fever $>38^{\circ}\text{C}$ that continues for greater than two weeks after common causes are excluded.

**Contact history with TB patient : a newly diagnosed TB case (within the past one year) in the close contact or household member

***the swelling and discharging wound in the neck or armpit not due to injury, and staying for a duration of more than one month should not be due to injury of any kind.

**** SAM/MAM: severe or moderate acute malnutrition classification from the assessment and classification table for malnutrition.

***** X-Ray suggestive of TB: however x-ray is not commonly available in health centers and primary hospitals but if it is available

SIGNS	CLASSIFY AS	TREATMENT (Urgent pre-referral treatments are in bold print)
• Contact with a known MDR TB patient	Suspected MDR TB	▶ Advise mother on the need of referral ▶ Refer Urgently to Hospital for MDR TB investigation and Treatment
• Contact with TB patient And two or more of the signs / One or more of the signs if known HIV+ And/ Or • A sign AND AFB/ GeneXpert +ve Or • A sign AND Chest X ray suggestive of TB (eg. miliary pattern)	TB	▶ Counsel the mother on DOT5 principle ▶ Advise mother to bring any other contacts ▶ Do provider initiated HIV testing and Counseling ▶ Link to TB clinic for initiation of treatment and follow up
• Contact to TB patient (non-MDR) and no other finding	TB Exposed Child	▶ Counsel the mother on the diagnosis of TB exposure and the need for INH prophylaxis ▶ Link to TB clinic for INH prophylactic-treatment initiation and follow up
• No conclusive sign and • No Contact with TB patient • AFB/GeneXpert -ve And • Chest X- ray not suggestive	No TB Infection	▶ Look and treat for other causes for the main complaint ▶ Counsel the mother on the need for INH prophylaxis in the presence of HIV infection for HIV +ve children ▶ Link to TB clinic for INH prophylactic-treatment initiation and follow up for HIV +ve children ▶ Follow up in 30 days



Save the Children

SICK YOUNG INFANT AGE UP TO 2 MONTHS



Save the Children

Check The Newborn For Birth Asphyxia

IF YOU ARE ATTENDING DELIVERY or
BABY BROUGHT TO YOU IMMEDIATELY AFTER BIRTH
Assess and check for Birth Asphyxia while drying and wrapping with dry cloth
If there is Birth Asphyxia, complete resuscitation within the 1st minute "Golden Minute of life"

Assess, Look

- Look the breathing
 - Is baby not breathing?
 - Is baby gasping?
 - Is baby breathing poorly (<30 breaths/minute)?
 - Is baby breathing normally (crying or ≥30 breaths/minute)?

Classify ALL Newborns

SIGNS	CLASSIFY AS	TREATMENT (Urgent pre-referral treatments are in bold print)
If any of the following sign: • Not breathing, OR • Gasping, OR • Breathing poorly (<30 breaths/minute)	BIRTH ASPHYXIA	▶ Start resuscitation immediately ▶ Clear mouth first, then nose with bulb syringe If not breathing ▶ Clamp/tie and cut the cord immediately ▶ Position the newborn supine with neck slightly extended ▶ Ventilate with appropriate size bag & mask ▶ If the baby start breathing regularly continue giving essential newborn care ▶ If the baby remains weak or is having irregular breathing after 20 minutes of resuscitation; refer urgently to hospital while continuing to resuscitate on the way ▶ Stop resuscitation after 20 minutes if no response (no spontaneous breathing) ▶ Monitor continuously for 6 hours baby with the mother ▶ Follow after 12 hrs, 24 hrs (in the facility), 3 days, 7 days and 6 weeks
• Breathing normally (crying or ≥30 breaths/minute)	NO BIRTH ASPHYXIA	▶ Give cord care ▶ Initiate skin-to-skin contact ▶ Initiate breastfeeding ▶ Give eye care ▶ Give Vitamin K ▶ Apply Chlorhexidine gel ▶ Give BCG and OPV 0 ▶ Advise mother when to return immediately ▶ Follow after 6 hrs (in the facility), 3 days and 7 days and 6 weeks

Assess The Newborn For Birth Weight And Gestational Age

Assess, Look

- Ask the gestational age
- Ask for birth weight or
- Weigh the baby (within 7 days of life)

Classify
ALL
Newborn Babies

SIGNS	CLASSIFY AS	TREATMENT (Urgent pre-referral treatments are in bold print)
• Weight < 1,500gm OR • Gestational age < 32 weeks	VERY LOW BIRTH WEIGHT AND/OR VERY PRETERM	▶ Continue breastfeeding (if not sucking feed expressed breast milk by cup) ▶ Start Kangaroo Mother Care (KMC) ▶ Give Vitamin K 0.5mg IM on anterior mid thigh, if not already given ▶ Refer URGENTLY with mother to hospital with KMC position
• Weight 1,500 - 2,500 gm OR • Gestational age 32-37 weeks	LOW BIRTH WEIGHT AND/OR PRETERM	▶ KMC if <2,000gm (see page 17) ▶ Counsel on optimal breastfeeding ▶ Counsel mother on prevention of infection ▶ Give Vitamin K 1mg IM on anterior mid thigh if not already given ▶ Provide follow-up for KMC ▶ If baby ≥ 2,000 gms follow-up visits at age 6-24 hrs, 3 days, 7 days & 6 weeks ▶ Give 1st dose of vaccine ▶ Advise mother when to return immediately
• Weight ≥ 2,500 gm OR • Gestational age ≥ 37 weeks	NORMAL BIRTH WEIGHT AND/OR TERM	▶ Counsel on optimal breastfeeding ▶ Counsel mother/family on prevention of infection ▶ Provide follow-up visits at age 6-24 hrs, 3 days, 7 days & 6 weeks ▶ Give 1st dose of vaccine ▶ Give Vitamin K 1mg IM on anterior mid thigh if not already given ▶ Advise mother when to return immediately

Check For Very Severe Disease And Local Bacterial Infection

Ask

Look, Listen, Feel:

Classify
all young
infants

- Is the infant having difficulty in feeding?
- Has the infant had convulsions?

- Count the breaths in one minute
 - Repeat the count if $\geq 60/\text{min}$
- Look for severe chest indrawing
- See if the young infant is not feeding
- See if the infant is convulsing now
- Look at the umbilicus. Is it red or draining pus?
- Measure temperature (or feel for fever or low body temperature)
- Look for skin pustules
- Look at the young infant's movements.
 - Infant move on his/her own
 - Infant move only when stimulated
 - Infant doesn't move even when stimulated

YOUNG
INFANT
MUST BE
CALM

AND if
temp. is
from
35.5 - 36.4°C

* These thresholds are based on axillary temperature. The thresholds for rectal temperature readings are approximately 0.5°C higher.

N.B. In a young infant with fever, malaria should be considered based on other associated symptoms.

If referral is not possible, see page 53-63 "Where Referral is not Possible".

SIGNS	CLASSIFY AS	TREATMENT (Urgent pre-referral treatments are in bold print)
- Not feeding well, OR - History of convulsions/convulsing now, OR - Fast breathing (≥ 60 breaths per minute), OR - Severe chest indrawing, OR - Fever ($\geq 37.5^{\circ}\text{C}^*$ or feels hot), OR - Low body temperature ($< 35.5^{\circ}\text{C}^*$ or feels cold), OR - Movement only when stimulated or no movement even when stimulated.	VERY SEVERE DISEASE	- Give first dose of intramuscular Ampicillin and Gentamycin - Treat to prevent low blood sugar - Warm the young infant by skin-to-skin contact if temperature is less than 36.5°C (or feels cold to touch) while arranging referral - Advise mother how to keep the young infant warm on the way to the hospital - Refer URGENTLY to hospital
- Red umbilicus or draining pus, OR - Skin pustules	LOCAL BACTERIAL INFECTION	- Give Amoxicillin for 5 days - Teach the mother to treat local infections at home - Advise mother when to return immediately - Follow-up in 2 days
None of the signs of very severe disease, or local bacterial infection	SEVERE DISEASE, OR LOCAL INFECTION UNLIKELY	Advise mother to give home care for the young infant
Temperature from 35.5°C – 36.4°C (both values inclusive)	LOW BODY TEMPERATURE	- Treat to prevent low blood sugar - Warm the young infant using skin-to-skin contact for one hour and reassess. If temperature remains same or worse, refer. (Advise mother to continue feeding and keep the infant warm on the way to the hospital). - Advise mother when to return immediately - Follow-up in 2 days



Save the Children

Check For Jaundice

- Look for jaundice:
 - Is skin on the face or eyes yellow?
 - Are the palms and soles yellow?

Classify
all
infants

SIGNS	CLASSIFY AS	TREATMENT (Urgent pre-referral treatments are in bold print)
• Palms and/or soles yellow, OR • Skin and eyes yellow and baby is < 24 hrs old, OR • Skin and eyes yellow and baby is ≥14 days old	SEVERE JAUNDICE	▶ Treat to prevent low blood sugar ▶ Warm the young infant by skin-to-skin contact if temperature is less than 36.5°C (or feels cold to touch) while arranging referral ▶ Advise mother how to keep the young infant warm on the way to the hospital ▶ Refer URGENTLY to hospital
• Only skin on the face or eyes yellow, AND • Infant aged 24 hrs -14 days old	JAUNDICE	▶ Advise mother to give home care for the young infant ▶ Advise the mother to expose and check in natural light daily ▶ Advise mother when to return immediately ▶ Follow-up in 2 days
• No yellowish discoloration of the eye and skin	NO JAUNDICE	▶ Advise mother to give home care for the infant



Save the Children

Does The Young Infant Have Diarrhoea?

THEN ASK:

Does the Young Infant Have Diarrhoea?

Ask

- For how long?
- Is there blood in the stool?

Look and Feel:

- Look at the young infant's general condition.
 - Infant moves only when stimulated
 - Infant does not move even when stimulated
 - Infant restless and irritable
- Look for sunken eyes
- Pinch the skin of the abdomen. Does it go back:
 - Very slowly (> 2 sec.)?
 - Slowly?

Classify DIARRHOEA

for dehydration

and if diarrhoea 14 days or more

and if blood in stool

* What is diarrhoea in young infant?

If the stools have changed from usual pattern: many and watery (more water than fecal matter). The frequent and loose stools of a breastfed baby may be normal and are not always diarrhoea

SIGNS	CLASSIFY AS	TREATMENT (Urgent pre-referral treatments are in bold print)
Two of the following signs: • Movement only when stimulated, or no movement even when stimulated • Sunken eyes • Skin pinch goes back very slowly	SEVERE DEHYDRATION	► If infant has another severe classification: - Refer URGENTLY to hospital with mother giving frequent sips of ORS on the way - Advise mother to continue breastfeeding more frequently - Advise mother how to keep the young infant warm on the way to hospital ► If infant does not have any other severe classification; give fluid for severe dehydration (Plan C).
Two of the following signs: • Restless, irritable • Sunken eyes • Skin pinch goes back slowly	SOME DEHYDRATION	► If infant has another severe classification: - Refer URGENTLY to hospital with mother giving frequent sips of ORS on the way - Advise mother to continue breastfeeding more frequently - Advise mother how to keep the young infant warm on the way to hospital ► If infant does not have any other severe classification; - Give fluid for some dehydration and Zinc supplement (Plan B) - Advise mother when to return immediately - Follow-up in 2 days
• Not enough signs to classify as some or severe dehydration	NO DEHYDRATION	► Advise mother when to return immediately ► Follow-up in 5 days if not improving ► Give fluids to treat diarrhoea at home and Zinc supplement (Plan A)
• Diarrhoea lasting 14 days or more	SEVERE PERSISTENT DIARRHOEA	► Give first dose of IM Ampicillin and Gentamycin ► Treat to prevent low blood sugar ► Advise how to keep infant warm on the way to the hospital ► Refer to hospital
• Blood in stool	DYSENTERY	► Give first dose of IM Ampicillin and Gentamycin ► Treat to prevent low blood sugar ► Advise how to keep infant warm on the way to the hospital ► Refer to hospital



Save the Children

Then Check For HIV Infection

ASK:

- What is the HIV status of the mother?
 - Positive
 - Negative
 - Unknown
- What is the HIV status of the young infant?

Antibody:

- Positive
- Negative
- Unknown

DNA PCR:

- Positive
- Negative
- Unknown

Classify
by
Test Result

SIGN	CLASSIFY AS	TREATMENT
• Young infant DNA PCR positive	HIV INFECTED	▶ Start Cotrimoxazole Prophylaxis from 6 months of age ▶ Assess feeding and counsel ▶ Advise on home care ▶ Refer to ART clinic for immediate ART initiation and care ▶ Assess for TB infection (Refer table on page 32) ▶ Ensure mother is tested and enrolled for HIV care and treatment
• Mother HIV positive, AND young infant DNA PCR negative/unknown OR • Young infant HIV antibody positive	HIV EXPOSED	▶ Start Co-trimoxazole Prophylaxis from 6 weeks of age ▶ Assess feeding and counsel ▶ If DNA PCR test is unknown, test as soon as possible starting from 6 weeks of age ▶ Ensure mother is tested & enrolled in mother-baby cohort follow up at ANC/ PMTCT clinic
• Mother and young infant not tested	HIV STATUS UNKNOWN	▶ Counsel the mother for HIV testing for herself and the infant ▶ Advise on home care of infant ▶ Assess feeding and counsel
• Mother or young infant HIV antibody negative	HIV INFECTION UNLIKELY	▶ Advise on home care of infant ▶ Assess feeding and counsel ▶ Advise the mother on HIV prevention



Save the Children

Check For Feeding Problem Or Low Weight For Age

Ask

- Is there any difficulty of feeding?
- Is the infant breastfed? If yes?
How many times in 24 hours?
- Do you empty one breast before switching to the other?
- Do you increase frequency of breastfeeding during illness?
- Does the infant receive any other foods or drinks?

Look and Feel:

- Determine weight for age
- Look for ulcers or white patches in the mouth (thrush)

Classify
**FEEDING &
UNDER-
WEIGHT**

IF AN INFANT

- Has no indication to refer urgently to hospital, and
 - Infant is on breastfeeding
- Assess Breastfeeding**

ASSESS BREASTFEEDING: Has the infant breastfed in the previous hour?

- If the infant has not fed in the previous hour, ask the mother to put her infant to the breast. Observe the breastfeeding for 4 minutes.
- If the infant was fed during the last hour, ask the mother if she can wait and tell you when the infant is willing to feed again.

• Is the infant well positioned?

To check the positioning, look for:

- Infant's head and body straight
- Facing her breast
- Infant's body close to her body
- Supporting the infant's whole body (all of these signs should be present if the positioning is good)

• Is the infant able to attach?

To check the attachment, look for:

- Chin touching the breast
- Mouth wide open
- Lower lip turned outward
- More areola visible above than below the mouth (all of these signs should be present if the attachment is good)

• Is the infant suckling effectively (that is slow deep sucks, sometimes pausing)?

Not suckling at all Not suckling effectively Suckling effectively
Clear blocked nose if it interferes with breastfeeding

SIGN	CLASSIFY AS	TREATMENT
If any of the following signs: • Not well positioned or • Not well attached to breast or • Not suckling effectively or • Less than 8 breastfeeds in 24 hours or • Switching the breast frequently or • Not increasing frequency of breastfeeding during illness or • Receives other foods or drinks or • The mother not breastfeeding at all or • Underweight or • Thrush (ulcers or white patches in mouth)	FEEDING PROBLEM OR UNDERWEIGHT	▶ Advise the mother to breastfeed as often and for as long as the infant wants, day and night ▶ If baby not sucking, show her how to express breast milk • If not well positioned, attached or not suckling effectively, teach correct positioning and attachment • If breastfeeding less than 8 times in 24 hours, advise to increase frequency of feeding • Empty one breast completely before switching to the other • Increase frequency of feeding during and after illness ▶ If receiving other foods or drinks, counsel mother on exclusive breast feeding. ▶ If not breastfeeding at all: • Counsel on breastfeeding and re-lactation • If no possibility of breastfeeding,* Advise about correct preparation of breast milk substitutes and using a cup ▶ If thrush, teach the mother to treat thrush at home ▶ Advise mother to give home care for the young infant ▶ Ensure infant is tested for HIV ▶ Follow-up any feeding problem or thrush in 2 days ▶ Follow-up for underweight in 14 days
• Not UNDERWEIGHT and no other signs of FEEDING PROBLEM	NO FEEDING PROBLEM AND NOT UNDERWEIGHT	▶ Advise mother to give home care for the young infant ▶ Praise the mother for feeding the infant well

*If no possibility of breastfeeding use the chart on page 12, "Check for feeding problem or underweight when an HIV positive mother has made informed decision not to breastfeed OR no chance of breast feeding by any reason."

NB. If the young infant has visible severe wasting or oedema, use the sick child acute malnutrition assessment box to classify for Severe Acute Malnutrition.



Save the Children

Check For Feeding Problem Or Low Weight For Age

Ask

- Is there any difficulty in feeding?
- What milk are you giving?
- How many times during the day and night?
- How much is given at each feed?
- How are you preparing the milk?
 - Let the mother demonstrate or explain how a feed is prepared, and how it is given to the infant
- Are you giving any breast milk?
- What foods or fluids in addition to the replacement feeding is given?
- How is the milk being given? Cup or bottle?
- How are you cleaning the utensils?

Look, Feel:

- Determine weight for age
- Look for mouth ulcers or white patches in the mouth (oral thrush).

Classify
FEEDING &
UNDER-
WEIGHT

SIGNS	CLASSIFY AS	TREATMENT
If any of the following signs: • Milk incorrectly or unhygienically prepared or • Giving inappropriate replacement milk or other foods/fluids or • Giving insufficient replacement feeds or • Mother mixing breast milk and other feeds or • Using a feeding bottle or • Underweight or • Thrush (ulcers or white patches in mouth)	FEEDING PROBLEM OR UNDERWEIGHT	▶ Counsel on optimal replacement feeding ▶ Identify concerns of the mother and the family about feeding. Help the mother gradually withdraw other foods or fluids ▶ If mother is using a bottle, teach cup feeding ▶ If thrush, teach the mother to treat thrush at home ▶ Advise mother how to feed and keep the young infant warm at home ▶ Follow-up any feeding problem or thrush in 2 days ▶ Follow-up underweight in 14 days
• Not UNDERWEIGHT and no other signs of FEEDING PROBLEM	NO FEEDING PROBLEM & NOT UNDERWEIGHT	▶ Advise mother to give home care for the young infant ▶ Praise the mother for feeding the infant well



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Summary

- IMNCI aims to reduce death, illness and disability, and to promote improved growth and development among children under five years of age
- The IMNCI case management process involves a stepwise approach consisting of the following elements: **assessment, classification, treatment, counselling and follow-up**



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Summary

- After checking the general danger signs, you should assess the child for cough/difficult breathing, diarrhoea, fever, ear problems, malnutrition, anaemia and HIV.
- The presence of any one of the general danger signs indicates a severe classification
- A child with a general danger sign or a severe classification should be referred immediately after giving appropriate pre-referral treatments



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Follow Up Visits For 2 Months Up To Five Years

If the child has:

Return for Follow-up in:

- **Pneumonia**
- **Some dehydration**
- **Dysentery**
- **Malaria, if fever persists**
- **Fever, if fever persists**
- **Fever no malaria, if fever persists**
- **Measles with eye or mouth complications**

2 days

- **Persistent diarrhea**
- **Acute ear infection**
- **Chronic ear infection**
- **Feeding problem**
- **Any other illness, if not improving**
- **MAM, if feeding problem**

5 days



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Follow Up Visits For 2 Months Up To Five Years

If the child has:	Return for follow-up in:
•Uncomplicated severe acute malnutrition	7 Days
•Anemia	14 Days
•Moderate acute malnutrition	30 Days



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Follow Up Visits For Young Infants

2 days

- LBW/preterm
- Local bacterial infection
- Low body temperature
- some dehydration
- Feeding problem
- Thrush

14 days

- underweight



Return Immediately If The Young Infant Has Any Of These Signs

- Breastfeeding or drinking poorly
- Vomiting after each feeding
- Convulsion
- Reduced activity
- Fast or difficult breathing
- Develops a fever or feels cold to touch
- Blood in stool
- Becomes sicker
- Palms and soles appear yellow



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Return Immediately

- **Advise the mother to come immediately if the child has any of these signs.**
 - **Any sick child**
 - Not able to drink or breastfeed
 - Becomes sicker
 - Develops a fever
- If child has COUGH OR COLD, also return if:**
- Fast breathing
 - Difficult breathing
- If child has diarrhoea, also return if:**
- Blood in stool
 - Drinking poorly



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MALNUTRITION



Save the Children

Contents

- Learning Objectives
- Overview of malnutrition
- Prevalence of malnutrition
- Types of malnutrition
- Risk factors for malnutrition
- Determinants of malnutrition
- Nutritional Assessment
- PEM (kwashiorkor, marasmus)
- Management of SAM



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Warm up questions

- What are the causes of under-five mortality?
- How can you define malnutrition?
- Why children are at risk for malnutrition?
- How do you assess nutritional status?
- kwashiorkor VS Marasmus



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Learning Objectives

- **At the end of this topic, students will be able to:**
 - ✓ Describe the epidemiology of malnutrition among children
 - ✓ Explain the risk factors of malnutrition among children
 - ✓ Analyze the causes malnutrition
 - ✓ Describe nutritional assessment methods
 - ✓ Explain classification of malnutrition



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Learning Objectives

- ✓ Describe PEM (kwashiorkor an marasmus)
- ✓ Recognize the complication of malnutrition
- ✓ Analyze the effect of malnutrition on the body
- ✓ Recognize the signs of SAM
- ✓ Outline the admission and discharge criteria of SAM
- ✓ Manage a child with SAM



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Overview Of Malnutrition

- **WHO** defines malnutrition as "the cellular imbalance between supply of nutrients and the body's demand for them to ensure growth, maintenance, and specific functions"
- Malnutrition is one of the most common causes of morbidity and mortality among children under the age of 5 years worldwide.
- Early onset of malnutrition can also have lasting effects on growth and functional status.



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Overview Of Malnutrition

- Severe acute malnutrition (SAM), often associated with infection contributes to high child mortality in underprivileged communities.



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Prevalence

- More than one-fourth of under-five children worldwide (about 150 million) are underweight; 27% are stunted while 10% are wasted.
- 2019 EMDHS shows
 - ✓ Stunted (37%)
 - ✓ Wasted (7%)
 - ✓ Underweight 21 %



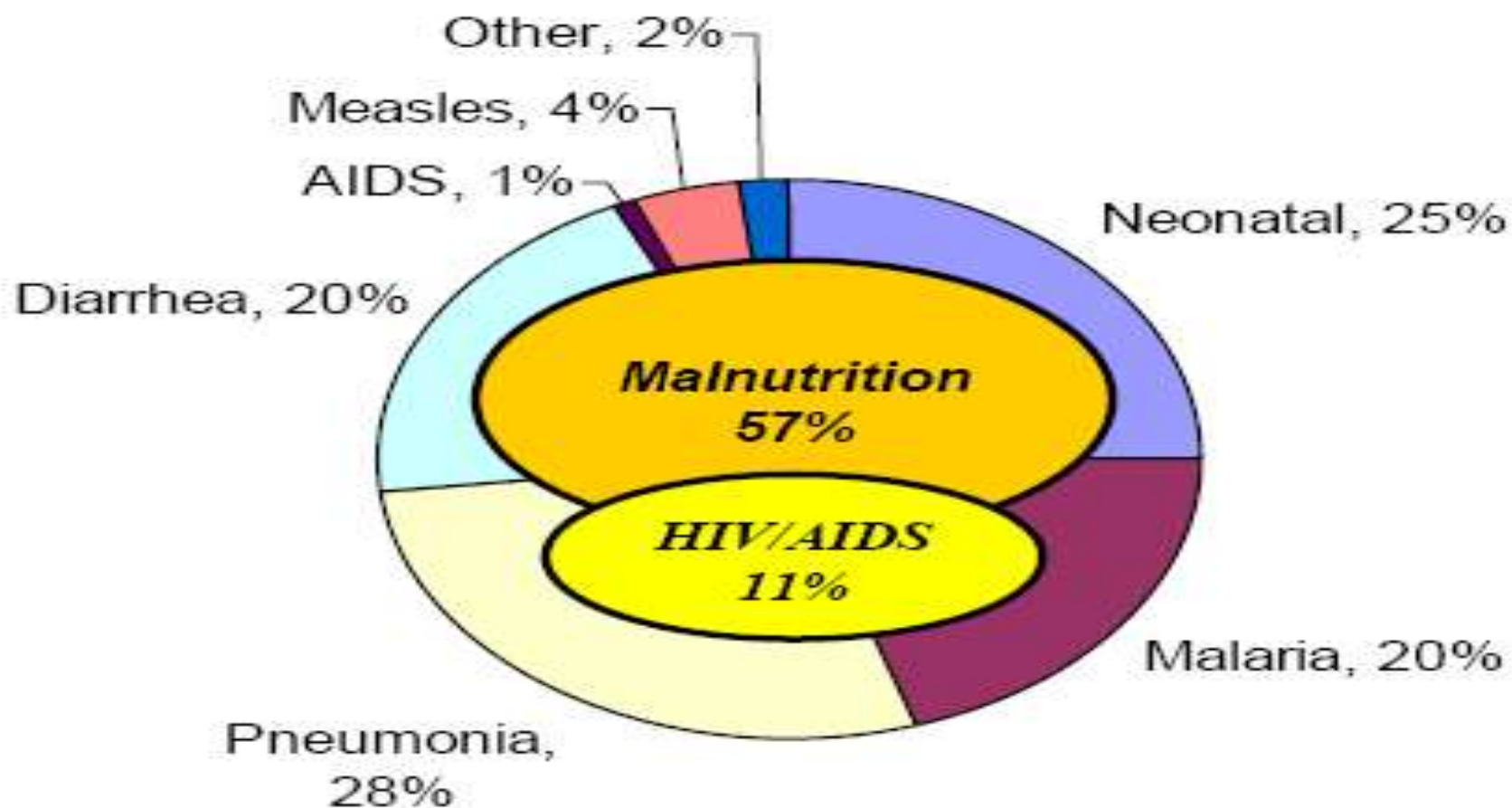
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Prevalence...

- Stunting is highly prévalent in sub-saharan Africain and in South Asia
- Increasingly, many countries are witnessing increasing public health problems posed by the double burden of malnutrition – both **under- nutrition and over weight**
- In the Ethiopian context, the commonest nutritional problems are **deficiency diseases**
- Malnutrition is the underlying cause of 57- 60% of child deaths in Ethiopia



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Types Of Malnutrition

❑ Malnutrition includes:

- Macronutrient deficiency
- Micronutrient deficiency (**Hidden Hunger**)
- Over nutrition-obesity
 - + Children with acute malnutrition appear wasted, whereas
 - + children with chronic malnutrition have stunted linear growth



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Types Of Undernutrition

**Based on
duration**

UNDERNUTRITION

```
graph TD; A([UNDERNUTRITION]) --> B([ACUTE UNDERNUTRITION]); A --> C([CHRONIC UNDERNUTRITION]); B --> D[•Wasting<br/>•Kwashiorkor<br/>•Marasmus<br/>•Marasmic- kwash]; C --> E[▪Stunting<br/>▪Underweight];
```

**ACUTE
UNDERNUTRITION**

- Wasting
- Kwashiorkor
- Marasmus
- Marasmic- kwash

**CHRONIC
UNDERNUTRITION**

- Stunting
- Underweight

Types...

- ❑ **Malnutrition also classified in to (based on etiology)**
 - **Primary malnutrition** resulted from inadequate food intake
 - **Secondary malnutrition** resulted from increased nutrient needs, decreased nutrient absorption, and/or increased nutrient losses



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Risk Factors Of Malnutrition

➤ There is high nutritional needs of infants and young children as;

- More energy expenditure (i.E., Basal metabolism, metabolic response to food, and physical activity),
- Rapid rate of growth
- New growth of tissues
- Physiological changes (e.G., Puberty).
- Infection is common during child hood period (particular at early age)

 **leads to nutritional disruption**



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Determinants Of Child Malnutrition

Generally:

- ❖ **No Single cause:**
- ❖ **Can have many causes**
 - ✓ **Basic causes**
 - ✓ **Underlying causes**
 - ✓ **Immediate causes**



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Underlying causes

- **Lack of knowledge about diet**

- ◆ Parents often withhold food supplements/fluids during episodes of diarrhea
- ◆ Affluent families may also have irrational beliefs about the nutritional needs of infants and

- **Economic factors (poverty)**

- **Social factors**

- Repeated pregnancies, inadequate child spacing, food taboos, broken homes and separation of a child from his parents `

- **Biologic factors** → maternal malnutrition → LBW (prematurity)

- ◆ Malnourished mothers have a high incidence of LBW and growth retarded babies with poor nutritional reserve



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Underlying causes

■ Feeding habits :

- ◆ Lack of EBF for first 6 months makes the child prone to **early onset malnutrition**
- ◆ Artificial feeding is disastrous for the baby because of the **poor quality** of the substitute milk, **excessive dilution and use of unhygienic feeding bottles and nipples**
- ◆ Delayed introduction of complementary feeding



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Underlying causes

■ Infections:

- Diarrhea, pneumonia, malaria, measles, and tuberculosis precipitate acute malnutrition and aggravate the existing nutritional deficit
- During infections, child's appetite is impaired
- The patient catabolizes his own tissues to produce the additional heat energy
- Metabolic demands for protein during infections are higher
- Protein may be lost because of tissue breakdown and in pus and exudates.
- Also, malnutrition may adversely affect the immune status and make the malnourished individuals more vulnerable to infections
- This sets up a vicious cycle of malnutrition-infection malnutrition



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Underlying causes

- Religious customs (people of certain religions avoid non-vegetarian diet which has high-quality proteins)
- Environmental factors such as overcrowded and unsanitary living conditions, drought, earthquakes.
- Insufficient health services
- Inadequate care for children and women
- Wars and civil unrest exaggerate pre-existing malnutrition in the community
- Natural disasters such as floods, earthquakes, and droughts



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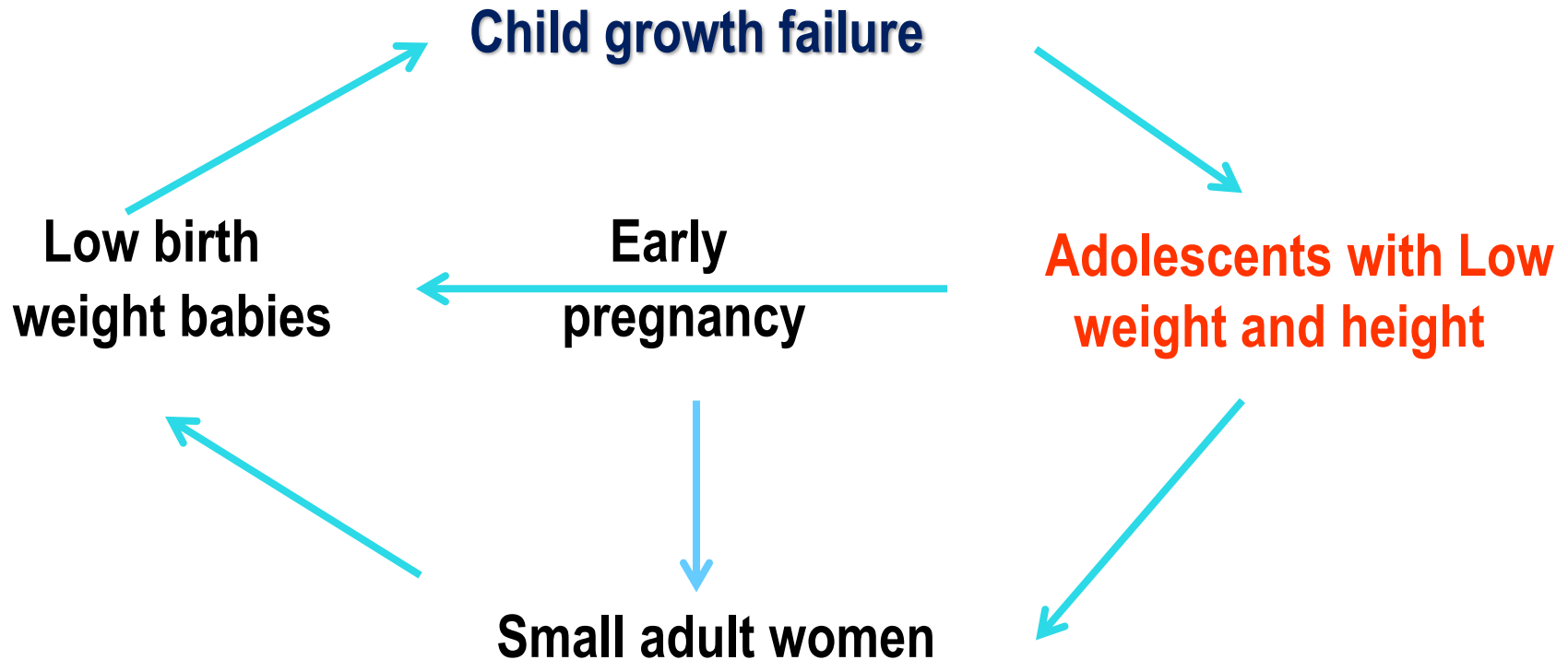
Strategies to Prevent and Control Under nutrition

- Improve household food security.
- Improve diversity of diet.
- Improve maternal nutrition and health care.
- Improve child feeding practices.
- Ensure child health care (immunization, medical care, growth monitoring).
- Provide nutrition rehabilitation.



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Intergenerational cycle/effect of malnutrition



How can we break the cycle?

- Can we conclude that teenage pregnancy is likely to play a big role in the cycle of growth failure and stunting?



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Nutritional Assessment



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Definition

- ❑ Nutritional assessment is an interpretation of **anthropometric, biochemical (laboratory), clinical and dietary survey data** to tell whether a person/group of people is well nourished or malnourished (Over nourished or under nourished).



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INTRODUCTION



- The nutritional status of an individual is often the result of many inter-related factors.
- It is influenced by food intake, quantity & quality, & physical health.
- The spectrum of nutritional status spreads from severe under nutrition to obesity

Nutritional Assessment -Why?

The purpose of nutritional assessment is to:

- ☀ Identify individuals or population groups at risk of becoming malnourished
- ☀ Identify individuals or population groups who are malnourished
- ☀ To develop nutrition and health care programs that meet the community needs which are defined by the assessment
- ☀ To measure the effectiveness of the nutritional programs & intervention once initiated



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Nutritional Assessment

- There are **direct and indirect** methods of assessing Nutritional status
- **Direct:** involve the direct measurement of body dimensions and proportions, dietary intake, appearance of the clinical symptoms and signs related to a specific nutrient.

➤ Abbreviated as the ABCDs

- ☐ **A=Anthropometry**
- ☐ **B= biochemical/Biophysical,**
- ☐ **C= Clinical**
- ☐ **D= Dietary**



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Cont...

- **Anthropometry:** height, weight, BMI-for - age, MUAC
- **Biochemical:** analysis of blood, urine, and other body tissues
- **Clinical:** complete physical examination, and a medical and psychosocial history
- **Dietary:** foods and quantities eaten, eating habits, accessibility of food, and cultural and socioeconomic factors that affect selection of food



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Anthropometric Assessments

- Assessment of nutritional status by **anthmpometry** is the simplest and most useful tool for assessing the nutritional status of children
- Weight, height, and mid-arm circumference should be compared to the **reference norms for the corresponding age in the well nourished and healthy children of the community**
- International references for anthropometric measurements have been advocated



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Anthropometric Assessments

- These include Harvard standards (UK), NCHS/CDC standards (US). WHO
- However, the recent reference values provided by WHO provide the most recent and widely acceptable international growth standards for comparison



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Purposes of Anthropometric measurements

- Anthropometric measurements are performed with two major purposes in mind:
 - ❑ **IN CHILDREN:** to assess physical growth
 - ❑ **IN ADULTS:** to assess changes in body composition or weight



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Anthropometric Measurements Of Growth #1

- ❑ Growth performance of children is an excellent reflection of their underlying nutritional status.
- ❑ Children adapt to the chronic nutritional insult by either reducing their rate of growth or by totally failing to grow.
- ❑ Therefore, assessment of growth performance of children is one very important purpose of anthropometric measurements.
- ❑ The following body measurements are good indicators of growth performance of children at different ages when combined with the cut-off points.



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Head Circumference (HC)

- ❑ Measured using flexible measuring tape
- ❑ It is the circumference of the head along the **supra orbital ridge anteriorly and occipital prominence posteriorly.**
- ❑ **HC is useful in assessing chronic nutritional problems in under two children.**
- ❑ But after 2 years as the growth of the brain is sluggish it is not useful.



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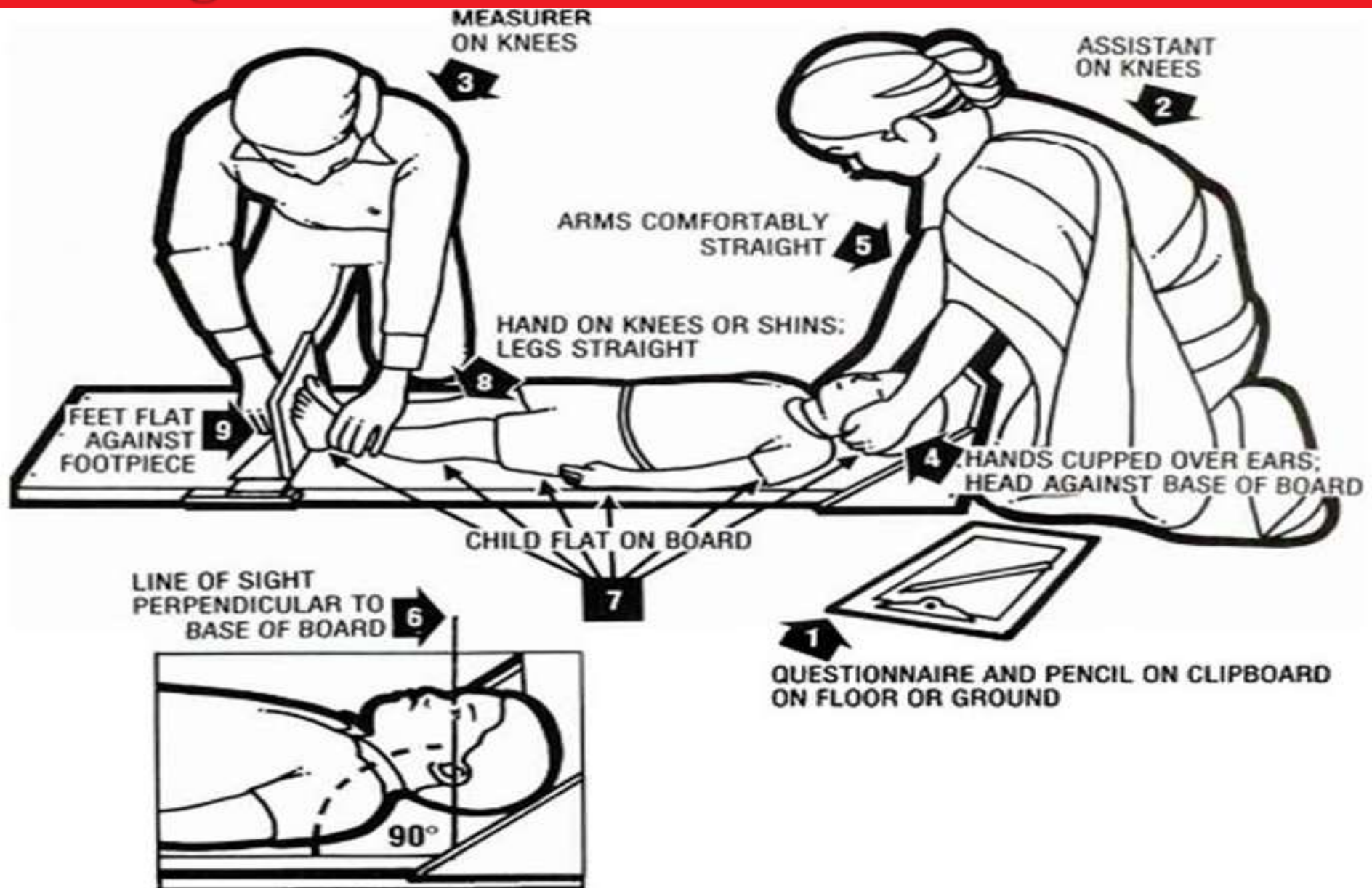
Length

- ☐ A wooden measuring board **(also called sliding board)** **is used for measuring length.**
- ☐ It is measured in recumbent position in children <2 yrs old
- ☐ It is always > height by 1-2cm.
- ☐ One assistant is needed in taking the measurement
- ☐ Measurement is read to the nearest mm



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Length...



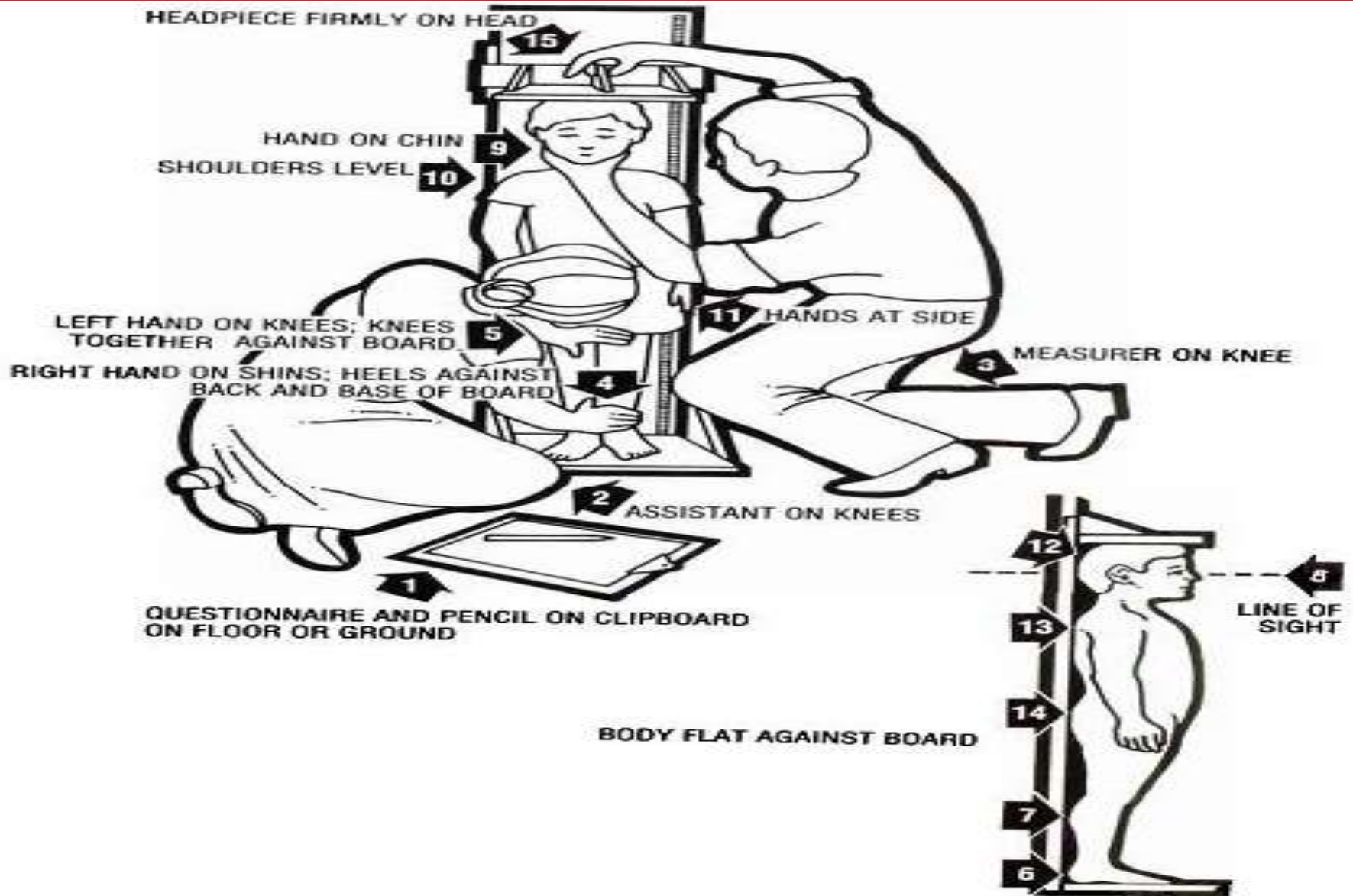
Height

- ☐ Is measured in children > 2 yrs and adults in standing position
- ☐ The head should be in the Frankfurt plane
- ☐ During measurement, knees should be straight and the heels buttocks and the shoulders blades, should touch the vertical surface of the stadiometer (anthropometer) or wall.
- ☐ **Stadiometer or portable anthropometer can be used for measuring.**
- ☐ There is also a plastic instrument called acustat Stadiometer that is cheaper than the conventional Stadiometer.



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Height...



Weight

- ❑ Weighing sling (spring balance) also called **salter scale** is used for measurement of weight in children **< 2 years**.
- ❑ In adults and children **>2 years**, **beam balance** is used



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Weight...



PUT HANDS THROUGH LEG HOLES



GRASP FEET

➤ **What is an index?** It is a combination of two measurements or a measurement plus age.

➤ The following are few of them:

❑ **Head circumference-for age**

❑ **Weight -for-age**

❑ **Height-for age**

❑ **Weight for height**



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Meanings Of The Indices Derived From Growth Measurements

Weight for Age = $\frac{\text{Weight of the child}}{\text{Weight the normal child of the same age}} \times 100$

Weigh for height = $\frac{\text{Weight of the child}}{\text{Weight of the normal child of the same height}} \times 100$

Height for age = $\frac{\text{Height of the child}}{\text{Height of the normal child of the same age}} \times 100$



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Mid Upper Arm Circumference (MUAC)

- Used to screen for malnutrition - time and equipment are very limited, such as famines and refugee crises
- Used to identify children for further evaluation
- It is independent of age and gender
- Between six month and five years of life



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MUAC...

- MUAC is measured on the upper left arm
- To locate the correct point for measurement,
 - ✓ The child's elbow is flexed to 90°, with the palm facing upwards
 - ✓ A measuring tape is used to find the midpoint between the end of the shoulder (**acromion**) and the tip of the elbow (**olecranon**);
 - ✓ This point should be marked
- ❖ The arm is then allowed to hang freely, palm towards the thigh, and the measuring tape is placed snugly around the arm at the midpoint mark
- ❖ The tape should not be pulled too tight or too loose
- ❖ Read the measurement to the nearest 0.1 cm.



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Indicator

- An indicator is an **index + a cut-off point**.

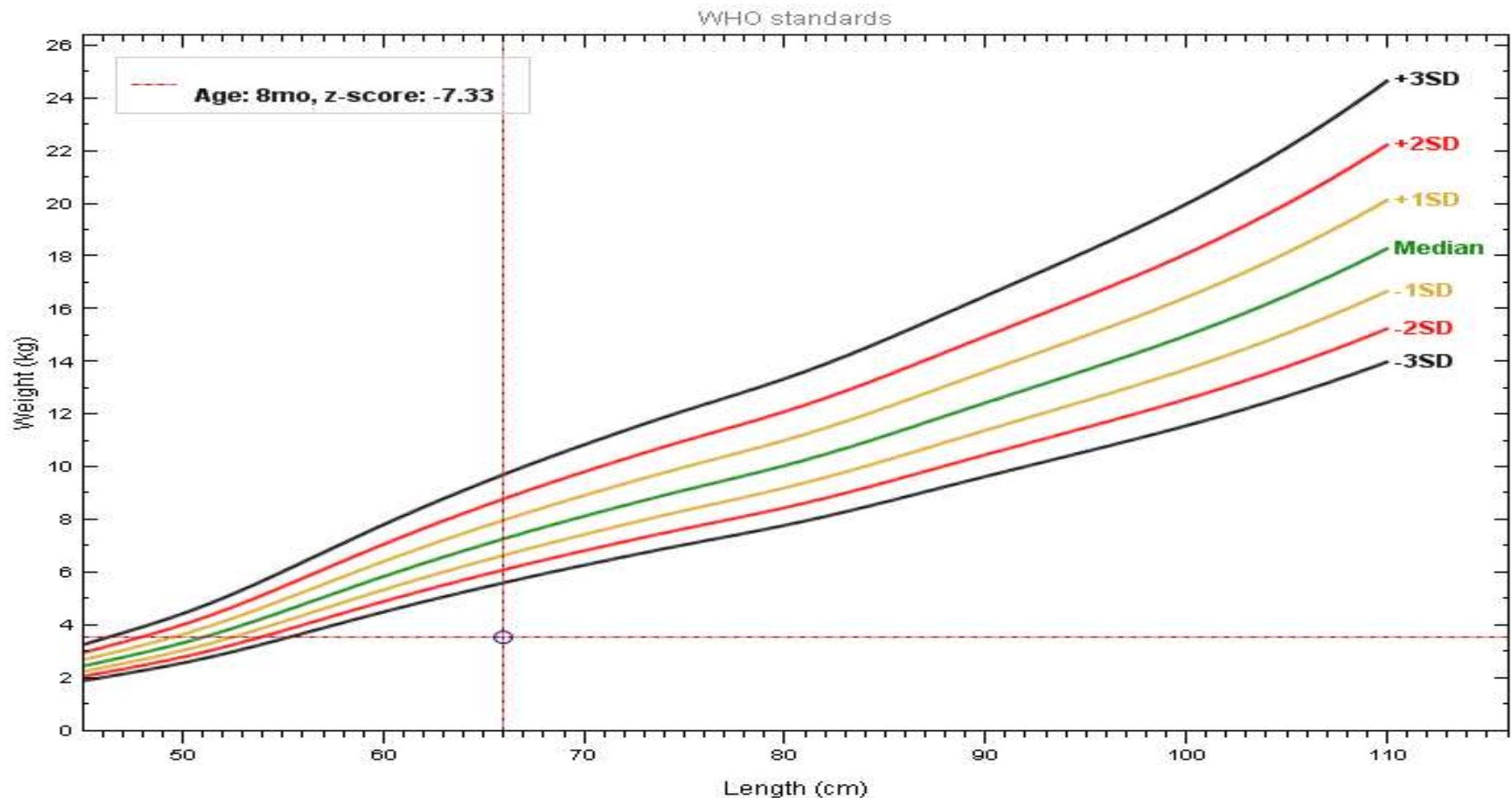
E.g.

- ❑ $W F A < 60\%$ = is indicator of severe malnutrition
- ❑ $BMI < 16 \text{ kg/m}^2$ = indicator of severe chronic energy deficiency
- ❑ $W F H < 70\%$ = is indicator of severe wasting



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We use SD in Ethiopia



Classification of PEM: WHO Classification

- WHO definitions — has developed criteria for the classification of moderate or severe malnutrition
- This classification is based on two anthropometric indicators (stunting and wasting) and one clinical indicator (edema).
- The child's weight for height, and the height for age are expressed as Z-scores
- Wasting and stunting are defined by the following



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Classification: WHO Classification

- **Wasting** : measured by deficit in weight-for-height.
 - Indicates a deficit in tissue and fat mass and may result either from failure to gain weight or from actual weight loss
 - It usually signifies acute onset malnutrition
 - A child is considered wasted if the weight for height/length is less than -2SD (Z) score or less than 85% of the expected
 - Moderate wasting – wt/ht Z-score <-2 to -3
 - Severe wasting – weight/height /length Z-score <-3 or less than 70% of expected



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Classification: WHO Classification

- **Stunting** (deficit in height-for-age)
 - Signifies accumulated consequences of retarded growth over a prolonged period of time
 - points towards a chronic course of malnutrition
 - It may also be a residue of past malnutrition or could be due to non-nutritional disorders like genetic short stature or endocrinal disorders.
 - The diagnosis of stunting is based on a height for age less than -zsd (z) score or less than 90% of the expected
 - Moderate stunting – height/length for age z-score <-2 to -3
 - Severe stunting – height/length for age z-score <-3 or less than 85% of expected



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Classification: WHO Classification

- ❑ Children are classified as having malnutrition:
 - Moderate malnutrition – moderate wasting or stunting
 - Severe malnutrition – severe wasting, severe stunting
 - Presence of edema along with stunting or wasting indicates severe under nutrition.



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Classification: WHO Classification

■ Underweight

- Is a composite index of weight-for-height and height-for-age and
- Thus does not distinguish between acute malnutrition (wasting) and chronic malnutrition (stunting)
- Children can be underweight for their age because they are stunted, wasted, or both
- Weight-for-age is an overall indicator of a population's nutritional health.



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Classification



Normal
Normal weight
and height

Wasted
Thinner than
normal

Stunted
Shorter than
normal

Wasted and stunted
Thinner and shorter
than normal



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Severe Acute Malnutrition (SAM)

- The term SAM has been coined by international agencies (WHO and UNICEF) to identify malnourished children at highest risk of death, and for the
 - Purpose of therapeutic feeding in the hospital or community
- SAM is defined by
 - A very low weight for height (below -3Z scores of the median WHO growth standards)
 - By visible severe wasting or
 - By the presence of nutritional edema
- Using mid-arm circumference, the cut-off point to define SAM is 11.5 cm for children aged between 6 months and 5 years



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Classification

- The severity of malnutrition can be calculated based on the median/ average value and classified as;

Grade	W/a (Gomez)	Ht/age (Waterlaw)	W/ht (Waterlaw)
Normal	>90	>95	>90
Mild	75-90	90-95	81-90
Moderate	60-74	85-90	71-80
Severe	<60	<85	<70



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Classification: Welcome

- This classification is also based on deficit in body **weight for age and presence or absence of edema**
- Children weighing between 60-80% of their expected weight for age with edema are classified as **kwashiorkor**.
- Those weighing between 60-80 percent of expected without edema are known as having **under weight**.
- Those without edema and weighing less than 60 per cent of their expected weight for age are considered to be having **marasmus**.
- **Marasmus- kwashiorkor** is applied to children with edema and body weight less than 60 per cent of expected



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Classification

The Welcome Classification

Indicator: Weight-for-age and edema

% expected Weight for Age	OEDEMA	
	Present	Absent
60 - 80	Kwashiorkor	Underweight
< 60	Marasmic-Kwashiorkor	Marasmus

Advantage: Useful for classifying more severe forms.

Limitations: Does not take account of height differences.

Age of the child must be known.

Malnutrition Indicators

NUTRITION INDICATOR	MEASUREMENT INDICATOR	CLINICAL INDICATOR
Acute Malnutrition (SAM & MAM)	Weight-for-Height	Wasting, kwashiorkor
	Mid-upper arm circumference	
Chronic Under nutrition	Height-for-Age	Stunting
Underweight (composite indicator)	Weight-for-Age	Underweight
Over nutrition	Body Mass Index (weight/Height)	Overweight/Obesity
Micronutrient Deficiencies	Biochemical indicators	Xerophthalmia, stomatitis



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Protein-energy Malnutrition (PEM)

- The term protein-energy malnutrition (PEM) applies to a group of related disorders that include **marasmus, kwashiorkor, and intermediate states.**
- Nutritional deficiency of macronutrients is a major health problem in the developing countries
- Usually associated with micronutrient deficiencies.



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PEM

- Worldwide, severe protein energy malnutrition is a leading cause of death among children younger than five years of age
- Severe protein-energy malnutrition is associated with
 1. Marasmus (wasting syndrome) and
 2. Kwashiorkor, or
 3. With manifestations of both
- Lack of protein in the diet is more often due to low food intake, rather than a qualitative defect in the diet
- Nutritional marasmus and kwashiorkor are two extreme forms of malnutrition.



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Kwashiorkor

- Cecilly Williams, a British nurse, had introduced the word Kwashiorkor to the medical literature in 1933.
- The word is taken from the Ga tribe in Ghana & used to describe : “The sickness of weaning” “The sickness the older child gets when the next baby is born”



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Kwashiorkor ...

- **In this form of severe undernutrition, the child's muscles are wasted, but the wasting may not be apparent due to **generalized oedema** (swelling from excess fluid in the tissues).**



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Kwashiorkor ...

- **Incidence is more in:**
 - Low birth weight
 - Age >6 months
 - Broken families
 - In children with whose parents are unemployed
 - Large families



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Clinical features

- ❑ Essential features of kwashiorkor include
 - (I) Edema
 - (II) Markedly retarded growth
 - (III) Psychomotor changes
- Other features that help in making a diagnosis include
 - ✓ Skin and hair changes
 - ✓ Hepatomegaly
 - ✓ Impaired appetite
 - ✓ Associated nutritional deficiencies.



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Clinical features

- ❖ Edema and preserved subcutaneous tissues **mask muscle wasting** which becomes apparent when the child recovers from edema
- ❖ Infections (eg, diarrhea, acute respiratory infections, skin infection) and infestations (eg, scabies, lice) are present in majority of cases



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Edema

- Is the clinical manifestation of expansion of extracellular water (ECW) volume caused
 - ✓ By pre-renal diversion of fluid from the capillary bed to extracellular space
- Edema may be caused by
 - I. Hypo-albuminemia
 - II. Retention of fluid and water due to increased capillary permeability as a result of infection, potassium deficiency
 - III. Free radical induced damage to cell membranes



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Edema

- The edema starts in the lower extremities and progresses to involve the upper limbs and the face
- The face appears moon-shaped and puffy
- The trunk is affected to a lesser extent, and ascites is rare unless there is a co-existent cause such as tuberculosis
- In a previously malnourished child, edema is precipitated by debilitating illnesses such as measles or diarrhea
- Presence of edema may mask features of severe dehydration secondary to diarrhea in a severely malnourished child



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How To Check For Edema?

- Grasp both feet so that they rest in your hand with your thumbs on top of the feet
- Press your thumbs gently for three seconds or count 101,102,103
- The child has edema if a pit (dent) remains in both feet when you lift your thumbs
- Edema must appear in both feet= sign of severe malnutrition



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Extent of edema

The extent of edema is commonly rated

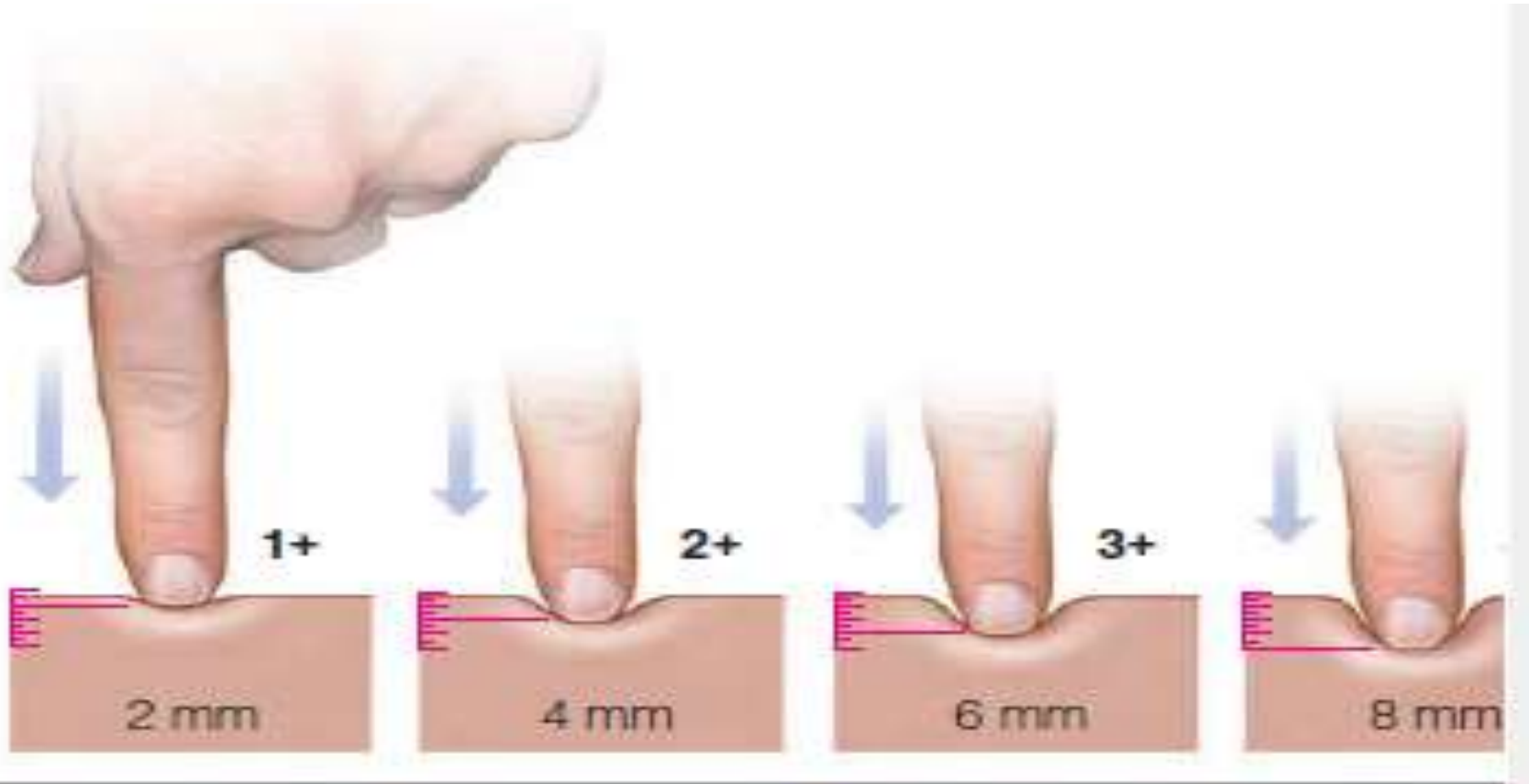
Grade	
Absent	Absent
Grade +	Mild: both feet/ankles
Grade ++	Moderate: both feet, plus lower legs, hands or lower arms
Grade +++	Severe: generalized bilateral pitting edema; including both feet, legs, arms and face



Save the Children

Edema

- Scale for grading edema



Save the Children

Psychomotor change

- Child with kwashiorkor is lethargic, listless, and apathetic
- Play activity and interaction with others is reduced
- Child resists attempts to examination by medical personnel
- Child loses appetite, and it is difficult to feed him orally in the initial phase of dietary management
- Child may need nasogastric feeding for nutritional rehabilitation
- Appetite improves once co-existent infection is controlled by antibiotics



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Skin changes

- A variety of skin changes have been described in kwashiorkor
- Dry and scaly skin with cracking is apparent over large areas of trunk and limbs giving **mosaic appearance**
- Most typical skin lesion is **crazy pavement dermatosis** which is characterized by **jet black patches** over pressure sites, especially of extremities which later exfoliate to expose raw **hypopigmented or reddish skin**
- Peeling plaques of this dermatosis give appearance of old paint peeling off the surface, and is termed **flaky paint dermatosis**
- Multiple punctate pinkish areas over the legs are seen in few cases



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Skin changes

- Perioral lesions occur frequently which are caused by riboflavin deficiency and Zinc deficiency
- Deep and indolent ulcerations and fissures in flexures or over pressure points may also occur
- Skin changes regress rapidly following successful dietary management of kwashiorkor



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Skin change

Dermatosis

- It is more common in children who have edema than in wasted children
- A child with dermatosis may have patches of skin that is abnormally light or dark in color
- Shedding of skin in scales or sheets, and ulceration of the skin of the perineum, groin, limbs, behind the ears, and in the armpits



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Skin change

Dermatosis...

- There may be weeping lesions
- There may be severe rash in the nappy area
- Any break in the skin can let dangerous bacteria get into the body
- When the skin is raw and weeping, the risk is very high.



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Skin change

Dermatosis...

- The extent of dermatosis can be described:
 - Mild (+): discoloration or a few rough patches of skin
 - Moderate (++): multiple patches on arms and/or legs
 - Severe (+++): flaking skin, raw skin, fissures (openings in the skin)



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Hair changes

- Hair become hypopigmented, lusterless, thin, and easily pluckable
- The discoloration may be seen only in some segments, often the growing ends
- During recovery, segmental discoloration (**flag sign**) is more apparent as the growing part of the hairs gets pigmented



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Changes In Body Composition And Function

Parameter	Features
Growth	Short statured
Mental faculties	Irritable, lack of interest in surroundings, apathetic, dull, lethargic, resents meddling
Hair	Lack of luster, easily pluckable, sparse , loss of natural curls, brownish discoloration, flag sign
Anterior fontanel	Delayed closure
Face	Moon face, a full well-rounded somewhat pendulous and blubbery cheeks
Eyes	Conjunctival pallor, signs of vitamin-A deficiency



Save the Children

Changes In Body Composition And Function

Parameter	Features
Oral cavity	Angular stomatitis: iron deficiency anemia, B-complex deficiency; oral thrush
Tongue	Oedema of tongue, scarlet and raw tongue (bright red), magenta tongue (purplish red), atrophy of papillae, geographic tongue (the tongue has irregularly distributed patchy areas of denudation and atrophy of epithelium)
Teeth	Caries, delayed dentition, enamel hypoplasia, enamel erosion
Gums	Spongy, bleeding gums, recession of gums



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Changes in Body Composition and Function

Parameter	Features
Cheeks	Baggy cheeks of Trowel
Neck	Neck appears to be long due to wasting of muscles and sub-cutaneous fat
Chest	All the ribs and spinal processes are prominently seen due to wasting of muscles and sub-cutaneous fat
Abdomen	Abdomen is distended due to intestinal atony, hepatomegally, rarely ascites

Changes in Body Composition and Function

Parameter	Features
Skin	Hypopigmented and hyperpigmented patches (crazy pavement dermatosis), flaky pavement dermatosis (peeling of skin), xerosis: generalized dryness with branny desquamation, petechiae, pellagrous dermatosis
Extremities	Are thin due to wasting, pedal oedema,
CNS	Patient is dull, lethargic, apathetic, hypotonia due to nutritional myelopathy



Figure 43-3 "Flaky paint" dermatosis on the thighs. (From Liu T, Howard RM, Mancini AJ, et al: Kwashiorkor in the United States. Arch Dermatol 2001;137:630-636.)

Changes in Body Composition and Function

Parameter	Features
CVS	Intensity of heart sounds is decreased due to decrease of cardiac output
Respiratory system	Features of secondary infection
GIT	Hepatomegaly Rectal prolapse is common Salivary glands atrophy



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Differential diagnosis

- Nephrotic syndrome
- Cirrhosis of liver
- Congestive heart failure



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Investigations

☐ Blood

- CBC
- Serum electrolytes, plasma protein estimation
- Blood culture & sensitivity for evaluation of septicaemia

☐ Urine

- Albumin, sugar, deposits, urine culture & sensitivity

☐ Stool

- Ova of parasite, culture & sensitivity if there is diarrhoea

☐ Chest X-ray: to r/o TB & other infections

☐ Mantoux test



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Complications

- Hypothermia
- Hypoglycaemia
- Infections
- Electrolyte disturbances
- Dehydration
- Anemia



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Prevention

- Health and nutrition education
- Early identification and treatment of the diseases
- Exclusive breast feeding up to six months
- Complementary feeding at the age of six months
- Family planning
- Safe and adequate water supply
- Immunization



Save the Children

Marasmus

- The term marasmus is derived from the Greek marasmos, which means wasting.
- A nutritional disorder due to deficiency of protein and calories, **particularly calories**, characterized by:
 - Growth failure
 - Gross wasting
 - Absence of oedema
 - Age 2- 4 yrs
 - Weight is more affected than height
- Marasmus represents the end result of starvation where both proteins and calories are deficient.



Save the Children

Marasmus

- Marasmus represents an adaptive response to starvation, whereas kwashiorkor represents a maladaptive response to starvation
- In Marasmus the body utilizes all fat stores before using muscles



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Clinical features

- The most characteristic feature of marasmus is **visible severe wasting** of muscle and subcutaneous tissues
- Child is usually alert, has an emaciated look, and has no edema
- Skin is thin, dry, scaly, and wrinkled with loose folds apparent over the gluteal region and thighs
- Subcutaneous fat gets depleted as it is used up for providing energy
- Buccal pad of fat is the last to disappear because of presence of saturated fats which are utilized at last



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Clinical features

- Atrophic muscles make a contour under the thin and wrinkled skin
- Bones appear prominent giving a skeleton like look
- Hairs may become hypopigmented
- Abdomen appears distended because of hypotonia of abdominal wall muscles
- Appetite may be voracious, especially in absence of infection or a secondary cause of malnutrition



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Clinical features: severe wasting

- A child with severe wasting has lost fat and muscle and has a “skin and bones” appearance.
- Severe wasting in infants of under 6 months is marked by visible severe wasting
- Whereas for children >6 month severe wasting is assessed objectively using anthropometric measurements (WFL/H, MUAC).



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Severe wasting...

- Visible severe wasting is an important sign of SAM for infants under 6 months of age
- It is used as a criteria to classify as SAM only for infants below 6 months age.
- To look for severe wasting, remove the child's clothes.
- **Look at the front view of the child:**
 - Is the outline of the child's ribs easily seen?
 - Does the skin of the upper arms look loose?
 - Does the skin of the thighs look loose?



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Severe wasting..

- Look at the back view of the child:
 - Are the ribs and shoulder bones easily seen?
 - Is flesh missing from the buttocks?
- When wasting is extreme, there are folds of skin on the buttocks and thighs
 - ✓ It looks as if the child is wearing “baggy pants”.
- Because a wasted child has lost fat and muscle, this child will weigh less than other children of the same height/length and will have a low weight-for-height/Length



Save the Children



ve the Children

Clinical features

Grossly underweight

No body fat

Gross muscle wasting



Save the Children

Changes in Body Composition and Function

Parameter	Features
Mental faculties	If there is no infection: • Patient is alert, playful, lovely look • The cry is vigorous • Appetite is vigorous If there is an infection: • The child is less active, apathetic, cries in a low tone, refusal to food
Head	• Hair is normal, delayed closure of anterior fontanel may be present

Changes in Body Composition and Function

Parameter	Features
Face	•Eyes: no signs of vitamin A deficiency •Mild to moderate anemia may be present •Sunken cheeks are present due to loss of buccal fat of fat •Child appears as a wizened little old man or monkey face
Oral cavity	•Oral thrush may be present •Stomatitis and angular stomatitis may be present
Chest	All the ribs and spinal processes are prominently seen due to wasting of subcutaneous fat & muscles

Changes in Body Composition and Function

Parameter	Features
Abdomen	•Abdomen is distended due intestinal atony •No hepatomegaly
Skin	•Thin •No skin changes
Extremities	Thin due to wasting of muscles & sub-cutaneous fat. There is no oedema
CVS	Size of heart decreased. Decreased intensity of heart sounds due to low cardiac output
Respiratory system	Features of secondary infection may be present



Save the Children

Investigations

- CBC
- Urine and stool culture
- Chest X-ray
- Mantoux test



Save the Children

Complications

1. Immediate complications

- Hypoglycaemia
- Hypothermia
- Septicaemia
- Electrolyte imbalance

2. Late complications

- Intellectual sub-normality
- Growth retardation



Save the Children

Mixed (marasmus-kwashiorkor)

- Mixed marasmus-kwashiorkor (edematous malnutrition) may occur:
 - In a child who has inadequate dietary intake of all nutrients and is triggered by a common infectious illness of childhood
- Children with mixed marasmus-kwashiorkor often have
 - ✓ Anorexia
 - ✓ Dermatitis and
 - ✓ Sometimes neurological abnormalities (depression and flat affect) and hepatic steatosis



Save the Children

Mixed (marasmus-kwashiorkor)

- The transition from marasmus to mixed marasmus-kwashiorkor is associated with particularly high morbidity and mortality
- Perhaps because of
 - ✓ The frequent association with acute infection or
 - ✓ Children with this clinical course are less well adapted metabolically than children with pure marasmus



Save the Children

End organ effects of malnutrition

- **Body composition**
 - Total body water increases(ECF)
 - Increased Na^+
 - Decreased K^+ and Mg^{++}
 - Muscle and fat loss
- **GIT**
 - Villus atrophy
 - Reduced enzymes
 - Bacterial over growth



Save the Children

End organ effects of malnutrition

- **Liver**
 - Reduced synthesis of protein
 - Impaired gluconeogenesis
 - Decreased metabolism of toxins and substances
- **Cardiac**
 - Myocardial atrophy, reduced cardiac out put , decreased BP
- **Hematology**
 - Anemia
- **Metabolic**
 - Hypoglycemia
 - Reduced metabolic rate



Save the Children

End organ effects of malnutrition

Renal impairment

- ✓ Glomerular filtration rate and renal plasma flow are reduced in severe PEM
- ✓ Capacity of the kidneys to excrete excess acid or water is greatly affected
- ✓ Urinary phosphate and sodium excretion are reduced.



Save the Children

End organ effects of malnutrition

Central nervous system

- ✓ Growth and development of the brain ; is adversely affected especially if malnutrition sets in first two years of life which is a period of rapid brain growth
- ✓ Associated micronutrient deficiencies (eg, iron, thiamine, vitamin B12, zinc, iodine) further impair the brain function resulting in lower developmental scores.
- ✓ There is impairment of sensory coordination and recognition of figures and shapes
- ✓ Lag in development and intelligence (IQ) may persist even after recovery from malnutrition.



Save the Children

End organ effects of malnutrition

Skin, Muscles, Exocrine glands

- The skin and subcutaneous fat are atrophied which leads to loose folds of skin and sunken eyes
- Many exocrine glands including the sweat, lacrimal (tear), salivary, and pancreatic glands are atrophied
- Many signs of dehydration (decreased skin turgor, dry oral mucosa, absent tears, sunken eyes) are unreliable in the severely malnourished child.



Save the Children

End organ effects of malnutrition

- **Immune System**

- **Immunity**

- Impaired especially the Cell mediated
 - Reduced IgA
 - Reduced phagocytosis

- **Inflammatory response**

- Impaired acute phase response
 - Reduced chemotaxis of WBC to site of infection/inflammation
 - Prone to infectious agents/subtle signs



Save the Children

End organ effects of malnutrition

- All aspects of immunity are diminished;
Cell-mediated immunity is depressed, IgA secretions are reduced, phagocytosis is impaired, lymph glands, tonsils and thymus are atrophied.
- Typical signs of infection like fever are frequently absent
- Relying on the presence of fever to suspect infection could therefore be misleading



Save the Children

End organ effects of malnutrition

- Severe malnutrition is a state of relative immuno-deficiency
- These children are more prone to infection: b/c of
 - ✓ There is impaired cell mediated immunity due to atrophy of thymus and lymphoid tissues
 - ✓ Sensitization to antigens and recognition of earlier sensitization is impaired
 - ✓ There is impaired fungicidal and bactericidal capacity of leukocytes



Save the Children

End organ effects of malnutrition

- Secretory IgA may be reduced resulting in **increased severity, prolonged infectivity, and delayed recovery from infections**
- **Leukocyte count** is generally normal in severe malnutrition, leukopenia may occur
- **Physical barriers** against infection such as skin and mucosal membranes are breached
- Skin is thin and ulcerated which is easily penetrated by microbes
- Mucosa are atrophic resulting in increased risk of gastrointestinal and respiratory tract infections



Save the Children

Management of Acute Malnutrition

- The comprehensive management of acute malnutrition, which is also called Community-Based Management of Acute Malnutrition (CMAM), consists of four main components:
 1. Community outreach/mobilization
 2. Inpatient Treatment/ care
 3. Outpatient Treatment Program (OTP)
 4. Targeted supplementary feeding program (TSF)

Inpatient Treatment/ care

- Children who are acutely malnourished with associated medical complications
- Poor appetite
- Infants less than 6 months with SAM need to be treated in inpatient care facility until they are well enough
 - To continue nutritional rehabilitation in OTP



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Outpatient Treatment Program (OTP)

- ❖ **Outpatient care is intended for children presenting SAM**
 - Without medical complications
 - Children that have recovered in inpatient care after they have recovered appetite
 - They receive ready-to-use therapeutic food (RUTF) and routine medicines, which are taken at home, and the child attends the outpatient care site every week



Save the Children

Recommended criteria for SAM and Admission to TFP

- All children who have SAM should be admitted to a Therapeutic Feeding Program (TFP)
 - ✓ In-patient care (TFU) or
 - ✓ Out-Patient Treatment Program (OTP)



Save the Children

Recommended criteria for SAM and Admission to TFP

- Children with SAM are treated as in-patient or OTP
- Depends on
 - The age of a child
 - Presence or absence of medical complications
 - Result of the appetite test determine whether a child with SAM should be treated in OTP or in-patient care (TFU, or stabilization center or SAM ward)



Save the Children

Criteria for SAM

1. Infants less than six months or less than 3 Kg:

- WFL less than 70% or $< -3Z$ score

OR

- Presence of pitting Edema of both feet

OR

- Visible Severe Wasting if it is difficult to determine WFL

- *Note: Infants below six months of age with SAM can be treated as in patient*



Save the Children

Criteria for SAM

Note:

- Visible severe wasting is an important sign of SAM for infants under 6 months of age
- It is used as a criteria to classify as SAM only for infants below 6 months age.



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Criteria for SAM

2. Children 6 months to 5 years:

- (WFL) / WFH less than 70 % or < -3 Z score **OR**
- Presence of pitting Edema of both feet **OR**
- MUAC < 11.5 cm
- ✓ Note: edema +3, WFL/H $< 70\%$ with edema +1,+2 or with medical complication, MUAC < 11.5 cm with edema +1,+2 or with medical complication, failed appetite test can be treated as in- patient



Save the Children

Admission Criteria (In-patient)

❖ Complicated Severe Acute Malnutrition

➤ WFL/H

✓ < 70% of median OR

✓ < -3Z score OR

➤ MUAC <11.5 cm OR

➤ Edema of both feet (+, ++) OR

➤ +++ Edema OR

➤ Marasmic - Kwashiorkor (WFL/H < 70% with edema OR

➤ MUAC <11.5cm with edema)

PLUS

○ Any one of the medical complications (see list below*)

○ Failed Appetite test



Save the Children

Assessment and Classification of Children for SAM

Children age 6 Months to 5 years		
Assess	Classify	Action to take
- WFL/H < 70% of median or < -3Z score OR MUAC < 11cm OR Edema of both feet (+, ++), PLUS - Any one of the medical complications (see list below*), or - Failed Appetite test - OR +++ Edema, OR - Marasmic Kwashiorkor (WFL/H < 70% with edema, OR MUAC < 11cm with edema)	Complicated Severe Acute Malnutrition	Admit for in-patient management
- WFL/H < 70% of median or < -3Z score OR MUAC < 11cm OR - Edema of both feet (+, ++) - AND - No medical complication AND pass appetite test	Uncomplicated Severe Acute Malnutrition	Manage in OTP using the OTP protocol or manage as in-patient if OTP service is not available
- WFL/H ≥ 70% to < 80% or ≥ -3Z to < -2Z score OR MUAC 11cm to < 12cm AND - No edema of both feet	Moderate Acute Malnutrition	Refer to supplementary feeding program if available, Counsel on infant and child feeding/care
- If WFL/H ≥ 80% or ≥ -2Z score OR MUAC ≥ 12 cm AND - No edema of both feet	No acute malnutrition	Congratulate and Counsel the mother on infant and child feeding/care



Save the Children

Admission Criteria (In- patient)

- **If they fulfil any one of the following:**
 1. Infants below six months of age with SAM **OR**
 2. Children 6 months to 5 years with SAM who have any one of the medical complications or failed appetite test **OR**
 3. Children with SAM and referred from OTP for in-patient care **OR**
 4. When OTP is not available in
 - ✓ Your working area or
 - ✓ Where the care taker lives or
 - ✓ If the care taker's choice is inpatient care, **all SAM children need to be admitted for in-patient care even if they do not fulfil the in-patient admission criteria**



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List of common medical complications:

1. Unable to breast feed, drink or feed
2. Vomiting everything
3. Convulsions
4. Very Weak, Lethargic or unconscious
5. Hypothermia: axillary temp $< 35^{\circ}\text{C}$ or rectal $< 35.5^{\circ}\text{C}$
6. Fever $\geq 38.5^{\circ}\text{C}$
7. Pneumonia/severe pneumonia
8. Shock
9. Dehydration (watery diarrhea with recent sunken eye balls.)
10. Dysentery
11. Persistent diarrhoea
12. Hypoglycaemia
13. Severe anemia (severe palmar pallor)
14. Jaundice
15. Bleeding Tendencies
16. Dermatositis +++
17. Corneal clouding or ulceration
18. Measles (now or with eye/mouth complication)



Save the Children

Medical complications

- **Edema grading:** bilateral edema below ankles (+); below the knees & the elbows (++); generalized edema involving the upper arms & face (+++)
- **Dermatosis grading:** few discolored or rough patches of skin (+); multiple patches on arms and/or legs (++); flaking skin, raw skin or fissures (openings in the skin) is grade +++ dermatosis
- **Measles:** If a child has generalized rash with one of the following: cough, runny nose or red eyes
- **Measles with eye or mouth complications:** If a child with measles has pus draining from the eye, clouding of cornea or mouth ulcers (deep or extensive ulcers)



Save the Children

Note

- Children with severe acute malnutrition that need in-patient care are **in danger of death** from
 - ✓ Hypoglycemia
 - ✓ Hypothermia
 - ✓ Fluid overload and
 - ✓ Undetected infections
- ❖ They cannot be treated like other children
- ❖ Their feeding and fluids must be carefully controlled, or they could die



Save the Children

Note

- ❖ To ensure the proper feeding and treatment routines, it is critical to keep these children in TFU/SC or separate rooms with in a ward
- ❖ Other health problems and infections should be treated in these rooms



Save the Children

Appetite Test

- An appetite test is done to children ages 6 months to 5 years who have no medical complications
- In order to determine whether the child can eat the RUTF (Plumpy'Nut)
- The test shows whether the child has appetite, accepts the RUTF's taste and consistency and can swallow
- If a child fails the appetite test, it reflects a severe disturbance of the metabolism and this child needs to be treated as an in-patient



Save the Children

Appetite Test

- A poor appetite shows that the child has significant infection or a major metabolic abnormality such as liver dysfunction, electrolyte imbalance, etc
- Such children with SAM are at immediate risk of death
- If a child passes the appetite test, he/she should be treated as out-patient if OTP service is available



Save the Children

Appetite Test

Why do we do appetite test?

- ✓ No signs of clinical infection in children with SAM
- ✓ No perfect correlation between metabolic malnutrition & anthropometric malnutrition
- ✓ **Metabolic malnutrition** is the common cause of death and often its sign is loss of appetite
- ✓ Appetite test is the most important criteria to admit SAM either to inpatient or OTP



Save the Children

APPETITE TEST

This is the minimum amount that malnourished patients should take to pass the appetite test

<i>Plumpy'nut</i>		<i>BP100</i>	
Body weight (Kg)	Sachets	body weight (Kg)	Bars
Less than 4 kg	1/8 to 1/4	Less than 5 kg	1/4 to 1/2
4 – 6.9	1/4 to 1/3	5 -9.9	1/2 to 3/4
7 – 9.9	1/3 to 1/2		
10 – 14.9	1/2 to 3/4	10 – 14.9	3/4 to 1
15 - 29	3/4 to 1	15 -29	1 to 1 1/2
Over 30 kg	>1	Over 30 kg	> 1 1/2

<i>RUTF paste</i>	
body weight (Kg)	Grams
3 - 3.9	15 - 20
4 - 5.9	20 - 25
6 - 6.9	20 - 30
7 - 7.9	25 - 35
8 - 8.9	30 - 40
9 - 9.9	30 - 45
10 - 11.9	35 - 50
12 - 14.9	40 - 60
15 - 14.9	55 - 75
25 - 39	65 - 90
40 - 60	70 - 100

How Does The Physiology Of Severe Malnutrition Affect Care Of The Child?

- The child with severe malnutrition must be treated differently because
 - The physiology is seriously abnormal due to **reductive adaptation**



Save the Children

What is reductive adaptation?

- The systems of the body begin to “**shut down**” with severe malnutrition
- The systems slow down and do less in order to allow survival on limited calories
- This slowing down is known as **reductive adaptation**
- As the child is treated, the body's systems must gradually "learn" to function fully again
- Rapid changes (such as rapid feeding or fluids) would overwhelm the systems, so feeding must be slowly and cautiously increased



Save the Children

How Does Reductive Adaptation Affect Care Of The Child?

- Reductive adaptation affects treatment of the child in a number of ways.
- Three important implications for care reductive adaptation for care includes:
 1. Presume and treat infection
 2. Do not give iron early in treatment
 3. Provide potassium and restrict sodium



Save the Children

Presume And Treat Infection

- Nearly all children with severe malnutrition have bacterial infections
- However, as a result of reductive adaptation,
 - ✓ The usual signs of infection may not be apparent, because the body does not use its limited energy to respond in the usual ways, such as inflammation or fever



Save the Children

Presume and treat infection

- Examples of common infections in the severely malnourished children are
 - ✓ Ear infection
 - ✓ Urinary tract infection and
 - ✓ Pneumonia
- ❖ Assume that infection is present and treat all severe malnutrition admissions with broad spectrum antibiotics
- ❖ If a specific infection is identified (such as Shigella), add specific appropriate antibiotics to those already being given.



Save the Children

Antibiotics for Severely Malnourished Children

- Give all severely malnourished children antibiotics for presumed infection even if they do not have clinical sign of systemic infections.
- Selection of antibiotics depends on the presence or absence of complications
- ❖ Complications include septic shock, hypoglycemia, hypothermia, skin infections or dermatosis, respiratory or urinary tract infections....



Save the Children

Antibiotics for Severely Malnourished Children

IF:	Give
No Medical Complications or Only failed appetite test	Oral Amoxicillin If Amoxicillin is not available, give Cotrimoxazole for 5 days
Medical Complications (Unable to feed , Vomiting everything, pneumonia...)	Gentamycin IV or IM once daily for 7 days, plus: Ampicillin IV every 6 hours for 2 days followed by oral Amoxicillin for 5 days If child is not seriously sick looking and can take oral medication, avoid IV Ampicillin and start with oral Amoxicillin and IM Gentamycin.
If the child has medical complications other than mentioned above, (dehydration, dysentery, persistent diarrhea, malaria...	Oral Amoxicillin and IM Gentamicin for 7 days. If child is seriously sick looking, instead of oral Amoxicillin & IM Gentamycin start with IV Ampicillin and IV Gentamycin

Antibiotics for Severely Malnourished Children

IF:	Give
If child fails to improve within 48 hours, ADD:	Chloramphenicol IV or IM every 8 hours for 5 days. (Give IV every 6 hours if suspect meningitis.) or ceftriaxone 100 mg/kg divided into 2 doses IV or IM + Gentamicin once daily for a total of 10 days.
If a specific infection requires an additional antibiotic, ALSO GIVE:	Specific antibiotic as indicated.



Save the Children

Antibiotics And Oral Dose

WEIGHT Kg	AMOXYCILLIN (50-100mg/Kg/d) Give two times/day			COTRIMOXAZOLE (SMX+TMP) 25mg SMX+5mg TMP/Kg/d Give two times/day		
	ADULT TABLET 500mg	PEDIATRI C TABLET 125 mg	SYRUP 125 mg/5ml	TABLET 240mg	TABLET 120mg	SYRUP 120mg/5ml
<5	1/4	1	5.0 ml	1/2	1	5ml
5-10	1/2	2	10ml	1	2	10ml
10-20	1	4	20ml	2	4	20ml
20-35	1 1/2			3		
>35	2			4		



Save the Children

Do Not Give Iron Early In Treatment

- Due to reductive adaptation, the severely malnourished child makes less hemoglobin than usual
- Iron that is not used for making hemoglobin is put into storage
- Thus, there is “extra” iron stored in the body, even though the child may appear anemic
- Giving iron early in treatment will not cure anemia, as the child already has a supply of stored iron



Save the Children

Do not give iron early in treatment

- Giving iron early in treatment can also lead to “free iron” in the body
- Free iron can cause problems in three ways:
 1. Free iron promotes bacterial growth and can make some infections worse
 2. Free radical formation
 3. The body tries to protect itself from free iron by converting it to **FERRITIN (a blood cell protein that contain iron)**
- This conversion requires energy and amino acids and diverts these from other critical activities
- Later, as the child recovers and begins to build new tissue and form more red blood cells, the iron in storage will be used and supplements will be needed.



Save the Children

Provide Potassium And Restrict Sodium

- Normally the body uses a lot of energy maintaining the appropriate balance of potassium inside the cells and sodium outside the cells.
- This balance is critical to maintaining the correct distribution of **water inside the cells, around the cells and in the blood**
- In reductive adaptation, the “pump” that usually controls the balance of potassium and sodium runs slower
- As a result, the level of sodium in the cells rises and the potassium leaks out of the cells and is lost (for example, in urine or stools)



Save the Children

Provide Potassium And Restrict Sodium

- Fluid may then accumulate outside of the cells (as in edema) instead of being properly distributed through the body.
- ❖ All severely malnourished children should be given potassium to make up for what is lost
- ❖ They should also be given magnesium, which is essential for potassium to enter into the cells and be retained
- ❖ The commercially prepared F-75 and F-100 have enough potassium and magnesium and there is no need to supplement.



Save the Children

Note

- Malnourished children already have excess sodium in their cells, so sodium intake should be restricted
- If a child has diarrhea, a special rehydration solution called ReSoMal should be used instead of regular WHO ORS
- ReSoMal has less sodium and more potassium than regular WHO ORS.



Save the Children

The NOT TO DOs

1. Do NOT give DIURETICS to treat Edema
2. Do NOT give IRON during the Initial feeding phase-add iron only after the child has been on F100 for 2days
3. Do NOT provide high protein formula, over 1.5gm/kg/day in initial phase
4. Do NOT give IV Fluids routinely
5. Don't give deworming tablets during phase I



Save the Children

The NOT TO DOs

■ Do not give diuretics to treat edema

- The edema is partly due to potassium and magnesium deficiencies that may take about 2 weeks to correct
- The edema will go away with proper feeding including a mineral mix containing potassium and magnesium
- Giving a diuretic will worsen the child's electrolyte imbalance and may cause death

■ Do not give iron during phase 1 and transition phase

- Add iron only when the child is in phase 2
- Giving iron early in treatment can have toxic effects and interfere with the body's ability to resist infection



Save the Children

The NOT TO DOs

- Do not give high protein formula (over 1.5 g protein per kg body weight daily)
 - Too much protein in the first days of treatment may be dangerous because the severely malnourished child is unable to deal with the extra metabolic stress involved
 - Too much protein could overload the liver, heart, and kidneys and may cause death
- Do not give IV fluids routinely
 - IV fluids can easily cause fluid overload and heart failure in a severely malnourished child
 - Only give IV fluids to children with signs of shock
 - Treatment will be described in Initial Treatment



Save the Children

A. Inpatient care

- ❑ In-patient care has three important phases of treatment**
 - Phase 1 (Stabilization phase)
 - Transition phase
 - Phase 2 (rehabilitation phase)



Save the Children

Phase I (Stabilization phase)

- Children with complicated SAM are initially admitted to an in-patient facility for stabilization
- These children are admitted to phase I room.
- Life-threatening medical complications are treated
- Routine drugs are given to correct specific deficiencies (antibiotics, folic acid, vit A..)
- Feeding with f-75 milk (low caloric and sodium)
- ❖ The children in phase 1 should be together in a separate room or section of the ward and not mixed with other patients



Save the Children

Transition Phase

- This phase is important for slow transition as the introduction of large amounts of RUTF or F100 could lead to
 - Imbalance of body fluids and severe medical complications
 - Important to avoid electrolyte disequilibrium.
- The diet used is f100(100ml=100kcal)
- Quantity of f100 given is equal to the quantity of f75 given in phase 1 or an equivalent amount of rutf
- Expected weight gain should be around 6 g/kg/day.
- Transition takes 3 days, during which f-100 or RUTF should be given



Save the Children

Transition Phase

- In this phase:
 - Routine drugs are continued
 - Feeding with RUTF or F100 is started



Save the Children

Phase 2 (Rehabilitation Phase)

- Whenever patients have good appetite and no major medical complication they enter Phase 2
- Many patients with a good appetite can be admitted directly into Phase 2
- Give RUTF or F100 according to **look-up tables**.
- Expected weight more than 8 g/kg/day.
- Start deworming and iron supplementation



Save the Children

During phase 2:

■ During phase 2:

- Routine drugs, deworming tablets and iron, are started
- Feeding with RUTF or F100 is increased in amount
- Child starts gaining weight



Save the Children

Criteria to move from Phase I to transition phase

- Return of appetite (easily finishes the F-75 feeds)
- Reduced edema or minimal edema (++ or less)
- Main medical complications are under control
- No IV line, no NGT
- ✓ The child may also smile at this stage.
- ✓ Children with +++ edema should wait in phase I at least until their edema has reduced to ++ edema
- ✓ These children are particularly vulnerable to fluid overload and heart failure



Save the Children

Move the child back to Phase 1

- If the patient gains weight more rapidly than 10g/kg/d
- If there is increasing edema
- If a child who does not have edema develops edema
- If there is a rapid increase in the size of the liver
- If tense abdominal distension develops
- If the patient gets significant re-feeding diarrhea - weight loss
- If patient develops medical complication
- If Naso-Gastric Tube is needed
- If patient takes less than 75% of the feeds



Save the Children

Criteria To Progress From Transition Phase To Phase II

- A good appetite, taking at least 90% of the RUTF or F100 prescribed for Transition Phase.
- Edematous patients (kwashiorkor) should remain in Transition Phase until there is a definite and steady reduction in oedema (now at + level).



Save the Children

Move Back From Phase II To Phase I

- Develops any signs of a complication
- Increase/development of edema
- Development of refeeding diarrhea sufficient to lead to weight loss
- Weight loss for 2 consecutive weighing
- Static weight for 3 consecutive weighing
- Fulfilling any of the criteria of “failure to respond to treatment”



Save the Children

Failure to respond

- A child is failing to respond if he or she:
 - ✓ Does not improve initially (primary failure); or
 - ✓ Deteriorates/regress after having progressed satisfactorily to phase 2 with a good appetite and
 - ✓ Weight gain in transition phase for in-patients and deteriorates after an initial response in out-patients (secondary failure).



Save the Children

Failure to Respond to Treatment

Criteria for failure to respond and to move back from out-patient to in-patient care	Time after admission
<i>Primary failure to respond</i>	
Failure to start to lose edema	14 days
Edema still present	21 days
Failure to gain any weight (non-edematous children)	21 days
<i>Secondary failure to respond</i>	
Failure of Appetite test	At any visit
Weight loss of 5% of body weight	At any visit
Weight loss for two successive visits	During OTP care
Failure to gain more than 2.5g / kg / d for 21days (after loss of edema (kwashiorkor) or after day 14 (marasmus))	During OTP care

Failure to Respond to Treatment

- **Possible causes :**
 - ✓ Inadequate feeding
 - ✓ Specific nutrient deficiencies
 - ✓ Untreated infection
 - ✓ Underlying systemic disorder
- A child may have a primary or secondary failure to respond to treatment.



Save the Children

Outpatient Treatment Program (OTP)

- Outpatient care is intended for children presenting SAM
 - Without medical complications
 - Passed appetite test and
 - For children that have transferred from inpatient care
- ❖ Most children with SAM have no medical complications and can be treated in outpatient care



Save the Children

Outpatient Treatment Program (OTP)

- ❖ During the OTP, the children with SAM receive:
 - Ready-to-use therapeutic food (RUTF) which are taken at home
 - Routine medicines which are taken at home,
 - Follow up care at the outpatient care site every one week
 - During follow up their progress in appetite and weight and presence or absence of complication will be assessed.



Save the Children

The 10 Steps for Management of Severe Malnutrition

- Sequential management of severe SAM recommended
- The two main aspects of management are
 - 1) Treatment of complications
 - 2) Dietary management and rehabilitation
- WHO describes the 10 steps of management of severe malnutrition
- These steps are accomplished in two phases:
 - An initial stabilization phase where the acute medical conditions are managed (steps 1-6);and
 - A longer rehabilitation phase (steps 7-10)



Save the Children

The 10 Steps for Management of Severe Malnutrition

■ The first six steps (Steps 1-6) are related to treatment of complications, and can be remembered with the mnemonic **SHIELDED**.

1. **S** - Sugar deficiency-ie, hypoglycemia
2. **H** - Hypothermia
3. **I** -Infection and septic shock
4. **EL** -Electrolyte imbalance
5. **DE** - Dehydration
6. **D** -Deficiencies of folic acid, vitamins, and other micro-nutrients

The 10 Steps for Management of Severe Malnutrition

■ Steps 7-10 are related to dietary management which can be remembered by the mnemonic **BEST** as described below:

7. **B**-Beginning of feeding: F-75

8. **E**-Energy dense feeding: F- 100

9. **S**-Stimulation of emotional and sensorial development

10. **T**- Transfer to home-based diets before discharge or transfer to nutritional rehabilitation centers

Common Incorrect Practices in Initial

Common Incorrect Practices in Initial Treatment – These cause deaths:	Correct Practice
Child not fed at night	During initial treatment ensure that the child is fed every 3 hours at night.
IV fluids given even though child is not in shock	Give IV only if signs of shock (cold hand plus slow capillary refill or weak/fast pulse).
Diuretics given to treat edema	Do not give. Edema will resolve with correct initial treatment using F-75 with correct minerals and vitamins.
High protein diet given immediately	Give F-75 until the child stabilizes; then start F-100.

Monitor case management practices

Common Incorrect Practices in Initial Treatment – These cause deaths:	Correct Practice
Antibiotics not given because no clinical signs of infection	Presume infection and give antibiotics to all severely malnourished children as described in the manual.
Standard ORS used instead of ReSoMal	Give ReSoMal to severely malnourished children with diarrhea.
Child left uncovered at night	Provide blanket and ensure the child is covered at night.
Anemia treated with iron from admission	Wait to start iron until the child has been on F-100 for 2 days.

F-75, F- 100, & RUTF



Save the Children

Learning Objectives

- Preparing F-75 and F-100
- Planning feeding for a 24-hour period for a child who is in:
 - ✗Phase I
 - ✗Transition Phase and
 - ✗Phase II
- Measuring and giving feeds to children
- Recording intake and output
- Planning feeding for a ward



Save the Children

Introduction

- Feeding is obviously a critical part of managing severe malnutrition
- Feeding must be started cautiously, in frequent, small amounts
- If feeding begins too aggressively, or if feeds contain too much protein or sodium,
 - The child's systems may be overwhelmed, and the child may die



Save the Children

F-75

- F-75 is the "starter" formula to use during phase 1
- Beginning as soon as possible and continuing for 2-7 days until the child is stabilized
- Severely malnourished children cannot tolerate usual amounts of protein and sodium at this stage,
- They may die if given too much protein or sodium
- They also need glucose, so they must be given a diet that is low in protein and sodium and high in carbohydrate



Save the Children

F-75

- F-75 is specially made to meet the child's needs **without overwhelming the body's systems in the initial stage of treatment**
- Use of F-75 prevents deaths
- F-75 contains 75 kcal and 0.9 g protein per 100 ml



Save the Children

F-100 Formulas

- As soon as the child is stabilized on F-75
- F-100 is used as a "catch-up" formula to rebuild wasted tissues during transition phase and phase 2
- F-100 contains more calories and protein: 100 kcal and 2.9 g protein per 100 ml.



Save the Children

RUTF

- Is high-energy, nutrient-dense food
 - ✓ Used for nutrition rehabilitation during phase 2 of in-patient care and outpatient care (OTP)
- Good for out-patient or home management of SAM
- It is similar in composition to F100 (except RUTF contains iron)
- The RUTF plumpy'nut **has a caloric value of 500 kilocalories (kcal) and 12.5 gm of protein per one sachet of product (92 gms)**
- There are two types:
 - ✓ Soft lipid-based paste (plumpy'nut®) or crushable nutrient bar (BP100).



Save the Children

RUTF

- The principle behind the recipes is to provide the energy and protein needed for catch-up

Note:

- The commercially manufactured F75 and F100 have all the necessary minerals and vitamins.
- Thus, there is no need to add mineral mix or vitamins or CMV in to these feeds.



Save the Children

Discharge Criteria

- The national protocol recommends that children with SAM should be discharged
 - ✚ As cured from therapeutic feeding program (TFP) if the child attains the following criteria:
- **Age 6 months to 5 years:**
- ✚ A weight-for-length or height $\geq 85\%$ on more than one occasion
 - ▶ Two days for in-patients
 - ▶ Two weeks for out-patients **and**
- ✚ No edema for 10 days (in-patient) or 14 days (out-patient) or
- ✚ A target weight gain reached for two consecutive measurements or visits if the child is admitted with MUAC



Save the Children

Discharge Criteria

- ⊕ Age less than 6 months or less than 3 kg being breast fed:
 - ▶ If he/she is gaining weight on breast milk alone after the **Supplemented Suckling technique has been used**
- ⊕ Age less than 6 months with no prospect of being breast-fed:
 - ▶ When they reach $\geq 85\%$ weight for length and switched to infant formula or other animal milk



Save the Children

Discharge Criteria

- It usually requires about 2 – 6 weeks for a child to achieve the target weight
- ✓ If the principles of care for SAM are followed
- If a child leaves the TFP before achieving 85 % weight-for-height/length, he/she is likely to get worse and have to return, or h/she may die at home
- If the caretaker decides to go home before fulfilling the discharge criteria and there is no OTP in the area, many preparations must be made to ensure that the caretakers can continue care at home
- Follow-up visits are essential



Save the Children

Managing Of Medical Complication



Save the Children

Learning Objectives

- Identifying and managing the severely malnourished child with:
 - Hypoglycemia
 - Hypothermia
 - Shock
 - Severe anemia
 - Corneal clouding and ulceration
 - Watery diarrhea and/or vomiting and dehydration
 - Heart failure
 - Infections
 - Abdominal distension



Save the Children

Introduction

- The focus of managing medical complication
 - ▶ Is to prevent death while stabilizing the child.
- The first step is to check the child for serious medical complications and provide emergency treatment as necessary
- Any child presenting to the hospital or health center should be checked for serious medical complications as part of standard procedure
- If serious medical complications are present, many procedures must be done very quickly, almost simultaneously
- Practice and experience is needed to perform efficiently in an emergency room as a team.



Save the Children

Daily Care

- ◆ Handle the child gently
- ◆ Give Routine medicines and other medications
- ◆ Give antibiotics as prescribed
- ◆ Give vitamin A
- ◆ Give Measles Vaccine
- ◆ Give Deworming tablets
- ◆ Give Iron
- ◆ Care for skin and bathe the child
- ◆ Care for the eyes



Save the Children

Handle the child gentle

- Children with SAM must be handled very gently, especially at the beginning of their care
- The body of a child with SAM is fragile and bruises easily
- The child needs all his energy to recover, so he must stay calm and not become upset
- It is important to speak quietly and handle children as little as possible at first
- Hold and touch children with loving care when feeding, bathing, weighing, and caring for them.



Save the Children

Folic Acid

- Folic acid is a vitamin important for treating and preventing anemia and repairing the damaged gut
- There is sufficient folic acid in commercially manufactured F75, F100 and RUTF to treat mild folate deficiency
- Folic acid: On the day of admission, one single dose of folic acid (5 mg) can be given to children with clinical signs of anemia.



Save the Children

Vitamin A

- Vitamin A should be given orally to all children with SAM on the day of admission (on day 1)
 - ✓ Except those with edema or those who received vitamin A in the past 6 months
 - ✓ However, for children with edema, it should be given once the edema subsides.
 - ✓ Children with measles and any clinical signs of vitamin a deficiency should be given vitamin a on admission even if edema is present
- Give a second dose on discharge if the child finishes the treatment in in-patient care



Save the Children

Timing and oral dosages of vitamin A

Targets	Timing	Age	Dosage
All children*	Day 1	<6 months	50 000 IU
		6 – 12 months	100 000 IU
		> 12 months and >8 Kg **	200 000 IU
All children who are not transferred to OTP	Day 15 or day of discharge	Same age-specific dose	
All children who will be transferred to OTP	On the 4 th week of treatment	Same age-specific dose	
Children with eye signs or has measles or had measles in the past 3 months	Day 1,2 and 15	Same age-specific dose	



Save the Children

Measles vaccine

- All children from 9 months who are not vaccinated should be given measles vaccine both on admission and discharge after Phase 2
- The first measles dose often does not give a protective antibody response
- It is given because it ameliorates the severity of incubating measles and partially protects from nosocomial measles
- The second dose is given to provoke protective antibodies.



Save the Children

Give Deworming tablets

- Worms are common in older children who play outside, and they can be a problem in severely malnourished children
- They can cause dysentery and anemia
- Albendazole or Mebendazole is given at the start of phase 2 for children greater than 1 year that will remain as in-patient
- For those transferred to outpatient Albendazole or Mebendazole is given at the second outpatient visit (after 7 days).



Save the Children

Age	Give a single dose if child is ≥ 2 years and didn't get within the previous 6 months			
	Mebendazole		Albendazole	
2 - 5 years	500mg tablet, or 5 tablets of 100 mg	Syrup, 100mg/5ml	400mg Tablet	Syrup, 100mg/5ml
	1 tablet (500mg)	5 tsp (25ml)	1 tablet	4 tsp (20ml)



Save the Children

Iron

- Even if the child is anemic, he/she should not be given iron until he is recovering and he/she is in phase 2
- If given earlier, iron can have toxic effects and reduce resistance to infection
- Provide iron to a child who is in phase 2
- Add 1 crushed tablet of ferrous sulphate (200 mg) to each 2 to 2.4 liters of F100
- Don't give iron to a child who is taking RUTF because it already has the required amount of iron.



Save the Children

Eye care

- Chloramphenicol or tetracycline eye drops are given for eye infection or possible eye infection
- Atropine eye drops are used to relax the eye when there is corneal involvement (i.e., corneal clouding or ulceration)
- In some cases both types of eye drops may be needed



Save the Children

If the child has:	Then:
Bitot's spots only (no other eye signs)	No eye drops needed
Pus or inflammation	Give chloramphenicol or tetracycline (1%) eye drops
Corneal clouding or Corneal ulceration	Give both: - Chloramphenicol or tetracycline (1%) eye drops and - atropine (1%) eye drops



Save the Children

IV. Summary table of systematic treatment of patients

	Direct admission to in-patient (Phase 1)	Direct admission to out-patient (Phase 2)
Vitamin A	- 1 dose at admission (conditional) - 1 dose on discharge - do not give when transferred to OTP management - it will be given in OTP	- 1 dose on the 4th week (4th visit)
Folic Acid	- 1 dose at admission if signs of anaemia	- 1 dose at admission if signs of anaemia
Amoxicillin	- Every day in Phase 1 + 4 more days in Transition	- 1 dose at admission + give treatment for 7 days at home
Malaria	- According to national protocol	- According to national protocol
Measles (from 9 months old)	- 1 vaccine at admission if no card - 1 vaccine at discharge	- 1 vaccine on the 4th week (4th visit)
Iron	- Add to F100 in Phase 2	- No - iron is already in all RUTF
Deworming	- 1 dose at the start of Phase 2	- 1 dose on the 2nd week (2nd visit)

Thank you!!!

